

VALE DO RIO BRANCO REDD+ GROUPED PROJECT



Project title	Vale do Rio Branco REDD+ Grouped Project
Project ID	Verra Project ID
Crediting period	01-January-2025 to 31-December-2064
Project lifetime	01-January-2025 to 31-December-2064; 40-year lifetime
(CCB) GHG accounting period	01-January-2025 to 31-December-2064; 40-year lifetime
Original date of issue	-
Most recent date of issue	-
Version	1.0
VCS Standard version	4.7
CCB Standards version	3.1
Project location	Brazil, Roraima
Project proponent(s)	ZEG Florestal S.A.

Validation/verification body	Undefined
History of CCB status	Not applicable
Gold Level criteria	Under development
Expected verification schedule	Undefined
Prepared by	ZEG Florestal S.A.

CONTENTS

1	SUMMARY OF PROJECT BENEFITS	4
1.1	Unique Project Benefits	4
1.2	Standardized Benefit Metrics	4
2	PROJECT DETAILS	5
2.1	Project Goals, Design and Long-Term Viability	5
2.2	Without-project Land Use Scenario and Additionality	70
2.3	Safeguards and Stakeholder Engagement	83
2.4	Legal Status and Property Rights	83
2.5	Additional Information Relevant to the Project	91
3	CLIMATE	92
3.1	Application of Methodology	92
3.2	Quantification of Estimated GHG Emission Reductions and Removals	100
3.3	Monitoring	100
3.4	Optional Criterion: Climate Change Adaptation Benefits	100
4	COMMUNITY	122
4.1	Without-Project Community Scenario	122
4.2	Net Positive Community Impacts	122
4.3	Other Stakeholder Impacts	122
4.4	Community Impact Monitoring	122
4.5	Optional Criterion: Exceptional Community Benefits	122
5	BIODIVERSITY	122
5.1	Without-Project Biodiversity Scenario	122
5.2	Net Positive Biodiversity Impacts	122
5.3	Offsite Biodiversity Impacts	122
5.4	Biodiversity Impact Monitoring	122
5.5	Optional Criterion: Exceptional Biodiversity Benefits	122
6	REFERENCES	123

1 SUMMARY OF PROJECT BENEFITS

1.1 Unique Project Benefits

This section is under development.

1.2 Standardized Benefit Metrics

This section is under development.

2 PROJECT DETAILS

2.1 Project Goals, Design and Long-Term Viability

2.1.1 Summary Description of the Project (VCS, 3.2, 3.6, 3.10, 3.11, 3.13, 3.14; CCB, G1.2)

The Vale do Rio Branco REDD+ Grouped Project is an AFOLU project located in Roraima State, in the Brazilian Amazon Biome. Developed by ZEG Florestal, it is a grouped project under Reduced Emissions from Deforestation and Degradation (REDD) category, within Avoided Planned Deforestation (APD) project type and, so far, is not covered by a jurisdictional REDD+ program. Its first instance covers an area of approximately 5,926 hectares in Caracaraí municipality, southern Roraima State.

Initially, the first instance belongs to the Vale Verde property, a 14,029 hectares property located along the Branco Riverbank, approximately 35 km from the BR-174 highway, that connects Manaus, Amazonas state capital, to Boa Vista, Roraima State capital, which provides the main land access to the city. The farm was previously managed by Madeireira Vale Verde (timber company) from 2006 to 2018, during which time Sustainable Forest Management was performed in the area. Operations ceased and the property was acquired by a rural producer, who is the current owner.

Promoting the conservation of forest land and reducing net greenhouse gas (GHG) emissions by preventing associated risks of degradation, deforestation, and forest fires are the project's main objectives. To achieve this, the project seeks to implement climate-focused strategies such as forest monitoring through remote sensing and local surveillance, as well as measures to prevent degradation, deforestation and firefighting efforts.

In terms of biodiversity, the project will introduce technologies for monitoring flora and fauna and establish a comprehensive biodiversity inventory and database. Additionally, the project will develop opportunities concerning bioeconomic development in the region, aiming to contribute to socioeconomic improvements and general well-being in local communities, with respect to their cultural traditions and ways of life. Also, the project intends to develop mechanisms to promote transparency for all information related to implemented activities. Further information will be provided in the project description document.

The project estimates an annual average of 133,515 tonCO_{2e} and a total of 3,871,957 tonCO_{2e} in reductions for the first instance over 29 years, within the 40-year project crediting period and 40-year project lifetime.

2.1.2 Audit History (VCS, 4.1)

This section is under development.

2.1.3 Sectoral Scope and Project Type (VCS, 3.2)

Table 2.1.3-A displays information about the sectoral scope and project type.

Table 2.1.3-A: Sectoral Scope and Project Type.

Sectoral scope	14: Agriculture, forestry, and other land use
AFOLU project category ¹	Reduced Emissions from Deforestation and Degradation (REDD)
Project activity type	Avoiding Planned Deforestation (APD)

2.1.4 Project Eligibility (VCS, 3.1, 3.6, 3.8, 3.18, 4.1; CCB Program Rules, 4.2.4, 4.6.4)

In accordance with the CCB Program Rules v3.1, the scope of the CCB Program encompasses all activities related to land-based projects aimed at generating positive net benefits for climate, community, and biodiversity. This includes projects that reduce greenhouse gas emissions from deforestation, such as Vale do Rio Branco REDD+ Grouped Project.

The project is classified under VCS Scope 14, which applies to Reduced Emissions from Deforestation and Degradation (REDD) projects. According to the VCS Standard v4.7, eligible activities within REDD projects are those that achieve a net reduction in GHG emissions by mitigating deforestation and/or forest degradation. This includes activities aimed at Avoiding Planned Deforestation (APD) and Avoiding Unplanned Deforestation and/or Degradation (AUDD). REDD projects are designed to prevent deforestation and/or degradation, whether they are planned (authorized and sanctioned) or unplanned (unauthorized). While activities categorized as degradation can be incorporated into REDD projects, such as unauthorized logging and fuelwood extraction, they must be combined with other REDD activities, as avoiding planned degradation is categorized under Improved Forest Management (IFM).

Therefore, project activities must adhere to the definitions provided in Appendix 1 of the VCS Standard v4.7:

“1) Avoiding Planned Deforestation (APD): This category includes activities that reduce net GHG emissions by permanently stopping or reducing deforestation on forest lands that are legally authorized and documented for conversion. Planned deforestation encompasses activities where a forest system would have been cleared and replaced by a different forest system with a lower carbon stock and where the recovery of timber was not the primary objective of the initial forest clearance.

¹ See Appendix 1 of the VCS Standard

2) Avoiding Unplanned Deforestation and/or Degradation (AUDD): This category includes activities that reduce net GHG emissions by stopping deforestation and/or degradation of degraded to mature forests that would have occurred in any forest configuration.”

In conformance with the VCS Standard v4.7, project activities must occur on forested land designated for conversion to non-forest land, as evidenced in Section 3.1.2 Applicability of Methodology.

Adhering to the VCS Program Definitions v4.5 requirements, this project defines forests as "land with woody vegetation that meets an internationally accepted definition (e.g., UNFCCC, FAO, or IPCC) of what constitutes a forest, which includes threshold parameters such as minimum forest area, tree height, and level of crown cover, and may include mature, secondary, degraded, and wetland forests." The land in the project area qualifies as forest based on the FAO (2023) definition: *“Land spanning more than 0.5 hectares with trees higher than 5 meters and a canopy cover of more than 10 percent, or trees able to reach these thresholds in situ. It does not include land that is predominantly under agricultural or urban land use.”* This qualification applies to the land as it was 10 years prior to the project start date (2014). The Brazilian Forest Service (SFB, 2019) considers the types of woody vegetation that align most closely with the FAO definition, including 13 vegetation categories described by the Brazilian Institute of Geography and Statistics (IBGE) Classification System. One such category is Dense Ombrophilous Forest, which covers the entire Instance #01 project area, as detailed in Section 2.2.1 Conditions Prior to Project Initiation and Land Use Scenarios without the Project.

The conversion of forest land to a deforested condition is legally permitted by Brazil's Forest Code, Law no. 12,651 of May 25, 2012 (Native Vegetation Protection Law). Generally, this law mandates that 80% of private property located in a forested area of the Legal Amazon must be preserved as Legal Reserve, along with Permanent Preservation Areas. The remaining unprotected area of the property is available for other land use, which can include legally permitted deforestation for production purposes. The Forest Code also stipulates that the Legal Reserve can be reduced to 50% when the state has approved Ecological-Economic Zoning (ZEE in Portuguese) and more than 65% of its territory is occupied by public domain Conservation Units, duly regularized, and by homologated indigenous lands. This condition applies to the Roraima State, where the project is located. Roraima State Decree No. 33,467-E, of October 31, 2022, approves the Legal Reserve reduction by 50% of the rural property area, without prejudice to the application of the rules on Permanent Preservation Areas, for properties located in forest areas in Roraima State. Consequently, it is legally permitted to deforest up to 50% of the property where the project is located.

Additional relevant information on VCS Program requirements and project eligibility is presented in Table 2.1.4-A, in accordance with VCS Standard v4.7.

Table 2.1.4-A: VCS Standard requirements and project conformance.

VCS Standard Section	Description	Project Conformance
3.1 General Requirements	3.1.1 Projects shall meet all applicable rules and requirements set out under the VCS Program, including this document. Projects shall be guided by the principles set out in Section 2.2.1.	As discussed at each reference section throughout this document, the project meets all applicable rules and requirements set out under the VCS Program and is guided by Relevance, Completeness, Consistency, Accuracy, Transparency and Conservativeness principles, as set out in VCS Standard, v4.7, Section 2.2.1.
3.1 General Requirements	3.1.2 Projects shall apply methodologies eligible under the VCS Program. Methodologies shall be applied in full, including the full application of any tools or modules referred to by a methodology, noting the exception set out in Section 3.14.1. The list of methodologies and their validity periods are available on the Verra website.	This grouped project applies VM0007 REDD+ Methodology Framework (REDD+MF), eligible under the VCS Program. More information about modules and tools applied can be found in Section 3.1.1 Title and Reference of Methodology and 3.1.2 Applicability of Methodology.
3.1 General Requirements	3.1.3 Projects shall apply the latest version of the applicable methodology in all cases unless a grace period applies to the project as set out in 3.22 below. Projects shall update to the latest version of the methodology when reassessing the baseline or renewing a crediting period.	The project applies the latest version of methodologies eligible under the VCS Program, as described on Section 3.1.1 Title and Reference of Methodology.
3.1 General Requirements	3.1.4 Projects and the implementation of project activities shall not lead to the violation of any applicable laws, regardless of whether or not the law is enforced.	The project and the implementation of the project activities do not lead to the violation of any applicable laws, as described on

VCS Standard Section	Description	Project Conformance
		Section 2.5.1 National and Local Laws.
3.1 General Requirements	3.1.5 Where projects apply methodologies that permit the project proponent its own choice of model (see the VCS Program Definitions for the definition of model), the model shall meet the requirements set out in the VCS Methodology Requirements, and it shall be demonstrated at validation that the model is appropriate to the project circumstances (i.e., use of the model will lead to an appropriate quantification of GHG emission reductions or carbon dioxide removals).	Not applicable, the project applies VCS VM0007 REDD+ Methodology Framework.
3.1 General Requirements	3.1.6 Where projects apply methodologies that permit the project proponent to choose a third-party default factor or standard to ascertain GHG emission data and any supporting data for establishing baseline scenarios and demonstrating additionality, such default factor or standard shall meet the requirements set out in the VCS Methodology Requirements.	Emission factors for biomass burning were sourced directly from IPCC documents, specifically the Guidelines for National Greenhouse Gas Inventories (2006) and Climate Change 2021: The Physical Science Basis (2021). Any other factors used will comply with the requirements set out in the VCS Methodology Requirements.
3.1 General Requirements	3.1.7 Where the rules and requirements under an approved GHG program conflict with the rules and requirements of the VCS Program, the rules and requirements of the VCS Program shall take precedence.	Not applicable, the project applies VCS VM0007 REDD+ Methodology Framework.

VCS Standard Section	Description	Project Conformance
3.1 General Requirements	3.1.8 Where projects apply methodologies from approved GHG programs, they shall conform with any specified capacity limits (see the VCS Program Definitions for the definition of capacity limit) and any other relevant requirements set out with respect to the application of the methodology and/or tools referenced by the methodology under those programs.	There are no capacity limits regarding the methodologies applied on this project.
3.1 General Requirements	3.1.9 Where Verra issues new VCS Program rules, the effective dates of these requirements are set out in Appendix 3 Document History and Effective Dates or equivalent for other program documents and are listed in a companion Summary of Effective Dates document which corresponds with each update.	Project was designed under VCS Standard v4.7, modules and tools are described in Section 3.2.1 Applicability of Methodology, and any update in these versions shall be implemented in the baseline reassessment.
3.6 Project Design	3.6.1 Projects may include multiple project activities where the methodology applied to the project allows more than one project activity and/or where projects apply more than one methodology.	The project applies VCS VM0007 REDD+ Methodology Framework, with Avoiding Planned Deforestation (APD) activity only.
3.6 Project Design	3.6.2 Where more than one methodology has been applied to a project with multiple project activities, the following applies: 1) Each project activity shall be specified separately in the project description, referencing the relevant methodology. 2) All criteria and procedures set out in the applied methodologies in relation to applicability conditions, demonstration of additionality, determination of baseline scenario, and GHG emission reduction and removal quantification shall be applied separately to each project activity,	The project applies VCS VM0007 REDD+ Methodology Framework, with Avoiding Planned Deforestation (APD) activity only, and if necessary, descriptions will be specified separately for each grouped project instance.

VCS Standard Section	Description	Project Conformance
	noting the following: a) A single set of criteria and procedures for the demonstration of additionality may be applied where the applied methodologies reference the same additionality tool and/or procedures, and where separate demonstration of additionality for each project activity is not feasible. For example, separate demonstration of additionality may not be practicable in project activities that are implemented at a single facility and therefore represent a single investment. The onus is on the project proponent to demonstrate to the validation/verification body that separate demonstration of additionality is not practicable, otherwise separate demonstration of additionality shall be provided. Where a methodology specifies requirements for demonstrating additionality in addition to those specified in the referenced additionality tool and/or procedures, such requirements shall be adhered to. b) The criteria and procedures for identifying the baseline scenario may be combined where the relevant methodologies or the referenced additionality tool and/or procedures specify criteria and procedures for combining baseline scenarios. 3) The criteria and procedures relating to all other aspects of the methodologies may be combined. 4) Where AFOLU projects are required to undertake non-permanence risk assessment and buffer withholding determination, this shall be done separately for each project activity.	

VCS Standard Section	Description	Project Conformance
3.6 Project Design	3.6.3 AFOLU projects that include multiple project activities shall conform with the respective project requirements of each included AFOLU category.	The project applies VCS VM0007 REDD+ Methodology Framework, with Avoiding Planned Deforestation (APD) activity only, conforming with the respective project activity requirements.
3.6 Project Design	Multiple Project Activity Instances (3.6.4 - 3.6.9)	Multiple project activity instances must adhere to each project activity criteria and requirements and could be combined in baseline determination or additionality demonstration.

2.1.5 Transfer Project Eligibility (VCS, 3.23, Appendix 2)

This section is under development.

2.1.6 Project Design (VCS, 3.6)

Indicate if the project has been designed as:

- ☐ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☒ Grouped project

2.1.6.1 Eligibility Criteria for Grouped Projects (VCS, 3.6; CCB, G1.14)

The eligibility criteria for the Vale do Rio Branco REDD+ Grouped Project are under development and will be fully detailed in the project description document, as follows:

- The Vale do Rio Branco REDD+ Grouped Project, implemented in Roraima State, Brazil, will encompass instances within the Roraima's Productive Use Zone (ZUP), which is the Project Zone. This zone is designated for agricultural productive land use in Roraima, providing a consistent legal framework to support its development and ensuring a homogeneous area for the project activities. Detailed information is in Section 2.1.16 - Project Zone Map and Project Location.

- New instances must be included in the Project Zone's geographic area.
- No new instances will be added if they overlap with any of the components of another AFOLU project's zone.
- New instances must not be, or have ever been, enrolled in any other VCS project.
- New instances shall use and apply the technologies and measures in the same manner that will be described in the project description document.
- The project applies VCS VM0007 REDD+ Methodology Framework, with Avoiding Planned Deforestation (APD) activity for its initial and future instances.
 - New instances shall meet the methodology and modules' applicability conditions as described in Section 3.1.2 - Applicability of Methodology.
 - Leakage assessment will be developed for initial instances following VCS VM0007 REDD+ Methodology Framework and will be reassessed when new instances are included in the project.
- New instances must have a start date that is the same as or later than the grouped project's start date.
 - For reference, the start date of Instance #01 is January 1, 2025.
 - New instances will only be eligible for crediting from the start date of the project activity instance or the start of the verification period in which they were added to the grouped project, up to the end of the total project crediting period.
- The determination of the baseline scenario and demonstration of additionality of new instances will be based upon the Project Zone's geographic area, for consistency with initial instances.
 - For reference, Project Zone's baseline scenario and additionality will be detailed in the project description document for Instance #01.
- New instances must be subject to the same community and biodiversity without-project scenarios; to the same processes for stakeholder engagement; and respect for rights to lands, territories, and resources including free, prior, and informed consent which will be detailed in the project description document.
- Non-permanence risk analysis will be assessed in Project Zone's geographic area. If necessary, the project area will be divided into sub-areas with similar risks and each area will have a risk rating determined individually as per the VCS Standard v4.7 recommendation.
- The inclusion of new instances must demonstrate evidence of project ownership by the project proponent from the respective start date of each project activity instance.
 - If necessary, new project proponents' inclusion will follow the requirements of VCS Registration and Issuance Process v4.5, Section 7.2 - Release and Accession of Project Proponents, and within five years of the project activity instance start date.

- The inclusion of new instances will be documented in the monitoring report, providing comprehensive technical, financial, geographic, and other relevant information to demonstrate compliance with the applicable eligibility criteria and facilitate evidence collection by the validation/verification body.
 - New instances shall have similar monitoring elements, which will be described in the project description document.
- If applicable, new instances must adhere to the clustering and capacity limit requirements as specified in sections 3.6.8 – 3.6.9 of VCS Standard v4.7.

2.1.7 Project Proponent (VCS, 3.7; CCB, G1.1)

Table 2.1.7-A displays information about the project proponent.

Table 2.1.7-A: Information about project proponent.

Organization name	ZEG Florestal
Contact person	Fernando de Barros Ferraz
Title	Technical Director
Address	Praça Prof. Sérgio Bonganhi, 120 Cj. 205, Piracicaba - SP
Telephone	+55 (11) 3074-5100
Email	fernando.ferraz@zegflorestal.com.br

2.1.8 Other Entities Involved in the Project

Table 2.1.8-A displays information about other entities involved in the project.

Table 2.1.8-A: Information about entities involved in the project.

Organization name	Ademar Luiz Bedin and Nilce Ferrari Bedin
Role in the project	Landowners
Contact person	Ademar Luiz Bedin
Title	Landowners

Address	Rua Milton dos Reis e Silva, Quadra 83 Lote 05, Cristalina - GO
Telephone	+55 (61) 9119-9201
Email	ademarbedin204@gmail.com

2.1.9 Project Ownership (VCS, 3.2, 3.7, 3.10; CCB, G5.8)

Instance #01 of the Vale do Rio Branco REDD+ Grouped Project is located in Caracaraí, Roraima, Brazil, specifically on the Vale Verde property, owned by Ademar Luiz Bedin and Nilce Ferrari Bedin.

In accordance with the CCB Standard v3.1, Section "G5 Legal Status and Property Rights," and the VCS Standard v4.7, Section "3.7 Ownership," an enforceable agreement set between the ZEG Florestal and the holder of the statutory, property, and contractual rights in the land that generates GHG emission reductions grants project ownership to the proponent. As the project proponent, ZEG Florestal assumes responsibility, in collaboration with the landowners, for defining the most effective management approach for the project. To this end, a Special Purpose Company (SPE in Portuguese) was created to comply with the guidelines established in the MOU and to manage the project's initiatives and financial administration matters.

The legal documentation establishing the land title and ownership is provided through the property's official land registration ("Matrícula do imóvel"), while the management arrangement between ZEG Florestal and the landowner is formalized in a Memorandum of Understanding (MOU). Both documents are available for auditor review during the validation process.

2.1.10 Project Start Date (VCS, 3.8)

Table 2.1.10-A displays the start date and provides justification.

Table 2.1.10-A: Project start date and justification.

Project start date	01-January-2025
Justification	The VCS Standard version 4.7 states that <i>"the project start date of an AFOLU project is the date on which activities that lead to the generation of reductions or removals is implemented."</i> On July 10, 2024, a Memorandum of Understanding (MOU) was signed by both parties, outlining their intentions to develop a REDD+ project on the property designated as the initial instance of the grouped project. The process began with a thorough evaluation of the property's compliance with environmental and land agency regulations, a process set to continue until December 31, 2024.

	During this period, studies and evaluations were conducted to understand the property's characteristics and to define an appropriate plan for maintaining forest protection within the property. Both parties agreed that once the land regularity assessment process is completed, activities promoting reductions in GHG emissions will be immediately implemented, starting on January 1, 2025.
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2.1.11 Benefits Assessment and Project Crediting Period (VCS, 3.9; CCB, G1.9)

Table 2.1.11-A describes the first crediting period, how it conforms with the VCS Program Requirements, as well as the CCB benefits assessment period.

Table 2.1.11-A: Benefits assessment and project crediting period.

Crediting period	The first project crediting period is 40 years, conforming with VCS Standard version 4.7, which states that <i>"the initial project crediting period shall be a minimum of 20 years up to a maximum of 100 years, which may be renewed at most four times, with a total project crediting period not to exceed 100 years"</i> .
Start date of first or fixed crediting period	01-January-2025 to 31-December-2064
CCB benefits assessment period	The CCB benefits assessment period is 40 years, following the minimum project longevity stated in VCS Standard version 4.7.

2.1.12 Differences in Assessment/Project Crediting Periods (CCB, G1.9)

The project will generate credits for 40 years and will maintain property protection and monitoring for the entire CCB benefits assessment period, which spans 40 years.

The primary factors explaining the differences between the Assessment and Project Crediting Periods are the planned land-use conversion activities and the credit generation needed to sustain a healthy income flow. This income is essential to support the initial infrastructure improvements and ongoing monitoring activities necessary to maintain property protection. All property monitoring and protection activities will be maintained throughout the period CCB benefits assessment period.

2.1.13 Project Scale and Estimated Reductions or Removals (VCS, 3.10)

The project is not a large project in its initial configuration. Its first instance will generate credits for at least 29 years, totaling 3,871,957 tCO₂e during the first instance crediting period, with an annual average of 133,515 tCO₂e. These numbers are under review and may be

modified in the project protocol. The Table 2.1.13-A contains the annual estimated tCO₂e reductions, the total estimated emission reductions, total number of years and the average annual reductions for the first crediting period.

☒ < 300,000 tCO₂e/year (project)

☐ ≥ 300,000 tCO₂e/year (large project)

Table 2.1.13-A: Estimated reductions (tCO₂e) for the project's first crediting period.

Calendar year of crediting period	Estimated reductions or removals (tCO ₂ e)
01-January-2025 to 31-December-2025	161,346.72
01-January-2026 to 31-December-2026	164,930.18
01-January-2027 to 31-December-2027	168,513.64
01-January-2028 to 31-December-2028	172,097.10
01-January-2029 to 31-December-2029	175,680.56
01-January-2030 to 31-December-2030	179,264.02
01-January-2031 to 31-December-2031	182,847.48
01-January-2032 to 31-December-2032	186,430.94
01-January-2033 to 31-December-2033	190,014.40
01-January-2034 to 31-December-2034	193,597.86
01-January-2035 to 31-December-2035	193,597.86
01-January-2036 to 31-December-2036	193,597.86
01-January-2037 to 31-December-2037	193,597.86
01-January-2038 to 31-December-2038	193,597.86
01-January-2039 to 31-December-2039	193,597.86
01-January-2040 to 31-December-2040	193,597.86
01-January-2041 to 31-December-2041	193,597.86

Calendar year of crediting period	Estimated reductions or removals (tCO ₂ e)
01-January-2042 to 31-December-2042	193,597.86
01-January-2043 to 31-December-2043	193,597.86
01-January-2044 to 31-December-2044	193,597.86
01-January-2045 to 31-December-2045	32,251.14
01-January-2046 to 31-December-2046	28,667.68
01-January-2047 to 31-December-2047	25,084.22
01-January-2048 to 31-December-2048	21,500.76
01-January-2049 to 31-December-2049	17,917.30
01-January-2050 to 31-December-2050	14,333.84
01-January-2051 to 31-December-2051	10,750.38
01-January-2052 to 31-December-2052	7,166.92
01-January-2053 to 31-December-2053	3,583.46
01-January-2054 to 31-December-2054	0.00
Total estimated ERRs during the first or fixed crediting period	3,871,957.22
Total number of years	29
Average annual ERRs	133,515.77

2.1.14 Physical Parameters (CCB, G1.3)

2.1.14.1 Location

The Vale do Rio Branco REDD+ Grouped Project, implemented in Roraima State, Brazil, will encompass instances within the state's Productive Use Zone (ZUP), which has been refined to establish the Project Zone. Detailed information is in Section 2.1.16 - Project Zone Map and Project Location. This zone is designated for agricultural productive land use in Roraima, providing a consistent legal framework to support its development and ensuring a homogeneous area for the project activities. Figure 2.1.14-A presents the Project Zone

location overview, which shows the project zone, the project Instance #01, the roads, and the boundaries of municipalities, states, and countries.



Figure 2.1.14-A: Project Zone Location Overview Map.

2.1.14.2 Topography

The region has three morphological domains, according to a survey carried out by the Ecologic-Economic Zoning - ZEE (SEADI, 2022), the Pharenozoic Watersheds and Sedimentary Covers, the Neoproterozoic Cratons, and the Quaternary Sedimentary Deposits,

which determine the main lithological unit's composition characteristics, potential weaknesses, slope, compaction degree, and vegetation cover of the relief forms. The state's north and northwest terrains, which constitute the geomorphological region of the Amazonas-Orinoco Interfluvial Plateau, present a moderate risk of erosion due to their extensive dissected relief structure and altitudes of up to 1,600 m, high slopes, and moderate vegetation cover.

The extreme north of the state, on the border with Guyana and Venezuela, is where the greatest concentration of areas considered vulnerable and moderately vulnerable to natural soil loss are concentrated, due to the combination of rugged relief, high slopes in steep forms ($>45^\circ$), semi- to unconsolidated sedimentary cover, the extensive presence of Neossolos Litólicos, and/or patches of Argissolo Vermelho-Amarelo. There are also Paleoproterozoic Cover geomorphological regions to the northwest with the same relief and soil characteristics mentioned above that indicate areas considered vulnerable and moderately vulnerable to natural soil loss (SEADI, 2022), however such areas are not included in the ZUP.

Most of the ZUP presents lithological units considered to be at low risk of erosion in the north, comprised of the geomorphological regions of the Amazonas-Orinoco Interfluvial Plateau and the Dissected Plateaus of Roraima. These lithological units present high to moderate altitudes, sometimes with slopes greater than 45° , dense vegetation cover and are conditioned by geological contacts with sedimentary rocks, found throughout the Rio Branco valley, from north to south of the state, comprising mainly the geomorphological units of the Cenozoic Cover, that is, the Boa Vista Depression, the Rio Branco-Rio Negro Pediplain and the Rio Branco-Rio Negro Depression. The most vulnerable areas of the ZUP combine Cenozoic covers of the Boa Vista Depression with a small extension of Yellow Latosols, and Fluvial Plains and Terraces with Quartzite Neosol.

The main cities are in depressions, where there is a predominance of reliefs with gentle undulations or flats, close to the Branco River, due to the ease of river transport to Manaus, a major economic hub in the region, following access to the sea.

ZUP is mostly covered by areas below 500 meters in altitude (97.3%), with only 2.6% between 500 and 1,000 meters above sea level. As shown in Figure 2.1.14-B, the Digital Terrain Model (DTM) of the region shows the topographical features and elevation variations across the state, highlighting the ZUP.

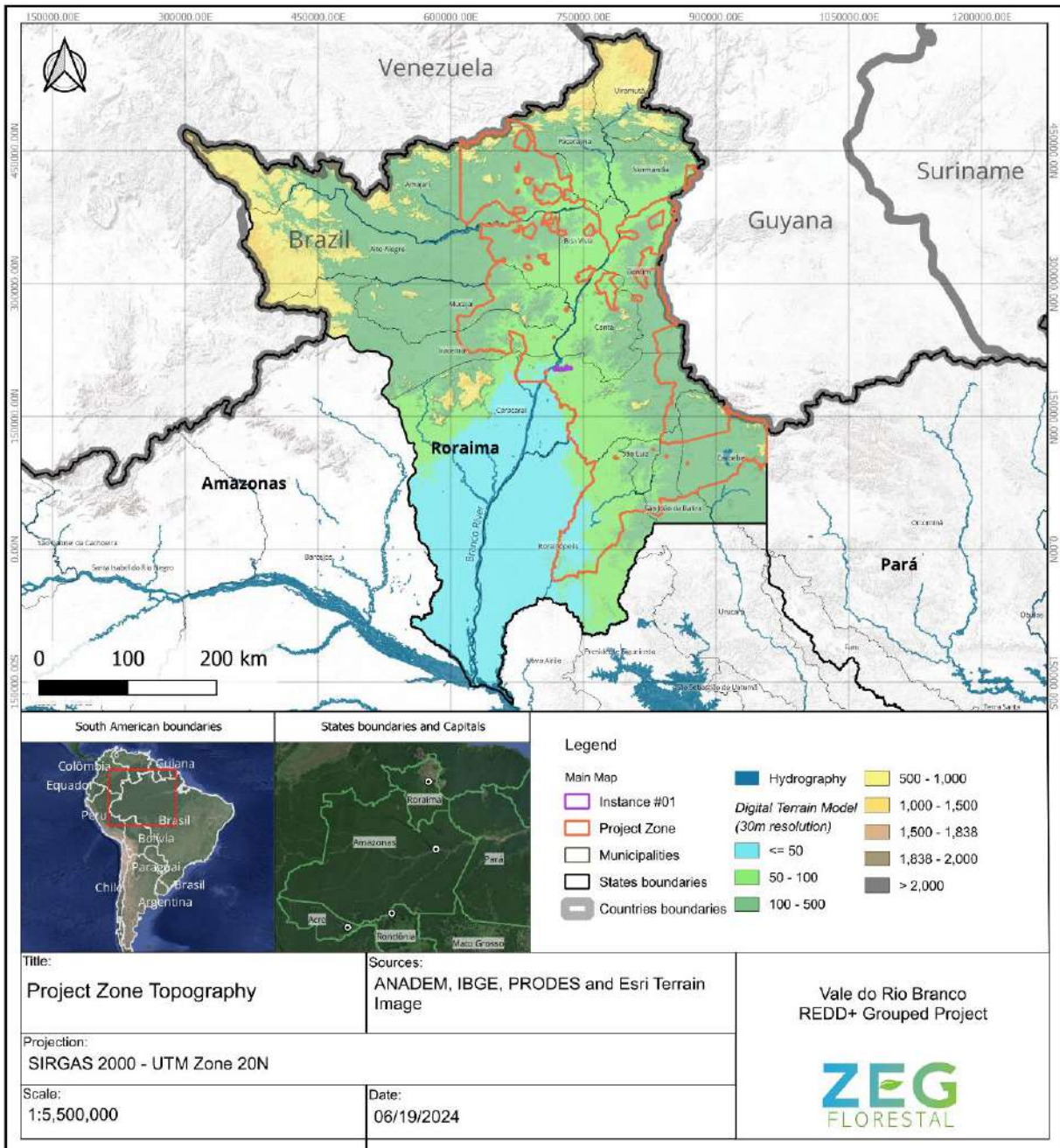


Figure 2.1.14-B: Project Zone Topographic Map.

With most of its area having flat to undulating terrain, the areas for agricultural development are vast. The categories of Flat (35.8%) and Gently Undulating (37.1%) occupy 72.9% of the total area of the zone, followed by the categories Moderately Undulating (13.4%), Undulating (6.6%), Strongly Undulating (5.9%), Mountainous (1.1%) and Rugged (0.1%). These slope categories, as defined by EMBRAPA (2018), can be observed in the region, with the ZUP highlighted, in Figure 2.1.14-C below.

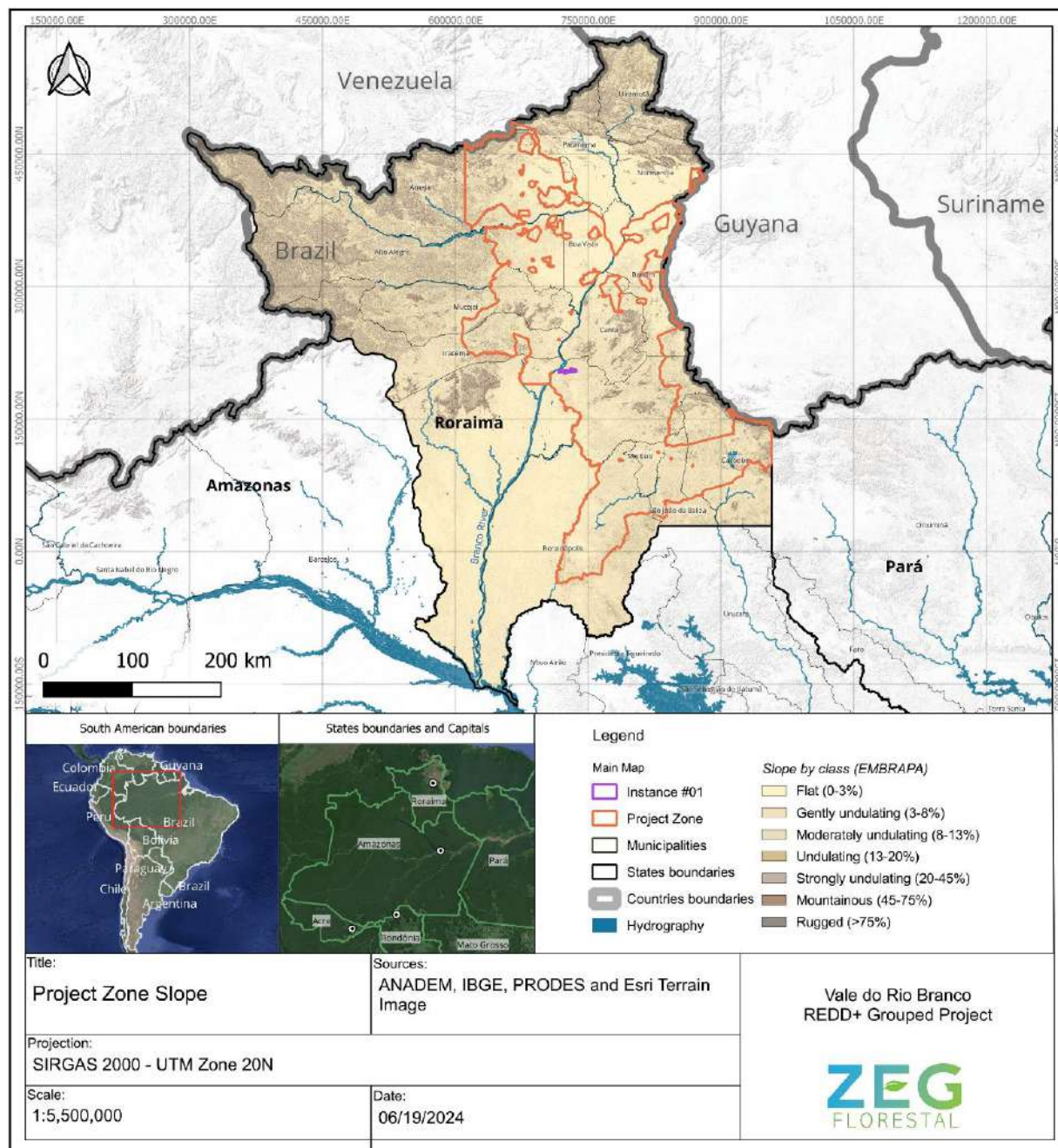


Figure 2.1.14-C: Project Zone Slope Map.

2.1.14.3 Soils

The state of Roraima presents great spatial variability in its soils, which can directly influence the land agricultural suitability, as well as its territorial development, which highlights the importance of studying its capabilities and vulnerabilities as a region. As shown in the Table 2.1.14-A below, there are 13 soil classes according to the Brazilian Soils Classification System - SiBCS (EMBRAPA, 2018), with the description of each class name. Given that the

Brazilian soil classification is frequently referenced and utilized in research, it has been maintained in Portuguese language, for clarity.

Table 2.1.14-A: Soil classes according to the SiBCS (2018).

Soil Class	Description
Argissolo	From Latin <i>argilla</i> , "clay"; denoting soils formed under influence of clay accumulation.
Cambissolo	From Latin <i>cambiare</i> , "change", "exchange"; denoting soils in formation (transformation). Incipient B horizon.
Chernossolo	From Russian <i>chorniy</i> , "black"; denoting soils rich in organic matter, with dark colors.
Espodossolo	From Greek <i>spodos</i> , "plant ashes"; denoting soils with an illuvial accumulation horizon of organic matter associated with the presence of aluminum. Spodic B horizon.
Gleissolo	From Russian <i>gley</i> , "mass of pasty soil"; denoting excess water. Gley Horizon.
Latossolo	From Latin <i>lat</i> , "brick"; denoting highly weathered soils. Latosolic B horizon.
Luvissolo	From Latin <i>luere</i> , "washing"; denoting clay translocation. Textural B horizon with high base saturation and high activity clay (Ta).
Neossolo	From Greek <i>neo</i> , "new"; denoting soils with poor pedogenetic development.
Nitossolo	From Latin <i>nitidus</i> , "brilliant"; denoting shiny surfaces on the structural units. Nitic B horizon.
Organossolo	From Latin <i>organicus</i> , "proper of or belonging to carbon compounds"; denoting soils with a strong or predominantly organic constitution. H or O horizon.
Planossolo	From Latin <i>planus</i> , "plain, flat"; denoting soils developed on plains or in depressions with seasonal flooding. Planic B horizon.
Plintossolo	From Greek <i>plinthos</i> , "tile"; denoting colorful clay materials that harden when exposed to air. Plinthic horizon.

Soil Class	Description
Vertissolo	From Latin <i>vertere</i> , "turn, invert"; denoting movement of soil material at the surface reaching the subsurface (expansion/contraction). Vertic Horizon.

As described in ZEE (SEADI, 2022), in the extreme north of the state, in the high mountainous area, Neossolos Litólicos and Cambissolos predominate, which are generally of low fertility and very shallow. The most fertile soils in the mountainous region, which occur in areas where rocks such as diabases and basalts outcrop, are mostly eutrophic, as they have a greater natural fertility and withstand production pressure for a longer period without requiring nutrient replacement.

Classes that present the greatest extension from north to south are Argissolo Vermelho-Amarelo, concentrated mostly in the northwest and southeast of the state, Latossolo Amarelo, which extends from the north to the south of the state, concentrated in the Branco River valley, and Latossolo Vermelho, which occurs mostly between the central part and the southwest part of the state.

The soils with the lowest agricultural suitability are of the Espodossolo and Neossolo Litólico classes. The Espodossolos are concentrated between the central and southern part of the state, while the Neossolo Litólico can be found on the extreme north of the state, in the mountain ranges between the borders of Venezuela and Guyana, as well as some patches in the central and southeastern part of the state (SEADI, 2022).

The central areas Savannah regions of Roraima contain Argissolos Amarelos and Latossolos Amarelos, which are used to cultivate corn, soybeans, and some fruits, however these regions have longer and more severe dry seasons (SEADI, 2022). SEADI (2022) still lists Latossolos Vermelhos, Argissolos Vermelhos, Nitossolos and Vertissolos (normally more fertile, productive and intensively cultivated soils) as found in small patches of basaltic volcanic rocks or richer igneous rocks dispersed in the Rio Branco-Rio Negro Pediplano, which extends east to west of the state, passing through a strip of the central part of the state.

The largest irrigated rice production areas in the Amazon are located along the banks of the Surumu, Tacutú, Maú, and Uraricoera rivers, where there is a predominance of Plintossolos Háplicos, Gleissolos, and Planossolos Nátricos, which have now been reduced by the demarcation of the Raposa-Serra do Sol Indigenous Land. These regions may have productivity equal to or greater than the major production centers in Brazil, including integrated crop-livestock (ILP) systems (SEADI, 2022).

The general soils composition within the ZUP is represented by 27.9% Latossolo Vermelho-Amarelo, 26.7% Argissolo Vermelho-Amarelo, accounting for approximately 54.6% of the soil composition in the ZUP. This is followed by Latossolo Amarelo (12.6%), Argissolo Amarelo (12%), Neossolo Litólico (9.3%), Espodossolo (3.5%), Latossolo Vermelho (2%), and Planossolo Háplico (1.7%). Collectively, these soil types represent more than 95% of the soil

composition in ZUP. The spatial distribution of these soil classes within the state, highlighting the ZUP area, is shown in Figure 2.1.14-D below.

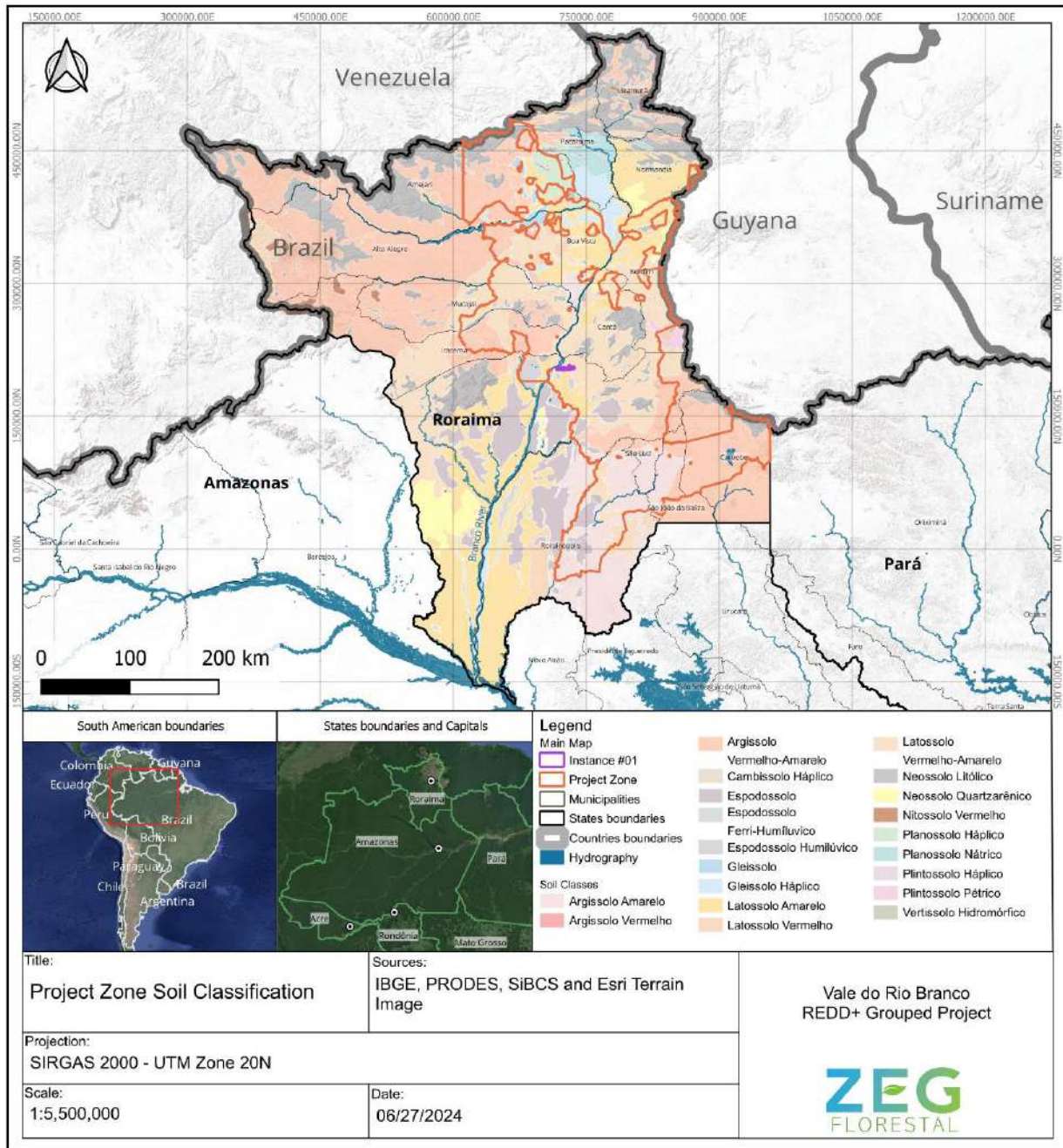


Figure 2.1.14-D: Project Zone Soil Classification Map.

2.1.14.4 Climate

According to the Köppen classification, Roraima's climate is predominantly a rainy tropical type without a dry season (Af), with average annual rainfall consistently exceeding 2,000 mm, reaching up to 2,600 mm. The thermal regime shows that average monthly temperatures in the state are always above 18°C, with annual average temperatures varying between 24 and 28°C. Maximum temperatures range between 30 and 32°C, and minimum temperatures range between 20 and 22°C. July is the coldest month, while November is the hottest. The dry season occurs from October to March, and the rainy season begins in April and extends until September. During the rainy season, the Branco River rises by approximately 6 to 8 meters in the city of Caracaraí (SEADI, 2022).

There are two additional climate types in the state of Roraima, both classified as tropical. Type Am – Tropical Monsoon or Humid – is the second most dominant climate type in the state, occurring in the north-central and central regions. This type experiences rainfall of 1,600 mm per year in the eastern portion and exceeds 2,000 mm per year in the western and southern portions. It represents the transition between the Superhumid climate areas (Af) and a transition strip to the third climate type, Semi-humid/Semi-arid (Type Aw – Tropical Savannah), which comprises the northeastern strip of Roraima and generally receives 70 to more than 80% of its annual precipitation between April and September.

Temperatures remain consistently high throughout the region due to its proximity to the Equator, with temperatures above 18°C throughout the year. The “Af” climate type is characterized by almost no dry months, whereas the transitional “Am” type may experience 1 to 3 dry months. The “Aw” type, featuring tropical savanna characteristics, typically has 4 to 7 dry months, significantly altering the rainfall pattern. Figure 2.1.14-E below illustrates the distribution of Köppen-Geiger climate classes across the state, with an emphasis on the ZUP area.

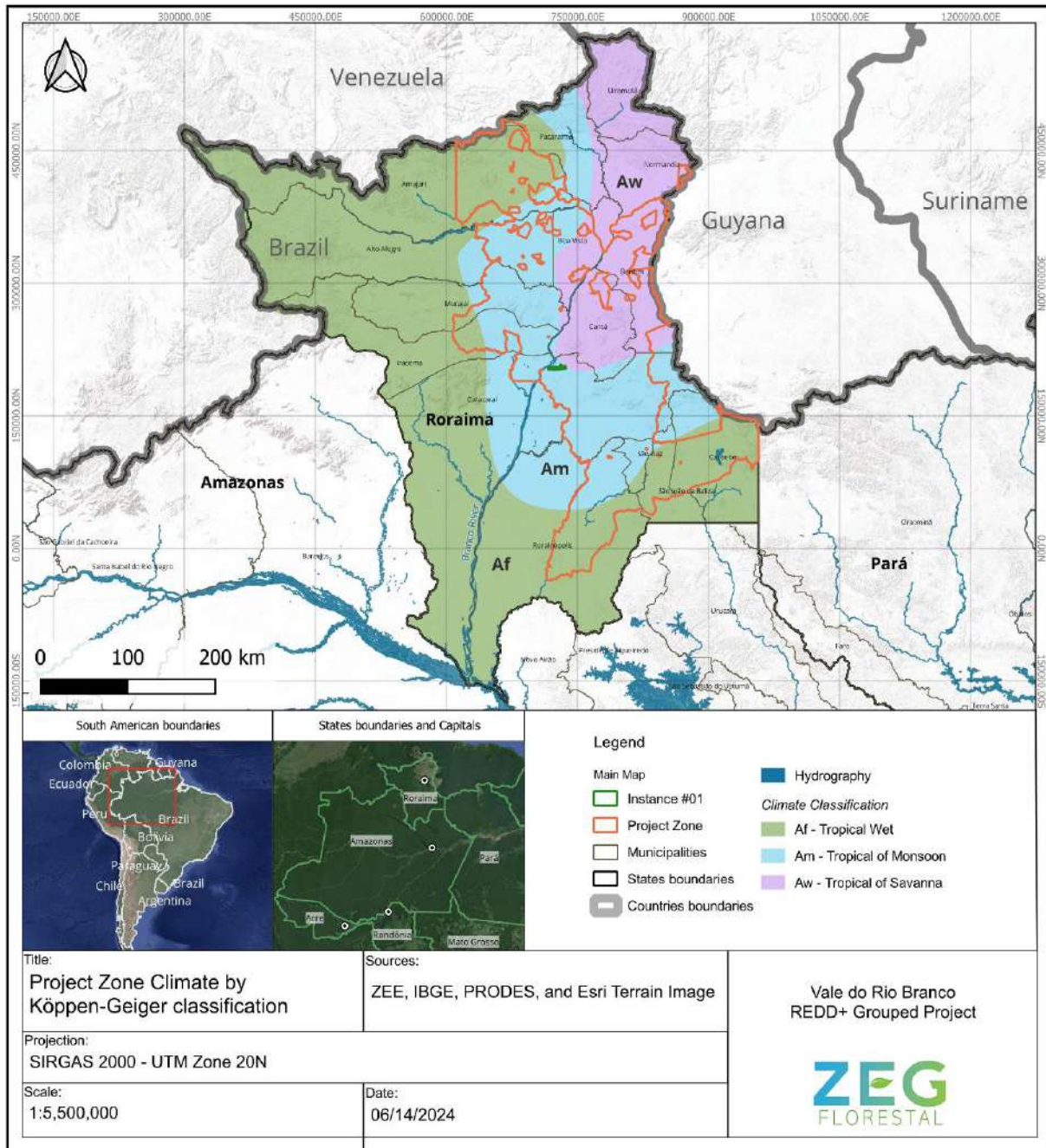


Figure 2.1.14-E: Project Zone Climate by Köppen-Geiger Classification Map.

2.1.14.5 Seasonality

Agricultural zoning is strongly influenced by the seasonality of rainfall, following the state's climate classification. In Roraima, the rainy season lasts from April to September, while the dry season lasts from October to March. However, this seasonal cycle is not uniform across the entire state, varying in terms of precipitation volume and the start and end of wet and dry seasons. The southern region of Roraima experiences less variation in precipitation between the wet and dry seasons, with accumulated rainfall of 1,400 mm during the rainy season and 1,100 mm during the dry season.

In contrast, along the northeastern border shared with Guyana, accumulated rainfall is 1,200 mm during the wet season and 200 mm during the dry season. The timing of rainfall varies by location. For instance, at Alalaú Base (in the south-central region, near the border with the state of Amazonas), the rainy season occurs from February to July. In Santa Maria do Boiaçu (along the lower Branco River) and Missão Surucucu (in the northwest region), rainfall is more concentrated from March to August. The Figure 2.1.14-F illustrates the spatial distribution and seasonality of precipitation in the state, with highlighted points featuring monthly precipitation graphs based on CHIRPS (FUNK, 2015).

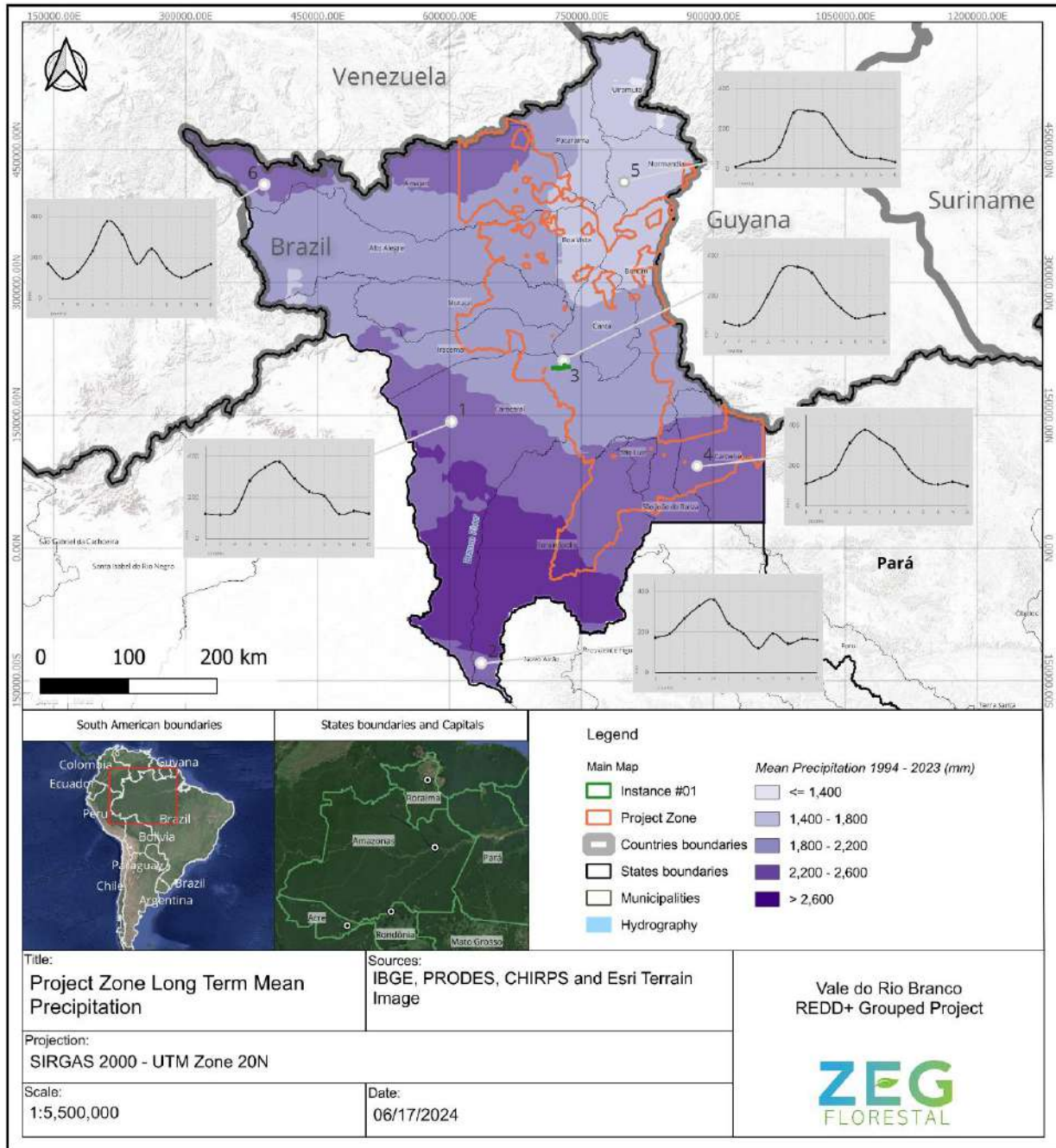


Figure 2.1.14-F: Project Zone Long Term Mean Precipitation Map.

2.1.14.6 Hydrology

According to Silva (2011), the main hydrographic network of Roraima is composed of the Branco River watershed, which extends over 45,530 km² in a north-south direction. This watershed is formed by rivers such as Tacutu and Uraricoera, totaling 548 km in length, divided into three parts: upper, middle, and lower Branco River. The municipality of Caracaraí is in the middle Branco River watershed, starting at the Bem-Querer rapids to Vista Alegre, covering 24 km.

The Branco River flows into the left bank of the Negro River, on the border between the states of Roraima and Amazonas. During its course, the Branco River passes through the municipalities of Boa Vista, Bonfim, Cantá, Mucajaí, Iracema, Caracaraí, and Rorainópolis (SGB, 2021). The hydrographic regime of the Branco River consists of a rainy period occurring between March and September, typically flooding in July, and a dry period between October and February. The possibility of navigating the river depends on the water level, so during the dry season, navigation may be impossible at certain times (Dias, 2017).

During the dry season, in the city of Caracaraí, the average flow rate is around 1,400 m³/s. On average, the Branco River transports approximately 45 mg/L of suspended sediments, with minimums during the dry season of around 10 to 20 mg/L and peaks during the flood of 70 to 80 mg/L, characterizing this river not only as a typical white water, but a mixed pattern of clear and white water. Regarding the flood pulse, it occurs once a year, with water levels above 5 meters typically persisting for about 82 days, using the Caracaraí gauge as a reference (Arouche *et al.*, 2013).

ZUP is mainly located within the Branco River watershed, encompassing the sub-watersheds of Tacutu River, Uraricoera River, Mucajaí River, Anauá River, and the Branco River itself. These rivers drain into the Branco River. Additionally, to the southeast of ZUP, there are the sub-watersheds of Jauaperi River and Jatapu River, which flow into the Amazon River but do not belong to the Branco River watershed.

The environmental characteristics of the urban environment contribute significantly to a high rate of surface runoff. The predominance of shallow and poorly developed litholic soils, with a clayey texture and very low porosity, results in severe restrictions on the infiltration potential, which consequently increases the surface runoff of rainwater (Monteiro, 1997 *apud* Arouche *et al.*, 2013).

As the main water resource in the state of Roraima, the Branco River is significantly impacted by human activities, particularly along its right bank where the capital, Boa Vista, is located. Boa Vista has a population of 413,486 inhabitants, which accounts for more than half of the state's total population of 636,707 inhabitants (IBGE, 2022a). The figure 2.1.14-G below depicts the watersheds that cover the ZUP, as well as the Branco River.

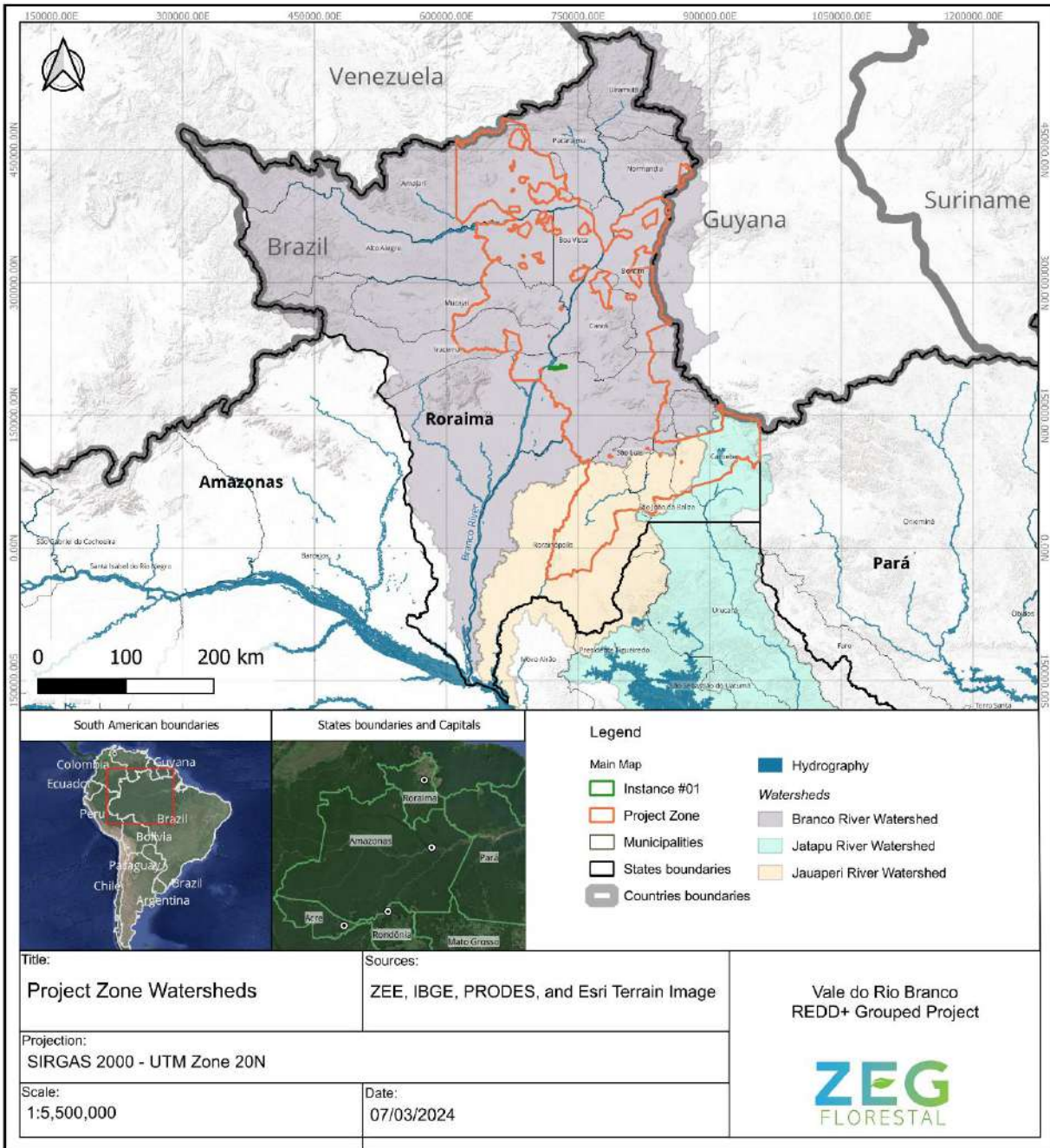


Figure 2.1.14-G: Project Zone Watersheds Map.

2.1.14.7 Vegetation

There are eight vegetation physiognomies described by the ZEE in Roraima: Dense Ombrophilous Forest, Open Ombrophilous Forest, Semi-deciduous Seasonal Forest, and Ecotone, classified as forest types, covering approximately 66% of the state's territory. The Campinarana, Savannah (Cerrado), and Steppe Savannah are classified as grassland types, comprising 33.27% of the state's territory. Additionally, Montane Refuge occupies less than 1% of the territory (SEADI, 2022).

The principal phytophysiognomy is Dense Ombrophilous Forest, which covers approximately 55% of the entire territory of Roraima with the most representative environments within this forest type being the Submontane (approximately 37%) and Montane (approximately 12%) regions. It extends from the eastern borders, adjacent to Guyana, to the western and northern borders of the state, adjacent to Venezuela, and encompasses much of the central, southern, and southeastern regions of Roraima. This region exhibits a wide diversity of features, ranging from montane plateaus up to 1,200 meters above sea level to alluvial areas along the lower Branco River, which are less than 70 meters above sea level (SEADI, 2022).

This forest type occurs in regions where excessive rainfall is the predominant factor in establishing a vegetation of large and tall trees in high densities, creating an environment with extensive floral diversity. The characteristics of this forest type, found in southeastern Roraima, have unique components. There is an abundance of *Lecythidaceae*, especially the castanheira (*Bertholletia excelsa*), and *Arecaceae* in more fertile soils, such as paxiúba (*Socratea exorrhiza*), mumbaca (*Astrocaryum gynacanthum*), tucumã (*Astrocaryum aculeatum*), and marajá (*Bactris sp.*) (SEADI, 2022).

The main threatened species of this vegetation type include garapeira (*Apuleia leiocarpa*), angelim-pedra (*Hymenolobium excelsum*), itaúba (*Mezilaurus itauba*), castanheira (*Bertholletia excelsa*), cedro-rosa (*Cedrela odorata*), ucuúba (*Virola surinamensis*), pau-amarelo (*Euxylophora paraensis*), pau-rainha (*Centrolobium paraense*), pau-santo (*Zollernia magnifica*), pau-rosa (*Aniba rosaeodora*), itaúba branca (*Mezilaurus navalium*), louro ferro (*Aniba ferrea*), and seringueira (*Hevea sp.*) (RADAMBRASIL, 1975).

The Open Ombrophilous Forest, covering approximately 3.51% of the state's territory, is the least common forest type. It differs from the Dense Ombrophilous Forest by having a more open canopy, allowing for the presence of palms, cipós (vines), and sororocas. This forest exhibits a highly variable forest density, ranging from areas with sparse tree coverage due to extensive liana growth to denser areas where some trees are covered by lianas while others are not (RADAMBRASIL, 1975). It is predominantly surrounded by Dense Ombrophilous Forest, mainly found in the central and eastern parts of the state. This forest type, characterized by “cipós”, is restricted to areas with adverse soil conditions, such as ferruginous lateritic soils, whereas the type with palm trees is associated with more eutrophic soils (SEADI, 2022).

The Semideciduous Seasonal Forest occurs between the Savannah and the Dense Ombrophilous Forest, predominantly in the central-northern part of the state, covering approximately 3.72% of the territory. This forest type is characterized ecologically by the shedding of leaves, which does not occur uniformly across all tree species. Therefore, two physiological responses are expected from trees individually: partial leaf shedding and complete leaf shedding. In Roraima, the stress on trees due to the climate is a result of water restrictions caused by the absence of rain for more than 60 consecutive days. Unlike Ombrophilous forests, this type consists of species adapted with some protection from drought. The forest is dominated by Amazonian genera distributed throughout Brazil, such as *Parapiptadenia*, *Peltophorum*, *Cariniana*, *Lecythis*, *Handroanthus*, *Astronium*, *Myrcarpus*, among others (SEADI, 2022).

When two or more vegetation types meet, they can create an area where the characteristics are mixed and not clearly defined, known as an ecotone. In Roraima, this occurs where the Seasonal Forest merges with the Dense Ombrophilous Forest and Savannah, characterized by intense rocky outcrops, irregular rainfall patterns, and transitional environmental traits. The Ecotone is concentrated in the central part of the state near the capital, Boa Vista, occupying approximately 3.89% of the state. It includes rocky outcrops, grassy areas, and forest features. Due to its location, it is heavily impacted by human activities, representing high rates of deforestation in proportion to other forest groups (SEADI, 2022).

As part of the grassland formations, Campinarana mainly occupies areas in the south-central region of the state, covering approximately 16.21% of the territory. This vegetation, typical of the Amazon region, does not exist in any other biome worldwide, and its conservation is highly recommended. It thrives on extremely sandy soils that are periodically inundated. Campinarana consists of grassy vegetation with occurrences of buriti palms (SEADI, 2022). It is an oligotrophic vegetation type, characterized by nutrient-poor soils, which influences the types of plants that can grow and thrive in these conditions. Located near the Branco River riverbed and influenced by the local rainfall regime, this formation is subject to periodic flooding (Ferreira, 2009). Campinarana exhibits elements of ombrophilous forests, arboreal elements of savannas, and other endemic features (Ferreira *et al.*, 2001). The fundamental distinction between Campinarana and Savannah lies in the former's ombrophilous nature, contrasted by the drought-resistant characteristics of the latter (SEADI, 2022).

Savannas in Roraima exhibit two distinct physiognomies: Savanna and Steppe Savanna. According to Aguiar & Camargo (2004 *apud* SEADI, 2022), the term 'Savanna' is synonymous with 'Cerrado' and shares similar characteristics. These include grassland vegetation on chemically poor soils with high aluminum saturation, as well as the presence of twisted tree elements (Almeida, 2005 *apud* SEADI, 2022). The Savannah occurs predominantly in the central-eastern region of the state, characterized by a seasonal climate with over 3 months of rainfall. These areas are typically flooded during the rainy season due to their location, altitudes lower than 100 meters above sea level. Soybean and cattle farming predominate these regions. The Steppe Savannah occurs in the north-northeastern regions of Roraima, characterized by reduced precipitation and higher altitudes, which give the vegetation a dry appearance. Despite this, perennial runoff from drainage networks supports the ecosystem. Found at an average altitude of 800 meters, it exists on plateaus that divide the watersheds of the Essequibo, Orinoco, and Amazon rivers. The landscape features tabular reliefs carved in sedimentary rocks.

The Montane Refuge differs in vegetation, in terms of its floristic and physiognomic-ecological aspects, from the dominant flora of the phytogeographic region. It constitutes a "relic vegetation," with endemic species that persist in very specific conditions, such as communities located at altitudes above 1,800 meters. It is highly sensitive to any type of intervention (IBGE, 2012).

The composition of ZUP includes the cited phytophysiognomies, although some are minor in proportion, such as the Montane Refuge. In this zone, the primary vegetation types are as follows: Dense Ombrophilous Forest (47.2%), Savannah (18.2%), Ecotone (10.6%), Campinarana (8.2%), Semideciduous Seasonal Forest (8.1%), Open Ombrophilous Forest

(4.8%), and Steppe Savannah (2.9%). The spatial distribution of these vegetation classes across the state can be seen in Figure 2.1.14-H below, with emphasis on ZUP.

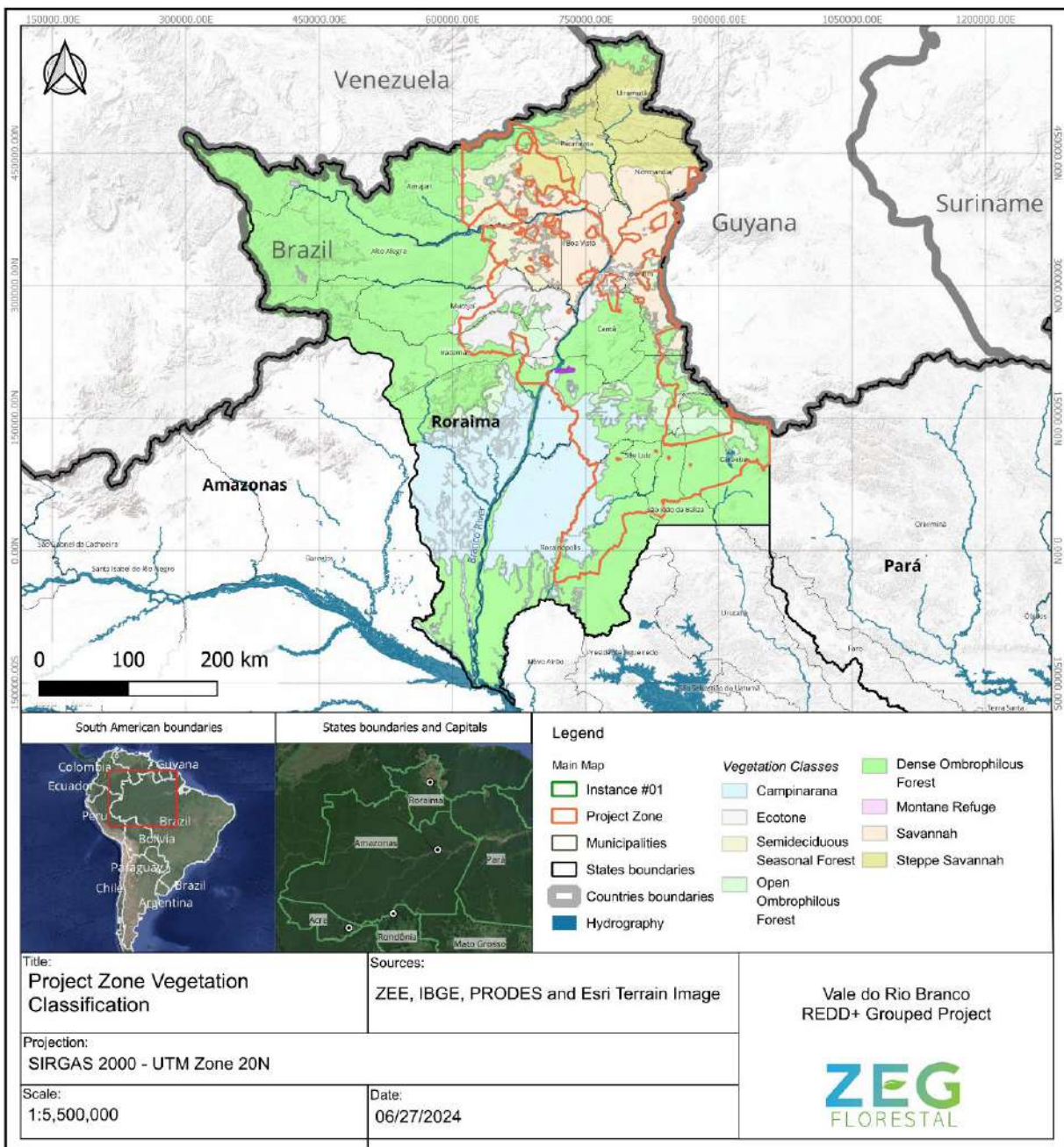


Figure 2.1.14-H: Project Zone Vegetation Map.

2.1.15 Social Parameters (VCS, 3.18; CCB, G1.3)

The following sections will cover the state's social information, focusing on the Project Zone, and later on the initial instance. It is important to highlight data such as land use,

economic activities, and major settlements, discussed specifically for the Project Zone (also referred to as the Productive Use Zone – ZUP, as detailed in the Section 2.1.16 Project Zone Map and Project Location). However, information on historical conditions, migration patterns, population, education, income, health, and other indicators are discussed for the state of Roraima due to the inability to territorially delineate this information for the Project Zone.

2.1.15.1 Roraima's Historical Conditions

The first colonizing incursions into the territory of Roraima occurred over a century after the discovery of Brazil. In 1660, Portuguese expeditions led by Captain Francisco Ferreira ventured into the Branco River Valley in search of “*drogas do sertão*” (native plants and spices used for medicine, food, and trade), as well as gold and precious stones (SEADI, 2022).

At the end of the 18th century, following the independence of the United States, the English began to occupy Amazonian lands to compensate for the loss of their colonies in what is now the American territory. Additionally, Spaniards, Dutch, and French also occupied territories in the northern part of the South American continent. It was during this time that the Portuguese constructed the São Joaquim Fort in 1775, with the aim of halting the incursions of Spaniards, English, French, and Dutch (SEADI, 2022). Freitas (1996) describes that by 1777, there were already six settlements: Nossa Senhora da Conceição, São Felipe, Nossa Senhora do Carmo, Santa Bárbara, Santa Izabel, and Santo Antônio, the latter rarely mentioned by historians. The settlement of Nossa Senhora do Carmo would later become the territory where Boa Vista is currently situated.

Various uprisings occurred in the region due to the Portuguese colonization practices, which led the indigenous population to refuse to work at São Joaquim Fort. This unrest also spurred migration to the Guianas, which subsequently promoted colonization and the introduction of cattle ranching in the region by the late 18th century. The expansion of cattle ranching continued through the 19th century with the establishment of farms and was driven by the influx of Northeastern Brazilian immigrants dedicated to this practice, which persisted until the mid-20th century (SEADI, 2022).

Santos (2000) reports that from 1929 onward, cattle ranching began to decline, and gold mining emerged as the primary economic activity in the region, particularly in the Tepequém mountain range, with the discovery of gold and diamonds intensifying migratory flows. The mining areas were predominantly temporary, shifting locations as soon as accessible mineral deposits were exhausted. Mining was conducted in a rudimentary manner, focused on individual sustenance. However, despite being an alternative economic activity in the region, it became the primary private economic activity of the state, disrupting agricultural activities as labor shifted to mining. There are reports that some indigenous communities specialized in mineral extraction.

Santos (2000) also reports that during this period there was a significant decline in food production, leading to shortages of agricultural products for the local population, who preferred to extract resources from the forest. It was only after the decline of mining years later that the economy shifted back to primary activities, such as animal husbandry and vegetable extractivism.

The definitive colonization came with the creation of the Federal Territory of Rio Branco in 1943, following the dismemberment of the state of Amazonas. The first governor of the Federal Territory, Ene Garcez dos Reis, initiated the establishment of public services to encourage rural production, such as the Agricultural Colonies of Mucajaí, which increased migratory flows in the mid-20th century. Additionally, he developed the Five-Year Territorial Development Plan, which outlined necessary infrastructure and equipment for development, especially the land connections between the main cities of the Territory, the Guianas, and Venezuela, for full regional integration (SEADI, 2022).

Starting in the 1960s, the military government intensified investments in the region under the motto “integrate to avoid surrendering” (*“integrar para não entregar”* in Portuguese). In the 1970s, the construction of highways such as BR-174, connecting Manaus to Boa Vista, BR-210, crossing the Federal Territory of Rio Branco transversely, and BR-401, linking the capital Boa Vista with the cities of Bonfim and Normandia on the border with Guyana, allowed access to practically the entire territory. This facilitated the development of new economic sectors, such as mineral and plant extractive industries, and new agricultural projects, significantly increasing migratory flows to the region. By the 1970s, cattle ranching once again became the primary activity in the Territory, reaching approximately 300,000 head of cattle (SEADI, 2022).

In 1962, the Federal Chamber renamed the Federal Territory of Rio Branco to the Federal Territory of Roraima through Decree-Law No. 1433. In 1982, the municipality of Caracaraí was divided, creating the municipalities of São Luiz, São João da Baliza, and Mucajaí, while the municipality of Boa Vista was also divided, resulting in the creation of the municipalities of Bonfim, Alto Alegre, and Normandia, bringing the total number of municipalities in the Territory to eight. Only in 1988 the National Constituent Assembly transformed the Federal Territories of Roraima and Amapá into states of the Federation. The current structure of the state of Roraima was completed with the creation of the municipalities of Caroebe and Iracema, in 1994, and the municipalities of Amajari, Cantá, Pacaraima, Rorainópolis, and Uiramutã, in 1995 (SEADI, 2022).

Currently, 74.69% of the territory of the state of Roraima consists of institutional areas, comprising 46.21% Indigenous Territories, 12.04% state Conservation Units, 6.59% Established Federal Conservation Units, and 3.13% Proposed Federal Conservation Units. Additionally, 5.49% are Settlement Projects, and 1.23% are areas under the Ministry of Defense. With all those percentages combined, a percentage of 25.31% remains for other types of development (SEADI, 2022).

2.1.15.2 Roraima's Migration Patterns and Urbanization

Migration patterns in Roraima are characterized by intense intra- and interstate mobility. Historically, Roraima has attracted migrants from various parts of Brazil, especially from the North and Northeast regions, in search of agricultural opportunities and land. During the 1970s and 1980s, the state received a considerable influx of migrants from Ceará, Maranhão, Pará, and Amazonas. Since then, Roraima has also attracted migrants from the

South and Southeast regions of Brazil, especially from Rio Grande do Sul (Diniz & Lacerda, 2015).

However, many migrants end up settling in urban centers due to the lack of infrastructure in rural areas. This rural-urban movement has exacerbated urban issues such as inadequate housing and lack of basic services (Diniz, 1997). Additionally, internal migration in the state, particularly from rural to urban areas, has been a persistent feature, reflecting the constantly changing economic and social dynamics.

2.1.15.3 Roraima's Productive Use Zone

The Project Zone encompasses the Productive Use Zone delineated by the Ecological-Economic Zoning (ZEE) of Roraima State, as detailed in Section 2.1.16 – Project Zone Map and Project Location. The ZEE is an essential tool for land use planning and environmental management. The ZEE of Roraima primarily aims to guide the sustainable development of the state, promoting the rational use of natural resources, ensuring environmental conservation, and improving the quality of life for the local population.

The Productive Use Zone (ZUP) covers approximately 31.2% of the state's total area, corresponding to about 70,266 square kilometers. This zone is delineated based on criteria that considers the productivity of the areas, their economic potential, and the diversity of fauna and flora present. The ZUP comprises areas that are already productive and in the process of consolidation, with potential for expansion, and are characterized by the exploitation of activities culturally and economically accepted by producers, often involving intensive use of irrigation and mechanization practices.

Boa Vista, the capital of Roraima, is the largest city in the Productive Use Zone and plays a central role in the state's economy and administration. With a population of approximately 400,000 inhabitants, Boa Vista houses the majority of Roraima's administrative, commercial, and health services. The city serves as a focal point for migrants from various parts of Brazil and even neighboring countries such as Venezuela. This constant migration contributes to the cultural and ethnic diversity of Boa Vista's urban population. Additionally, Boa Vista has experienced rapid urban growth, which poses significant challenges for the city's infrastructure and public services (Diniz, 1997).

Another significant urban center within the Productive Use Zone is the city of Rorainópolis. Located south of Boa Vista, Rorainópolis is the second-largest municipality in terms of population, with approximately 30,000 inhabitants. The city serves as a crucial support point for agricultural activities and is known for its agricultural settlement projects. Many of Rorainópolis' residents are migrants from other parts of Brazil who have been attracted by the availability of land and agricultural opportunities. However, the absence of adequate infrastructure and basic services within the municipality continues to pose a persistent challenge for the local population (Diniz & Lacerda, 2015).

Caracaraí is another important city in the Productive Use Zone, with a population of approximately 20,000 inhabitants. Caracaraí serves as a significant center for trade and transportation in the region. The city is strategically located along the banks of the Branco River, which facilitates river transport and the trade of agricultural and extractive products.

Caracarái also acts as a connection point for rural communities and settlements along the river, serving as a hub for the distribution of products and access to basic services (Hemming, 1990). In addition to the larger cities, the Productive Use Zone encompasses various villages and settlements that play crucial roles in the region's economy and social life.

It is important to note that in the northern region of the Productive Use Zone, there are territorial segments excluded from the ZUP due to the presence of an indigenous land complex, which is included in the Special Use Zone of Roraima's ZEE. This area is predominantly inhabited by the indigenous peoples of the Macuxi and Wapishana ethnic groups. The Macuxi and Wapishana have a historic presence in the state and continue to practice traditional agriculture and other subsistential activities. These indigenous groups face challenges, including the need to protect their lands and natural resources from invasions and environmental degradation (Melo, 2016).

The rural population of the Productive Use Zone is characterized by a combination of farmers and extractivists. Many of these residents rely directly on natural resources for their subsistence. Agricultural practices range from subsistence farming to commercial grain production and cattle ranching. The diversity of economic activities in the rural zone reflects the region's wealth of natural resources, including forests, rivers, and fertile lands. Despite this, the lack of access to basic infrastructure and social services remains a persistent issue encountered by rural communities (Vale, Schaefer & Costa, 2007).

2.1.15.4 Project Zone Land Ownership

In the Project Zone, land ownership is categorized into private properties, non-designated areas, and public lands designated for settlement. This classification arises because the Project Zone encompasses the Productive Use Zone of Roraima, which is classified as an area for state production, excluding Indigenous Lands and Conservation Units which follow specific legislation (Roraima State Complementary Law No. 323 of 08/02/2022).

2.1.15.5 Project Zone Main Settlements

The Project Zone comprises parts of the following municipalities, with their respective contributions to the Project Zone's total area: Alto Alegre (7%), Amajari (13%), Boa Vista (6%), Bonfim (9%), Cantá (10%), Caracarái (17%), Caroebe (8%), Iracema (6%), Mucajaí (7%), Normandia (less than 1%), Pacaraima (less than 1%), Rorainópolis (12%), São João da Baliza (3%), and São Luiz (2%). As it does not include urban areas (cities) or legally protected public areas (Conservation Units and Indigenous Lands), the main communities located within the Project Zone are rural clusters (villages) and settlement projects. Table 2.1.15-A presents the municipalities, settlements, and rural clusters identified in the geographic information databases of INCRA (2024a) and IBGE (2023a). Settlements are included under the municipality to which they legally correspond, with any identified overlaps in other municipalities, if present, noted in parentheses beside the name. The municipalities of Normandia and Pacaraima are not included in the table due to the absence of the communities included in the Project Zone.

Table 2.1.15-A: Municipalities, Settlement Projects, and Rural Clusters identified in the Project Zone.

Municipalities	Settlement Projects (PA)	Rural Clusters
Alto Alegre	PA Paredão	Vila São Silvestre; Vila Taiano; Vila do Paredão
Amajari	PA Amajari; PA Bom Jesus; PA Tepequem	Uraricoera; Vila do Trairão; Vila Tepequem
Boa Vista	PA Nova Amazonia; PA Nova Amazonia I	Depósito
Bonfim	PA Alto Arraia; PA Caju II; PA Renascer; PA Taboca (Cantá); PA Vilena (Cantá)	Nova Esperança; São Francisco; Vilena
Cantá	PA Pau Rainha (Caracaraí); PA Esperança; PA Jacamim; PA Pau-Brasil (Caracaraí); PA Quitaúau; PA Seringueira; PA Tatajuba; PA Terra Nova; PA União	Vila Central; Vila Serra Grande II; Vila São Raimunda; Serra Grande I; Santa Cecília; Vila do Aguiar; Rodrigao Confiança III; Vila União; Félix Pinto
Caracaraí	PA Arco-Íris; PA Cupiuba; PA Angelin (Cantá); PA Caferana (Cantá); PA Castanheira (Cantá); PA Caxias (Cantá); PA Cujubim; PA Itã; PA Jatoba (Cantá); PA Novo Paraíso; PA Rio Dias; PA RR-170 (Cantá); PA Serra Dourada (São Luiz)	Caxias; Vila São José; Petrolina do Norte; Povoado Baraúna; Vila Novo Paraíso
Caroebe	PA Jatapu (São João da Baliza); PA Serra Talhada	Vila Entre Rios
Iracema	PA Ajarani (Mucajaí); PA Japão; PA Maranhão (Mucajaí); PA Massaranduba (Mucajaí); PA São Jose	Vila São Raimundo
Mucajaí	PA Nova Floresta (Iracema); PA Samauma; PA Talismã (Iracema); PA Vila Nova	Vila Cachoeirinha; Vila Tamandaré; Vila do Apiaú; Serra Dourada; Da Penha; Nova; Povoado Samauma

Municipalities	Settlement Projects (PA)	Rural Clusters
Rorainópolis	PA Equador; PA Integração (São Luiz and São João da Baliza); PA Jundiá; PA Ladeiraao; PA Trairi; PAD Anauá (Caracaraí and São Luiz)	Martins Pereira; Nova Colina; Vila do Equador
São João da Baliza	PA São Luizão (Caroebe)	-
São Luiz	PA Bom Sucesso (São João da Baliza)	Vila Moderna; São Luiz

2.1.15.6 Project Zone Land Use

Land use in the Project Zone was analyzed using data from MapBiomass (2023), which maps land cover and use in Brazil using satellite images and geoprocessing technologies. Between 2013 and 2022, forest cover decreased by 4%, corresponding to a loss of approximately 260,000 hectares. Non-forest natural formations, including Savanna and Campinarana vegetation in Roraima, decreased by 1%, resulting in a reduction of about 66,000 hectares. In contrast, the area of pasture increased by approximately 269,000 hectares, representing a 4% growth in the Project Zone. This indicates a constant decline in native vegetation and a significant increase in pasture. Other land uses account for approximately 5% of the total area.

In 2022, land cover distribution in the Project Zone consisted of 60% forest (4,087,894 hectares), 20% non-forest natural formations (1,385,422 hectares), and 14% pasture (919,347 hectares). The chart in Figure 2.1.15-A illustrates the evolution of land use in the Project Zone as a percentage between 2013 and 2022.

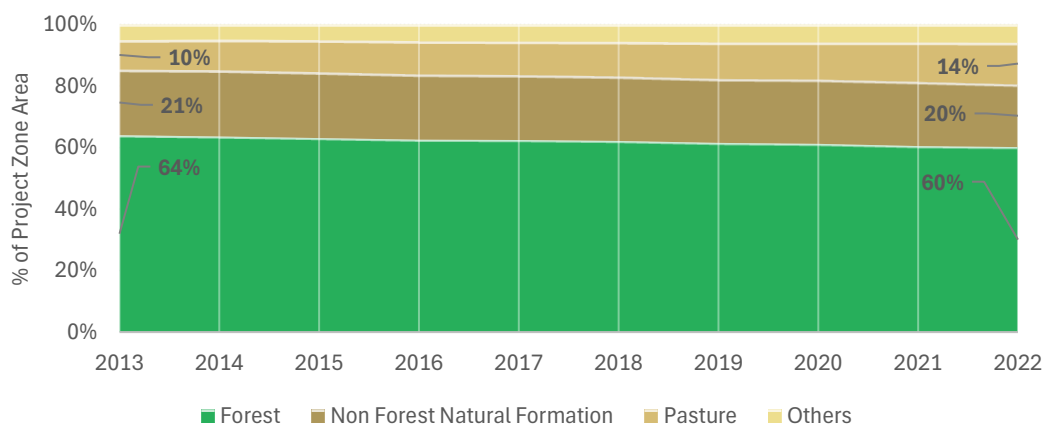


Figure 2.1.15-A: Land use in the project zone from 2013 to 2022.
Source: Adapted from MapBiomass (2023).

2.1.15.7 Project Zone Economic Activities

Agriculture is the primary economic activity in the Productive Use Zone, with a significant focus on subsistence and small-scale farming. Farming families typically cultivate cassava, corn, beans, and rice, using traditional cultivation techniques. However, these agricultural practices often face challenges due to the low fertility of the soils and the lack of adequate infrastructure (Urquiza *et al.*, 2018). The soil in the region is predominantly acidic and nutrient-poor, which limits agricultural productivity without the use of appropriate soil amendments and fertilizers. In many areas, the practice of burning is used to clear the land, which intensifies soil degradation and contributes to deforestation.

This trend continued until the mid-2010s, when regional production characteristics began to shift towards a significant expansion of soybean cultivation. The earliest records of soybean cultivation can be traced to 2004, covering approximately 12,000 hectares. Although the area decreased between 2009 and 2012, after 2013, the soybean area showed consistent growth, except for a decline between 2016 and 2017. The first major increase occurred in 2018, when the cultivated area reached 37,670 hectares. Data from the IBGE Automatic Recovery System – SIDRA (IBGE, 2024a), show that by 2022, the planted area exceeded 100,000 hectares, as demonstrated in the chart in Figure 2.1.15-B. However, this data does not account for the most recent years, during which new legislation has been enacted, permitting a higher proportion of planted area on each rural property.

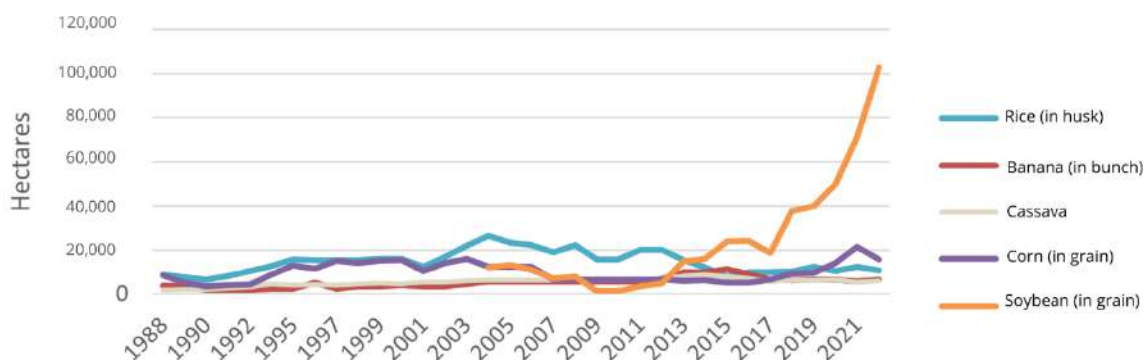


Figure 2.1.15-B: Total area of major agricultural crops in the state of Roraima between 1988 - 2022. Source: IBGE (2024a)

Cattle ranching also plays a crucial role in the economy of the ZUP. Cattle ranching is common, with pastures often established in deforested areas. However, this practice leads to soil degradation and forest fragmentation, increasing ecosystem vulnerability. The expansion of cattle ranching is driven by local and regional demand for meat, but it is often conducted without sustainable management practices, resulting in negative environmental impacts. With the rising demand and prices, both nationally and internationally, Roraima's cattle herd has

been increasing significantly since the mid-2010s, reaching its peak in 2022, with a count of 1,133,502 heads of cattle, as shown in the chart in Figure 2.1.15-C.

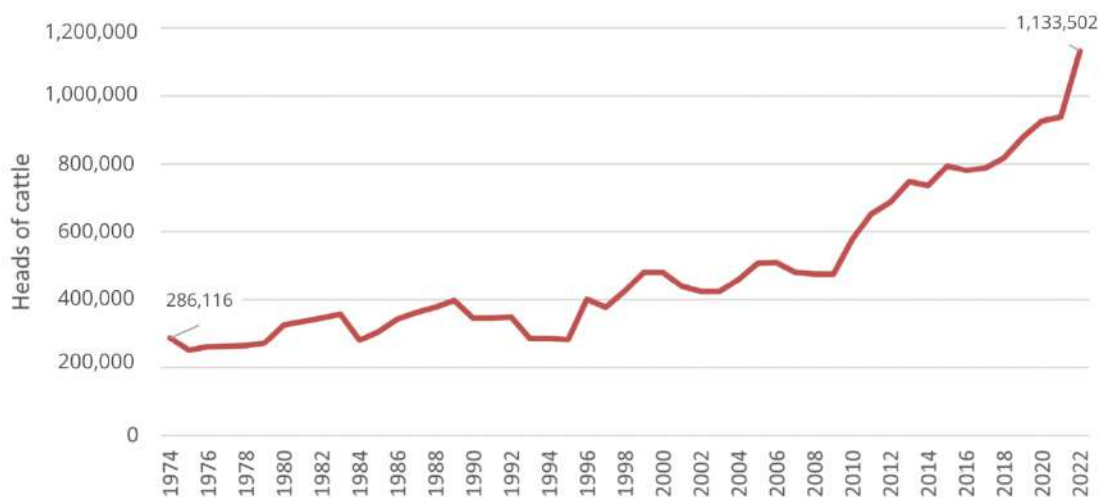


Figure 2.1.15-C: Total number of heads of cattle in the state of Roraima between 1974 - 2022. Source: IBGE (2024b)

Another significant economic sector is extractivism, particularly the timber extraction and non-timber products such as nuts and native fruits. Timber extraction is conducted by both large companies and small producers, and, in many cases, it is carried out unsustainably, contributing to illegal deforestation (McGregor & Eden, 1991). On the other hand, the extractivism of non-timber products represents an important source of income for local communities, promoting the sustainable use of forest resources when managed properly.

Mining, particularly gold extraction, is a significant economic activity in the ZUP. Areas such as Serra do Tepequém have historically been exploited by artisanal miners and mining companies, causing deforestation and water contamination due to the use of mercury in the extraction process (Almeida-Filho & Shimabukuro, 2000). Mining activities also generate social impacts, including land conflicts and public health issues in local communities.

The transportation infrastructure in the ZUP, primarily composed of unpaved roads, directly influences land use and economic development. The BR-174, which connects Manaus to Boa Vista, exemplifies how road construction can facilitate access to remote areas, encouraging the occupation and conversion of forests into agricultural lands and pastures (Barni *et al.*, 2015). However, the poor infrastructure and inadequate road maintenance hinder the transport of agricultural products and access to markets, limiting sustainable economic development.

2.1.15.8 Roraima's Population, Gender Distribution, and Age Structure

The state of Roraima has the smallest resident population among Brazilian states, with a total of 636,707 inhabitants according to the 2022 Census. Approximately 64.94% of this population (413,486 individuals) resides in the municipality of Boa Vista. The municipality with

the second-largest population is Rorainópolis, with 32,647 inhabitants. Since the last census, conducted in 2010, there has been a population increase of 41.34% (186,228 individuals) in the state, reflecting a geometric growth rate of 2.93%. Boa Vista also has the highest population density, with a rate of 72.71 inhabitants per square kilometer, significantly higher than the second municipality with the highest density, São Luiz, which has a density of 4.79 inhabitants per square kilometer (IBGE, 2022a).

The population of Roraima comprises a diverse mix of ethnic groups, including indigenous peoples, migrants from other regions of Brazil, and some international immigrants. The primary indigenous groups present in the state are the Macuxi, Wapishana, and Yanomami. These groups have a rich cultural heritage and traditions deeply rooted in the sustainable management of land and natural resources. The Macuxi and Wapishana are renowned for their traditional agricultural practices and ethnopedological knowledge, which are crucial for the sustainable management of natural resources in the region (Vale, Schaefer & Costa, 2007).

The state population is distributed among 320,392 men (50.32%) and 316,315 women (49.68%). In terms of gender distribution, the state's population exhibits a relatively balanced proportion between men and women. However, some indigenous communities may show variations due to their specific social structures (Souza & Onety, 2017). Additionally, according to ethnic distribution, the census recorded 364,494 “pardo” individuals (equivalent to mixed race, 57.25%), 131,260 white individuals (20.62%), 89,882 indigenous individuals (14.12%), 49,195 black individuals (7.73%), and 784 “amarelo” individuals (equivalent to Asian, 0.12%) (IBGE, 2022a).

The age range of the population is predominantly young, with a mean age of 26 years, while the average age of the indigenous population is 18 years. This population distribution can be observed in Figure 2.1.15-D, which presents Roraima's age pyramid. The age pyramid chart presents the distribution of men on the left in light blue and women on the right in dark blue. The gray bars in the background represent the Brazilian average.

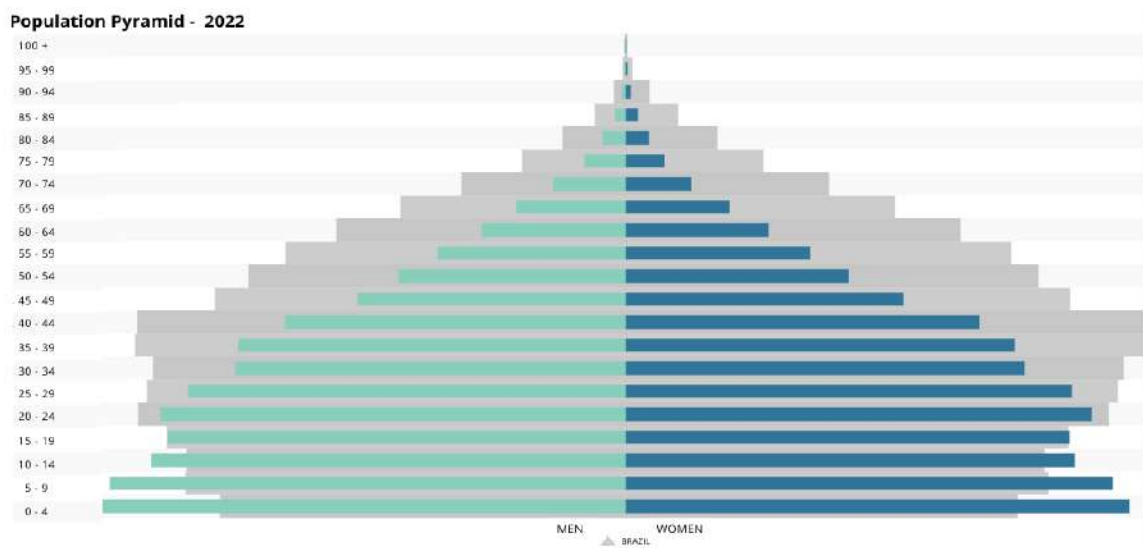


Figure 2.1.15-D: Age pyramid of Roraima compared to the Brazilian population, 2022.
Source: IBGE (2022a).

2.1.15.9 Roraima's Population Income

The Human Development Index (HDI) of Roraima is 0.699, ranked twentieth (20th) among the 26 states and the Federal District. The state's Gross Domestic Product (GDP) is R\$ 18.2 billion, with a per capita GDP of R\$ 27,877, varying from R\$ 41,131 in Bonfim to R\$ 13,069 in Uiramutã (IBGE, 2022a).

The monthly per capita household income is R\$ 1,425, which is significantly lower compared to the average earnings of employed individuals aged 14 years or older, which amounts to R\$ 3,115. Consequently, the average monthly salary of formal sector workers varies between 2.9 times the minimum wage in Boa Vista and 1.6 times the minimum wage in municipalities such as Caroebe, Iracema, and Uiramutã (IBGE, 2022a).

Household income in the Project Zone varies significantly, with a substantial portion of the population living in conditions of poverty. The local economy is predominantly based on subsistence agriculture, with many residents engaged in agricultural settlement projects. However, these projects often lack basic infrastructure and social services, leading to low migrant retention rates (Diniz & Lacerda, 2015).

2.1.15.10 Roraima's Education

In Brazil, small schools (with up to 50 enrollments) are mainly concentrated in the North and Northeast regions. Roraima is the second state with the highest percentage of small schools, with 42.5% of the state's schools classified this way (INEP, 2023).

According to the 2023 School Census data (INEP, 2023), there are 180,965 students enrolled in basic education in Roraima, covering preschool, elementary, and high school levels. Of these, approximately 137,000 are in urban areas and 44,000 in rural areas. In terms of gender, 51% of the students are men and 49% are women. Regarding race/ethnicity, 66% are black/mixed race, 13% are white, and the rest are composed of other races or not declared. As for teachers, Roraima has 8,299 teachers, with 3,062 in rural schools and 2,035 in schools with indigenous education.

The state has a total of 885 schools, of which 66% are in rural areas (581 schools) and 34% in urban areas (304 schools). Additionally, there are 421 schools with indigenous education. The average number of students per class ranges from 12 to 27 in elementary and high school. There are also about 3,560 enrollments in schools located in settlement areas and 28,100 enrollments in schools that offer indigenous education, both referring to basic education in 2023 (INEP, 2023). Regarding higher education, Roraima has a total of nine higher education institutions, of which six are private and three are public (INEP, 2022).

The availability of internet in schools in Roraima is limited, with overall access below 65%. Municipalities in the south and central region of the state have higher availability, exceeding 70%, while in the northernmost municipalities, such as Pacaraima, Normandia, and Uiramutã, at most 30% of schools have internet access (INEP, 2023).

Regarding the quality of education, the age-grade distortion rate in elementary school, which assesses the percentage of students who are behind in their schooling, is 21.3% in Roraima, compared to 11.5% in Brazil.

According to the Continuous National Household Sample Survey of 2022, the percentage distribution of the population aged 25 and over in Roraima reveals a variation in education levels. The majority, 37.9%, have completed high school, making this the most common educational level among adults. Additionally, 20% of the population has incomplete elementary education, and 6.5% have completed elementary education. A smaller percentage, 5%, have incomplete high school. Regarding higher education, 5.6% of the population started but did not complete a course, while 19.1% completed higher education. Finally, 5.9% of the adult population has no formal education (IBGE, 2022b).

2.1.15.11 Roraima's Health and Sanitation Indicators

Roraima has an infant mortality rate that ranges from 55.48 deaths per 1,000 live births in Alto Alegre municipality to 5.52 deaths per 1,000 live births in Caroebe municipality (IBGE, 2022a). From 2012 to 2022, the state of Roraima experienced a 13.8% increase in its infant mortality rate, which reached 18.8 per 1,000 live births, with 246 infant deaths recorded in 2022. This rate exceeds those of other states within the Legal Amazon and the broader national average, as described by the Amazônia Legal em Dados (2024), analysis based on data from the Unified Health System (SUS -Sistema Unificado de Saúde).

As a common disease indicator due to lack of sanitation, hospitalizations due to diarrhea vary between 158 per 100,000 inhabitants in the municipality of São João da Baliza and 7.2 per 100,000 inhabitants in the municipality of Bonfim, with five municipalities not providing data (IBGE, 2022a).

Among the vaccine-preventable diseases, in 2023, the state of Roraima recorded 44 cases of meningitis, 429 cases of chickenpox, 21 cases of mumps, 2,051 cases of COVID-19, and 42 cases of influenza. There were also 34,708 cases of malaria, 374 cases of American tegumentary leishmaniasis, and 17 cases of visceral leishmaniasis (SESAU, 2024).

In 2022, the life expectancy in Roraima was 73.2 years, compared to 73.4 years in the Legal Amazon and 77.8 years in Brazil. Between 2012 and 2022, the life expectancy of Roraima's inhabitants increased by 3 years. Despite this, the state remained 0.2 years below the average of the Legal Amazon and 4.6 years below the national average. These comparative metrics are provided by the Amazônia Legal em Dados (2024), based on the official IBGE database.

The General Hospital of Roraima "Rubens de Souza Bento", located in the capital Boa Vista, is regarded as the principal hospital in the state and a reference point for the regional community. It comprises two emergency units and a total of 403 beds (Hospital Geral de Roraima, 2024).

The average number of public basic health care units (UBS in Portuguese) per thousand inhabitants in the state of Roraima is 0.44 (IDSC, 2023). Table 2.1.15-B presents this indicator for each municipality in the state.

Table 2.1.15-B: Number of public basic health care units (UBS in Portuguese) per thousand inhabitants. Source: IDSC (2023).

Municipality	Number of UBS per thousand inhabitants
Alto Alegre	0.30
Amajari	0.47
Barcelos	0.16
Boa Vista	0.11
Bonfim	0.80
Cantá	0.83
Caracaraí	0.48
Caroebe	0.28
Iracema	0.27
Mucajaí	0.59
Normandia	0.39
Pacaraima	0.73
Rorainópolis	0.37
São João da Baliza	0.50
São Luiz	0.48
Uiramutã	0.29

Sanitation data is scarce in the region. The National Sanitation Information System (SNIS, 2024) Panel of Indicators shows that in 2023, all 15 municipalities had access to water, but only two had access to sanitation services. Considering only water supply, 13 municipalities were served by some form of contracted delegation (indicators GE001 and GE002), with one municipality having an expired delegation contract (indicator GE002), while two municipalities were served without a contracted delegation (indicator GE003). The total

water supply index (IN055) is 79.4% of the total population, varying from 96.45% in Boa Vista to 4.13% in Uiramutã, with 9 municipalities having less than 50% coverage.

Sewage service indicators show that the total coverage (IN056) in the state is 92.8%, but this figure is heavily influenced by Boa Vista, which accounts for the largest share of generation. Ten municipalities lack any sewage data, while the coverage in other municipalities ranges from 92.8% in Boa Vista to 51.37% in Bonfim for the year 2023 (SNIS, 2024).

Regarding solid waste, Boa Vista and Bonfim have the highest total service coverage rates (IN015), at 96.45% and 91.22%, respectively. Uiramutã and Amajari have the lowest coverage rates, at 13.60% and 15.95%, respectively (SNIS, 2024). Although 8 of the municipalities have coverage rates below 50%, at least the statistics are provided.

It is evident that Boa Vista, due to its large population and status as the capital, has the highest coverage rates for water, sewage, and solid waste services. While water supply coverage is extensive across the state, there are still gaps in service delivery to more remote areas. Sewage sanitation indicators are notably sparse, highlighting a critical need for improvement in this area. It is important to note that other municipalities have significantly smaller and more dispersed populations compared to Boa Vista, which poses challenges for implementing centralized collection and treatment infrastructure, often necessitating the use of septic tanks. Solid waste management is problematic even in smaller communities; however, the data reveals a substantial service gap across the state, with higher coverage levels concentrated in municipalities located in the central region.

2.1.15.12 Instance #01: Direct and Indirect Impact Region

The region directly impacted by this project consists of both small and large rural properties within a two-kilometer radius of the property. In this direct impact zone, many small properties are currently uninhabited, used only on weekends or at specific times by their owners. Those small properties that are still continuously inhabited are focused on small-scale production of cattle, cassava, and some regional products such as açaí and andiroba. The larger properties within the direct impact zone are primarily used for cattle ranching and cultivating fruit crops such as passion fruit. Expanding the area of influence to a 5-kilometer radius revealed two settlements officially recognized by INCRA: PA Jatobá and PA Cujubim. In the area affected, either directly or indirectly, by Instance #01, there are no officially recognized territories or villages that are subject to disputes or impacts. This is illustrated in Figure 2.1.15-E.

The current residents of the region, directly or indirectly influenced by the property, are migrants from the north and northeast of the country, the oldest of which have migrated to the region due to the encouragement of agricultural and extractive sectors in the 1940s. A second wave of migration was identified as being influenced by the mining, especially gold, and agricultural sectors in the 1970s. In recent decades, there has been a third wave of migration linked to the problems in Venezuela and migrants from the northeast and south of Brazil seeking opportunities in gold exploration and agricultural production.

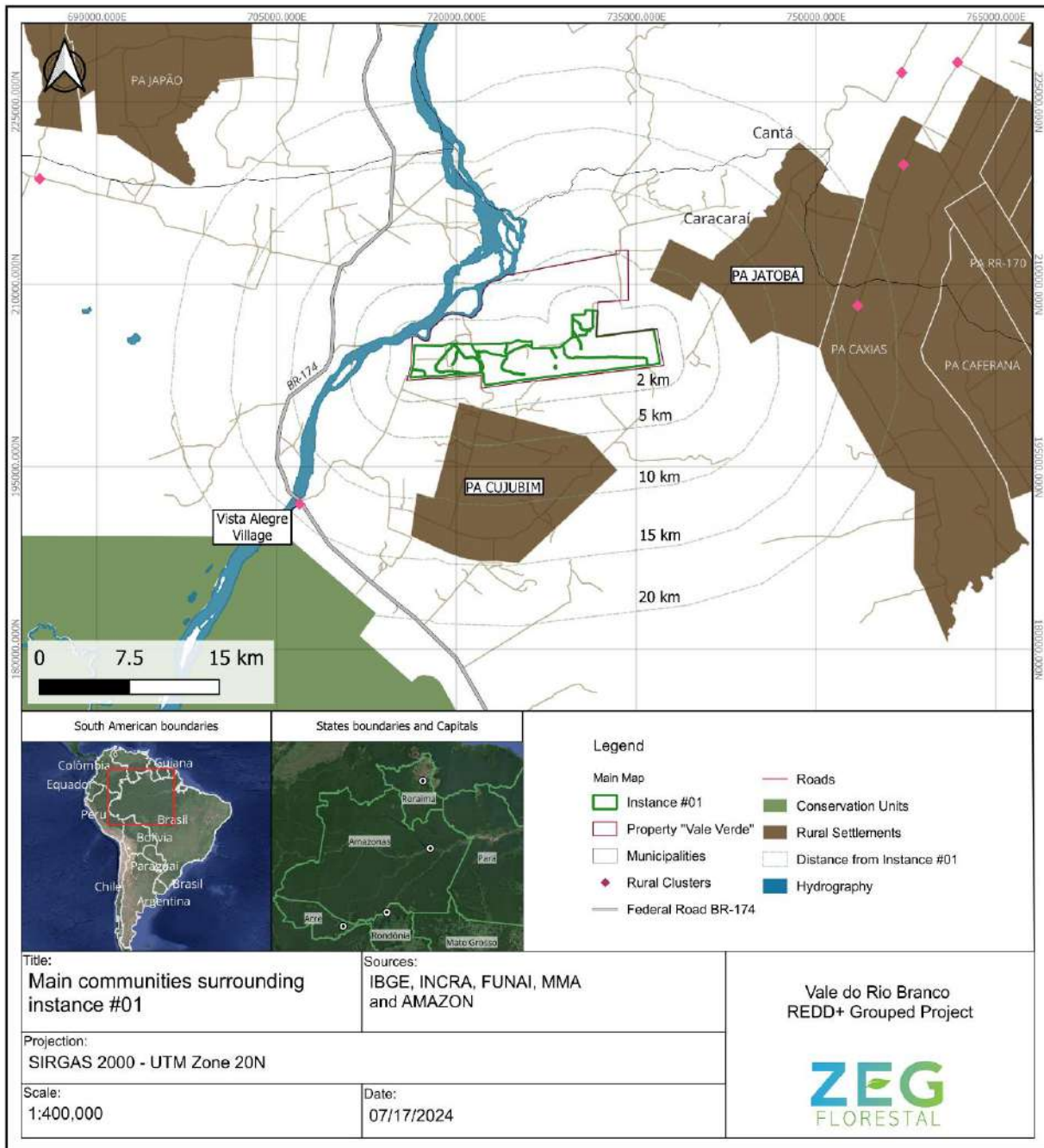


Figure 2.1.15-E: Main communities surrounding instance #01.

2.1.15.13 Surrounding Communities of Instance #01

Instance #01 is located near to two main communities, the Jatobá Settlement Project (PA Jatobá) and the Cujubim Settlement Project (PA Cujubim), as illustrated in Figure 2.1.15-E. Furthermore, during a field visit in February 2024, isolated dwellings were observed in the area north of instance #01 property and along the banks of the Branco River.

The rural settlement consists of a set of independent agricultural units established by the National Institute of Colonization and Agrarian Reform (INCRA) in an area where originally there was a single owner or public lands (INCRA, 2020). Roraima, like other states in the Amazon, was settled due to public policies that encouraged national migration, accompanied by the implementation of settlement policies. Family farming was also promoted as a means of subsistence for settlers and to stimulate products marketing to supply the capital, Boa Vista. The establishment of settlements in Caracaraí occurred in two periods: first in the 1990s and again in the 2000s, with six projects in each period (Bijou, 2020). PA Cujubim was established in 1995, covering an area of 14,195 hectares and capable of settling 243 families; PA Jatobá was established in 2005, covering 13,977 hectares with a capacity to settle 223 families. Individual plots, called “*lotes*”, within the settlements occupy 60 hectares each (INCRA, 2024b).

The roads delineated the routes that people traveled during the migratory process, primarily originating from the Northeast of the country and often destined for the southern region of Roraima. In anticipation of better living conditions, many workers began living in Caracaraí, drawn by the opportunity to acquire land and the social benefits of settlement programs like those created by INCRA. The unpaved rural roads, called “*vicinais*”, were organized in a pattern known as the “fishbone” pattern due to their design. The main road starts narrow and branches out on both sides, forming land plots. As it extends deeper into the forest to create new plots, it widens, resembling a fish skeleton. Figure 2.1.15-F presents a map as a visual example of the “fishbone” pattern in the southeastern region of Roraima.

Over the years, several settlements where family farming was promoted have failed to meet the population's needs in agricultural colonies (Bijou, 2020). As a result, many individuals have abandoned the land, migrating elsewhere, as income from farming alone was insufficient to alleviate poverty. Poverty can also be assessed in terms of limited access to other infrastructural conditions such as schools, hospitals, adequate sanitation, and internet access. Bijou (2020) also observed during field activities in Caracaraí, numerous land plots that have been abandoned or are advertised for sale. Others have been merged into large properties by local businessmen and politicians, diverging from their original purpose.

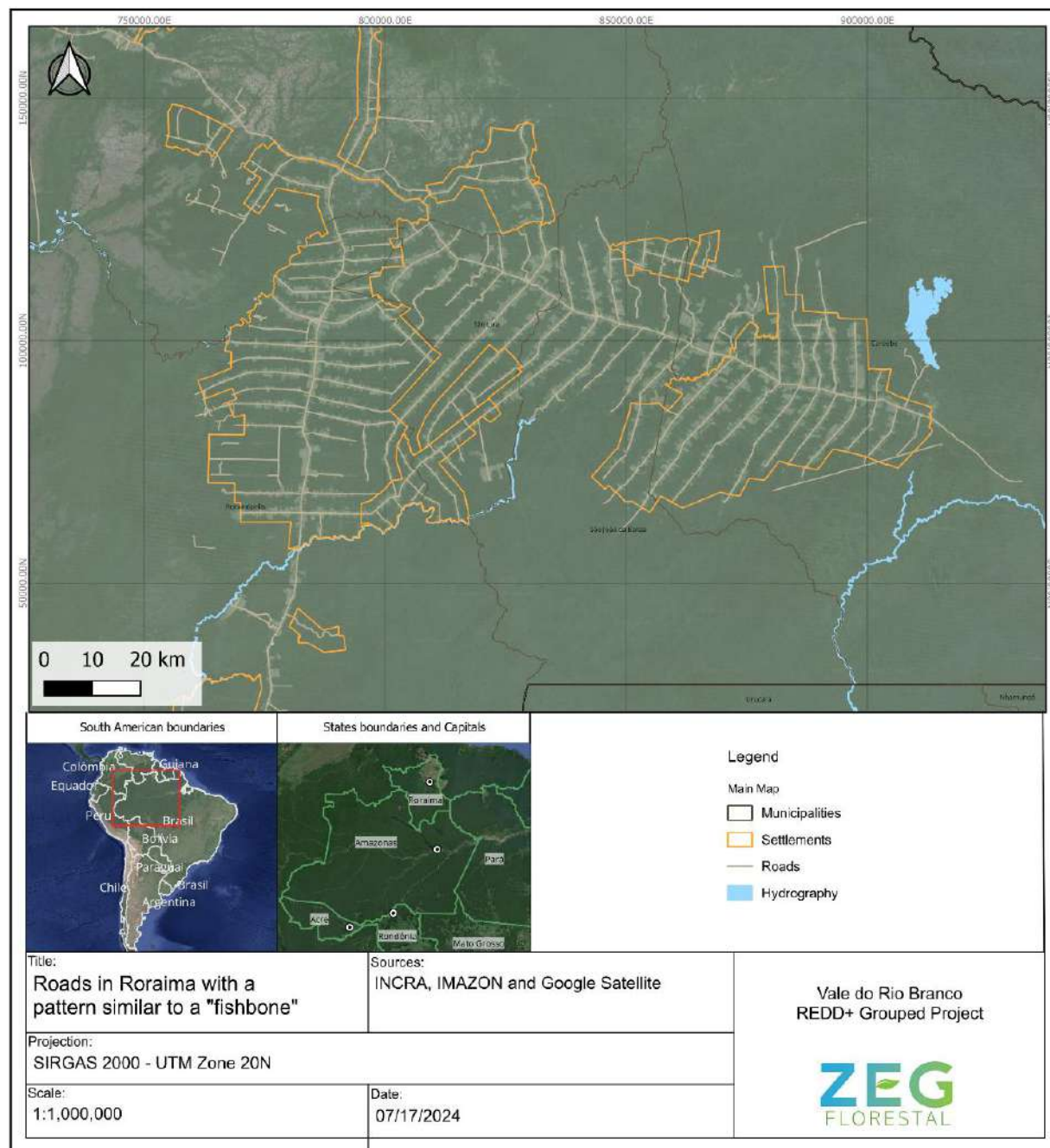


Figure 2.1.15-F: Roads in Roraima with a pattern resembling a "fishbone".

The main economic activities of social groups in Caracarái are fishing, agriculture, extrativism of forest products, such as Brazil nut extraction, which is an important activity in the region (Embrapa, 2023). During field visits in February and September 2024, residents reported cultivating cassava "macaxeira", corn, maxixe, rice, beans, cocoa (with support from the Institute of Technical Assistance and Rural Extension - IATER), passion fruit, cupuaçu, mango, and watermelon. There is also cattle ranching (with milk and cheese production), as well as the rearing of smaller animals, such as chickens and pigs. Additionally, they mentioned

the extraction of açai (Figure 2.1.15-G), tucumã, bacaba (Figure 2.1.15-H), peach palm “pupunha”, andiroba, sorva (historically exploited, similar to rubber tree and depicted in Figure 2.1.15-I), and primarily Brazil nuts (Figure 2.1.15-J). In February, due to the dry season, fewer fishermen were observed on the Branco River (Figure 2.1.15-K), which was at low water levels and difficult to navigate. In September, some fishermen were observed fishing along the Branco River. Fishermen in the region typically reside in the city, however, some of them who practice artisanal fishing, reside along the Branco River bank.



Figure 2.1.15-G: Açai extraction in the region of instance #01.



Figures 2.1.15-H and 2.1.15-I: Bacaba (left) and sorva tree (right) in the region of instance #01.



Figure 2.1.15-J: Brazil nuts in the region of instance #01.



Figure 2.1.15-K: Fishermen on the Branco River.

2.1.15.14 Cujubim Settlement Project

PA Cujubim is located 12 km from the municipal headquarters of Caracaraí, accessed via BR-174, beginning on vicinal 01, with its entrance across from Vista Alegre village (Rocha, 2019). Figure 2.1.15-L shows a vicinal sign for the settlement and Figure 2.1.15-M shows the community gathering building in the village of PA Cujubim. Figure 2.1.15-N shows the village market, and Figure 2.1.15-O shows a church and another community building in PA Cujubim.

The Associação dos Produtores Rurais e Vicinais do Projeto Cujubim (APRUV) was identified as the central residents' association based in the community. APRUV has been collaborating with the state government to include small producers in agricultural support and development projects within the state.

PA Cujubim is an established settlement in the region where all residents have already received titles to their land. Many properties in this settlement no longer have residents, and the process of selling these properties has already begun. Many of them have already been transferred, and some now form medium to large properties primarily focused on livestock production. The small properties that still have residents typically depend on government benefits, such as “Bolsa Família” Program, and produce minimal agricultural products.

According to APRUV, PA Cujubim has around 200 residents. The association has developed projects with the government to secure machinery for local agricultural development. In 2023, approximately 20 farmers from this community participated in a corn planting project alongside government initiatives. Additionally, the association received a donated truck to transport the community's production.



Figure 2.1.15-L: Vicinal sign in PA Cujubim.

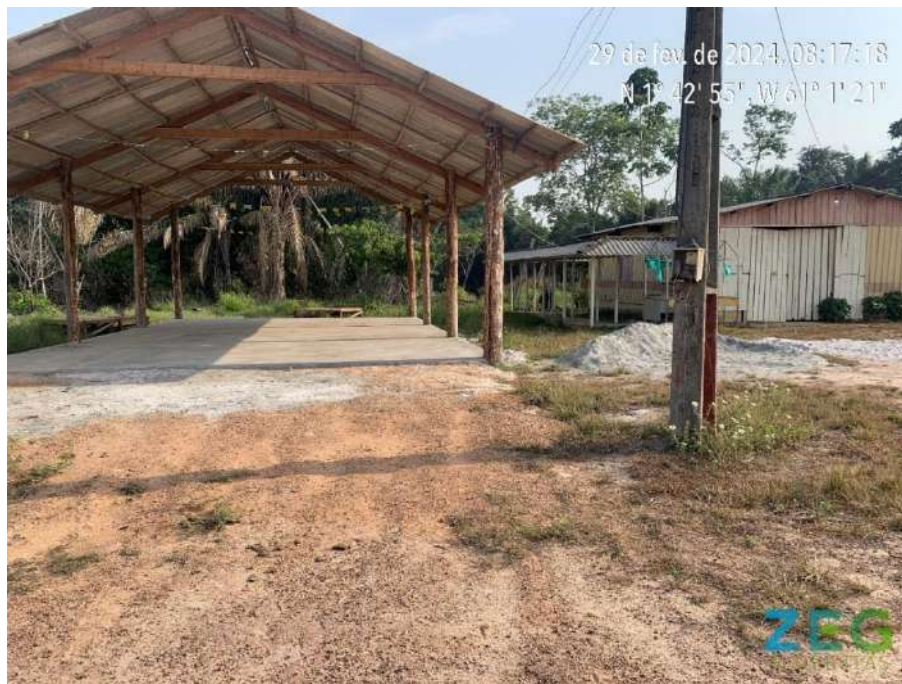


Figure 2.1.15-M: Community gathering building in PA Cujubim.



Figure 2.1.15-N: Village market in PA Cujubim.



Figure 2.1.15-O: Church and another community building in PA Cujubim.

In interviews conducted by Bijou (2020), both annual and perennial crops are predominantly cultivated by families. Corn cultivation stands out as the most cultivated species, followed by cassava. This can be attributed to the strategy adopted by families to produce staple foods, as evidenced by the presence of cassava flour in family meals. Direct sales also occur in the city of Caracaraí and the neighboring village of Vista Alegre (Rocha et

al., 2022). Brazil nut extraction, fishing, and hunting are also done to earn money and/or feed families. Agricultural activities and Brazil nut extraction present low productivity and economic return, making it difficult to survive solely through these practices (Bijou, 2020). These activities are typically conducted as a family group effort, involving the husband, wife, and children. Even when not directly involved in nut extraction, women and children contribute to agricultural tasks.

Brazil nut extraction is traditionally carried out using minimal technology, predominantly in extractive reserve areas. Within these reserves, designated areas known as placements “*colocações*” are where families gather nuts. These roles are typically passed down from father to son, transferring the necessary knowledge for the activity (Bijou, 2020). It is common for extractivists to camp in these forests for a period while gathering, as observed in Figure 2.1.15-P.



Figure 2.1.15-P: Brazil nut extrativism camp in the region of instance #01. Source: IFT (2024).

During a field visit in February 2024, it was common for artisanal fishermen in the region to engage in Brazil nut gathering to supplement their income during the “*defeso*” period (March to June), when fishing temporarily halts to preserve species during spawning.

In interviews conducted by Bijou (2020), residents highlighted Brazil nut extraction as a crucial economic activity, contributing significantly to their diets. However, this activity is seasonal, restricted to the harvest period from June to September, and thus not available year-round. The community's supply chain relies on intermediaries, known as “*atravessadores*”, whose operations are facilitated by the BR-174, both before and during the extraction. During the peak Brazil nut extraction season, agricultural activities decrease due to the flooding of arable lands.

The PA Cujubim region has access to a public electricity supply, benefitting from the federal "Luz para Todos" project, which aims to provide electricity to rural areas with limited access. However, the electricity supply in the state of Roraima is very irregular and often underperforms.

The population of this community is predominantly older, with most residents aged 40 or over and having adult children. There are few children in the community. The area lacks public infrastructure such as schools, health centers, or other public facilities, and all residents rely on the services available in Caracaraí.

The socioeconomic situation of PA Cujubim highlights the broader challenges faced by settlements throughout the state of Roraima: a population dependent on public benefits, lacking developmental structures, and suffering from inadequate improvements to their quality of life, as provided by the public sector.

2.1.15.15 Jatobá Settlement Project

PA Jatobá is the community closest to instance #01, located approximately 60 km from Caracaraí and about 107 km from Cantá via BR-174. Although PA Jatobá is predominantly located within Caracaraí area, its main access route is through the municipality of Cantá. According to Silva (2019), this unique situation creates uncertainties for settlers regarding the responsibility of municipal authorities for public assistance in the settlement. There is an ongoing debate among the settlers about the need to redefine the legal jurisdiction of the area belonging to Cantá, as it has more structured access. Most families seek Cantá's municipal offices for access to social policies like healthcare and education.

Most of PA Jatobá is now stable, with titles either in process or being issued to the settlers. The population primarily consists of migrants from the North and Northeast regions of Brazil who moved to the area at various times in the last century. Today, PA Jatobá continues to attract new migrants regularly.

During a field visit in February 2024, two residents' associations were identified in PA Jatobá: the Associação de Agricultores Familiares Nova Vida (headquarters shown in Figure 2.1.15-Q) and a Women's Association was being formed (construction of the headquarters shown in Figure 2.1.15-R).

The Associação de Agricultores Familiares Nova Vida has been actively involved in addressing the community's jurisdictional conflicts and improving access to public facilities. According to the association's president, the community consists of approximately 300 residents. The Women's Association, formed by local women farmers, aims to secure access to agricultural development projects with the state government. This association, which has about 20 members, is already approved and is working on a "cabruca" cocoa planting project in collaboration with the state government. Cabruca planting involves growing cocoa trees under the canopy of native forest trees, which helps preserve the environment and maintain biodiversity while producing cocoa.



Figure 2.1.15-Q: Headquarters of Associação de Agricultores Familiares Nova Vida in PA Jatobá.



Figure 2.1.15-R: Women's Association construction of the headquarters in PA Jatobá.

Historically, the establishment of the area into a settlement occurred in 2005. However, new families had to stay out of the current agrovillage for four years (2006–2010) due to poor road conditions that hindered access to the main site and the individual plots. The secondary roads constructed through the area for the new families to access their plots are difficult to pass through and interfere with the transportation of goods. This has been a major factor preventing consistent agricultural production in the settlement. Access and internal transportation within the settlement are challenging. The “*vicinais*” opened within the settlement are dirt roads with a few wooden bridges, lacking maintenance and generally in poor condition (Silva, 2019).

In a study conducted by Silva (2019), four types of families living in PA Jatobá were observed:

- Families engaged in agricultural and extractive activities exclusively for subsistence, due to limited resources for production and selling.
- Families producing for subsistence but able to extract surplus for commercial purposes.
- Families lacking sufficient resources or land to produce, relying on purchasing food and/or receiving donations.
- Families owning land and resources but engaging in other income-generating activities to purchase food, given the challenges in accessing their plots.

Families with a fixed income from social security, such as retirement or pension, often manage to sustain their annual agricultural production. Additionally, the Bolsa Família Program (government program that provides financial aid to low-income families to reduce poverty) serves as supplementary or primary income, crucially supporting the purchase of food, medicine, clothing, and school supplies.

In the study by Silva (2019), the most produced and/or acquired foods by families were: beans, rice, corn, flour, pasta, sugar, sweet potatoes, milk, coffee, meats (chicken, duck, beef, pork, fish), cassava, yams, pumpkin, okra, and maxixe. The research also identified a vulnerability regarding workplace safety for those working in agriculture. In cases of work accidents, guidance is needed to access basic rights like comprehensive medical care and welfare benefits.

Regarding basic access to healthcare through the public health system (Sistema Único de Saúde – SUS), the settlement receives assistance from a healthcare agent, although materials are limited, conditions are poor, and there are no on-site health clinics. Regular dental care suffers similar setbacks. The nearest health clinic is in Vista Alegre village. For more complex healthcare needs, settlers primarily travel to hospitals in the capital, Boa Vista.

PA Jatobá has a primary school, as observed in Figure 2.1.15-S. However, other public services have been provided inconsistently due to an ongoing jurisdictional dispute regarding whether the settlement falls under the municipality of Caracaraí or Cantá. This situation has created challenges for the community. Additionally, there have been issues with school transportation for children attending schools outside the settlement. This problem has led some families to move away from PA Jatobá to ensure their children receive a consistent

education (Silva, 2019). This issue is particularly concerning during the rainy season when difficulties in accessing the “vicinais” prevent the school bus from reaching the settlement.



Figure 2.1.15-S: School in PA Jatobá.

PA Jatobá already has access to electricity, thanks to the federal “*Luz para Todos*” project, which aims to provide electricity to rural areas that previously lacked access. However, the electricity supply in the state of Roraima is very irregular and often underperforms, similar to PA Cujubim conditions. The electrical grid suffers frequent power outages and distribution issues (Silva, 2019). Internet access is limited, but there is radio-based internet provided in the settlement, as reported during a field visit in February 2024.

Common practice for disposing of regular waste includes burning or burying it in holes, without recycling, though some families reuse plastic bags, PET bottles and glass. Some families use organic waste for composting. Regarding sewage disposal, a larger percentage use septic tanks, followed by a smaller percentage disposing of sewage in open areas or using rudimentary septic systems.

There is no piped water distribution system. The settlement initially utilized an artesian well and water tank during its installation phase (Figure 2.1.15-T), but these are non-functional due to insufficient electrical power. Many families rely on shallow wells and use the water, sometimes without proper treatment, for all their household consumption needs (Silva, 2019).



Figure 2.1.15-T: Non-functional water tank in PA Jatobá.

The socioeconomic situation of PA Jatobá faces the same challenges as PA Cujubim: a population dependent on public benefits, lacking developmental structures, and suffering from inadequate improvements to their quality of life, as provided by the public sector.

2.1.16 Project Zone Map and Project Location (VCS, 3.11, 3.18; CCB, G1.4-7, G1.13, CM1.2, B1.2)

The Project Zone is the area where project activities aim to generate or maintain net climate benefits (VCS Program Definitions, v4.5), and where implemented activities directly affect land and associated resources, including those related to the provision of alternative livelihoods and community development (CCB Standard, v3.1).

For grouped projects, the Project Zone also includes all potential project areas, as per the CCB Standard, v3.1. Within this project, Vale do Rio Branco REDD+ Grouped Project, the Project Zone is initially defined in accordance with the legal delimitation of the Ecological-Economic Zoning of the state of Roraima (Roraima State Complementary Law No. 323 of 08/02/2022). Specifically, the Project Zone is based on the Productive Use Zone (ZUP).

The ZUP is defined as the area of the state designated for production. For properties that have forested areas within this zone, a state law has reduced the Legal Reserve to 50% of the property, excluding Permanent Preservation Areas. Consequently, all forested lands within the Project Zone, where new project instances may be included, are subject to the same legislation. More information in Section 2.5.1 – National and Local Laws.

Furthermore, the Special Use Zone (ZUE) includes Indigenous Lands, Conservation Units, Military and Inalienable Areas. The Urban Zone (ZU) covers Urban and Peri-Urban Areas. ZUE and ZU, alongside the Conservation Units created or expanded after the 2022 Zoning and before the project start date, are excluded from the Project Zone. The Project Zone consolidated area comprises 6,905,775 hectares. If relevant for future additional project instances, the Project Zone will be updated to encompass necessary adjacent areas. The project first instance, belonging to Property “Vale Verde”, is in the Midwest region of the Project Zone. It does not overlap with the project area of any other VCS AFOLU projects, as there are no projects currently located in Roraima State.

The Figure 2.1.16-A displays the map of the project zone and location of the first instance, including the rural clusters and settlements, Indigenous Lands and Conservation Units in the region.

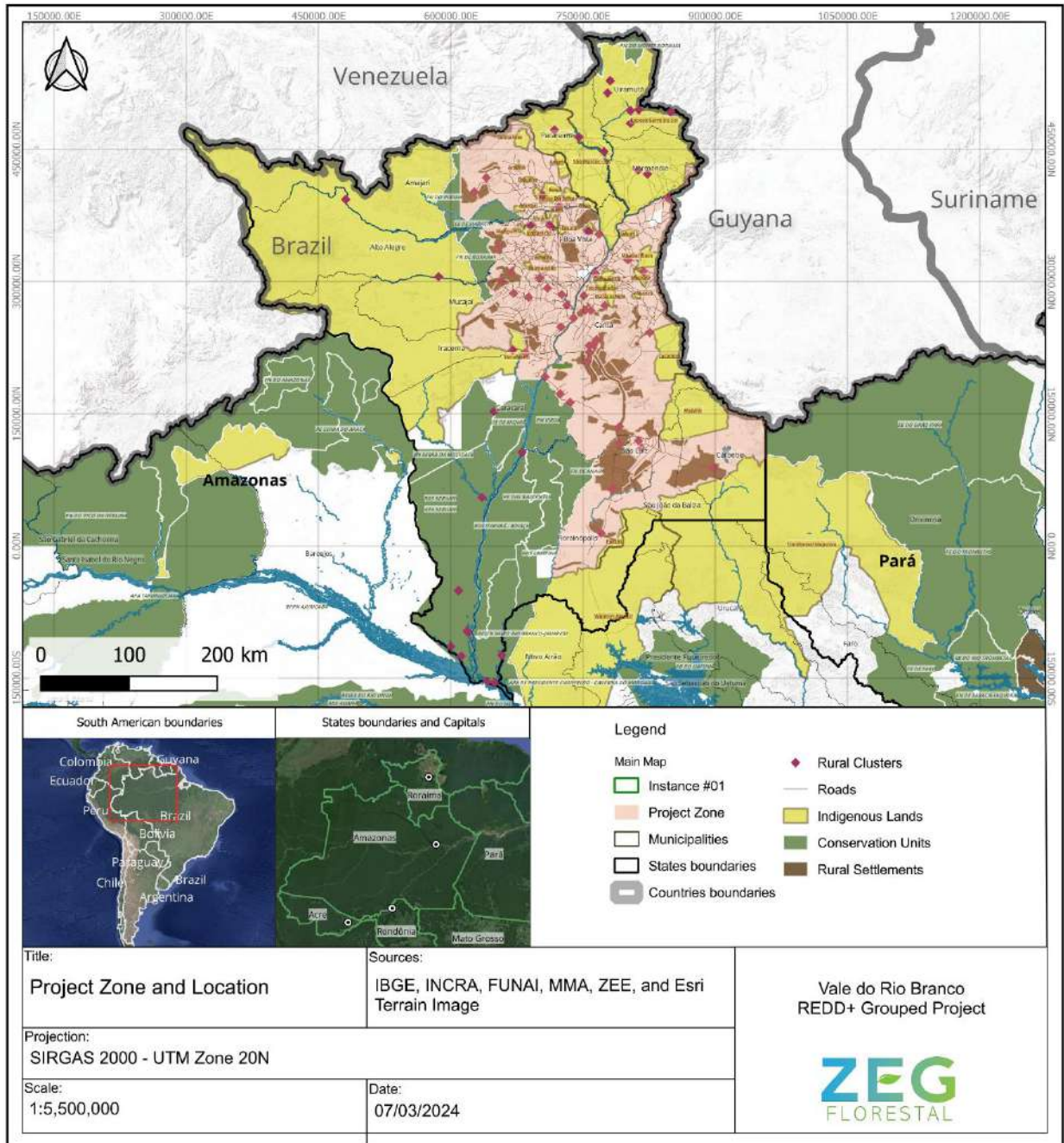


Figure 2.1.16-A: Project Zone and Location of Vale do Rio Branco REDD+ Grouped Project.

Vale Verde Property covers an area of 14,029 hectares and is located on the left bank of the Branco River in the municipality of Caracaraí, as seen in Figure 2.1.16-D below. The most common route to Vale Verde's headquarters is via the BR-174 highway in the Boa Vista-Caracaraí direction. After crossing the bridge over the Branco River, one should turn left onto vicinal 01, 02 and 03 of the Cujubim Settlement, and continue for approximately 35 km to reach

the farm's entrance (IFT, 2024). Figure 2.1.16-B shows an internal road on Vale Verde Property, and Figure 2.1.16-C shows a view of the Branco River bank.



Figure 2.1.16-B: Internal road on Vale Verde Property.



Figure 2.1.16-C: View of the Branco River bank.

The municipality of Caracaraí is located 134 km from the capital Boa Vista and is accessible via the BR-174 highway. The Branco River, which crosses the municipality, provides a significant waterway connection to Manaus in Amazonas. The city also features an airport with a 2,500-meter paved runway, facilitating air transportation.

Caracaraí is bordered to the north by the municipalities of Cantá and Iracema; to the south, by Rorainópolis, São João da Baliza, and São Luiz; to the east, by Caroebe and the international border with Guyana; and to the west, by the state of Amazonas. Covering a total area of 47,380 km² (IBGE, 2023b), Caracaraí accounts for approximately 21% of Roraima's territory and is in the Amazonas time zone (UTC-4).

In Figure 2.1.16-D below, an overview of Instance #01 Location can be seen, showing the Caracaraí City Hall, the rural settlements in the vicinity, the Federal Road BR-174, and the deforested areas in the region until 2023.

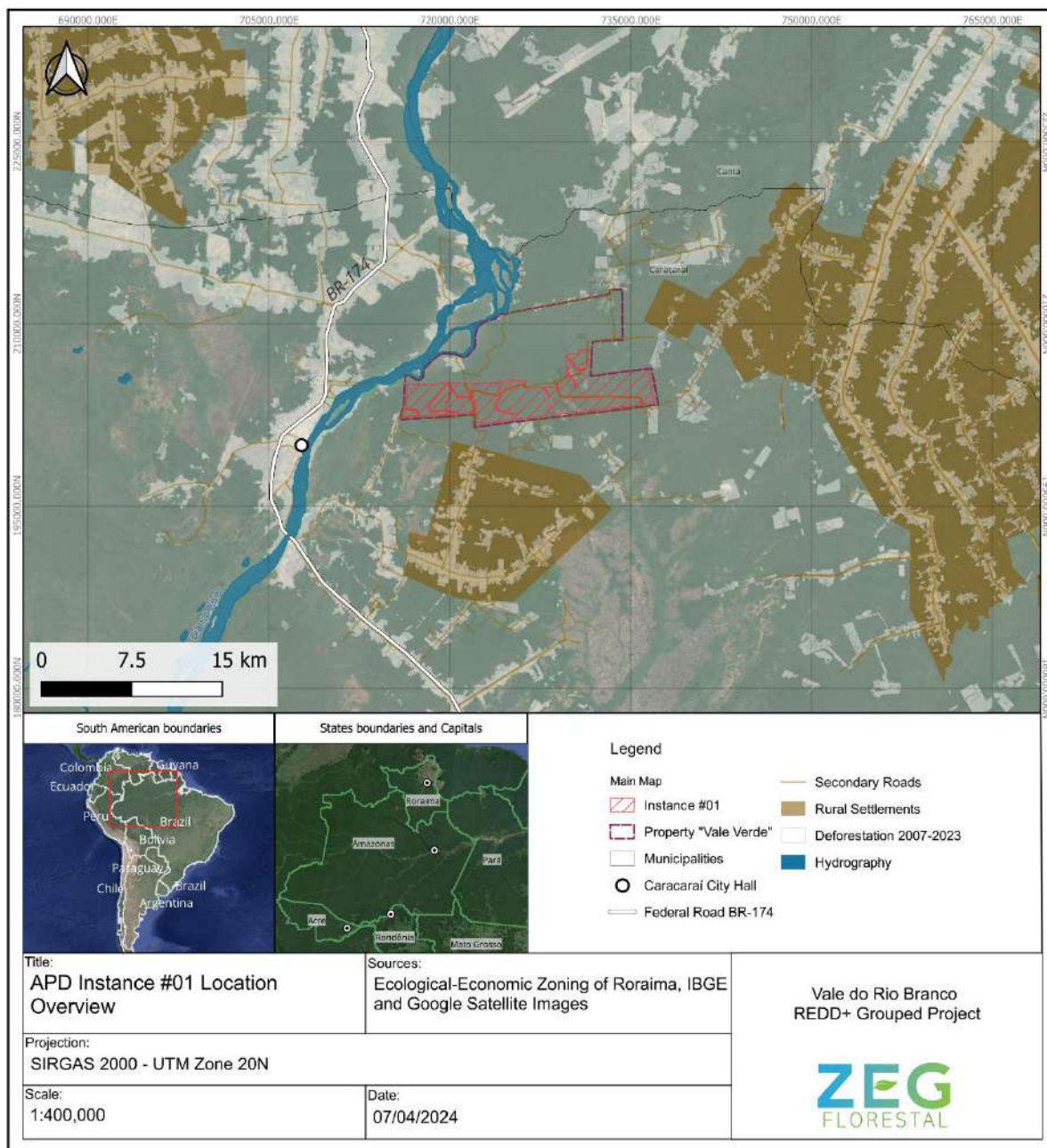


Figure 2.1.16-D: Instance #01 Location Overview of Vale do Rio Branco REDD+ Grouped Project.

2.1.17 Project Activities and Theory of Change (VCS, 3.6; CCB, G1.8)

Considering the Vale do Rio Branco REDD+ Grouped Project's Instance #01, the project will implement climate-focused strategies such as forest monitoring through techniques that use remote sensing techniques and local surveillance, as well as measures to prevent degradation and deforestation, including firefighting efforts. This will be accomplished through ground surveillance along internal and border roads. Orbital remote sensing monitoring requires periodic interpretation of satellite and radar imagery, and firefighting efforts will

encompass the planning and structuring of firefighting teams, along with cooperation agreements with municipal authorities. The project also aims to conduct weather condition analysis and alerts, such as daily fire index calculations in the project area. Forest inventory will be the main focus of the project activity.

Regarding infrastructure, the project aims to build and maintain firebreaks, project signs, and fences within the property, along with the regular maintenance of internal rural roads. It also seeks to improve other infrastructure aspects, such as well drilling, electricity, and internet connectivity to support project activities.

In terms of biodiversity, the project will introduce technologies for monitoring flora and fauna and establish a comprehensive biodiversity inventory and database. Environmental education activities will be promoted, and standard procedures will be defined in cases of unauthorized activities.

Additionally, the project will prospect opportunities related to bioeconomic development in the region, aiming to contribute to the socioeconomic development and general well-being of local communities. Through a social inventory and field survey, the project will assess needs and development opportunities, identify potential partnerships with relevant institutions to support technical assistance and rural development, and establish a community engagement framework.

2.1.17.1 Theory of Change

The development process of the Theory of Change for the Vale do Rio Branco REDD+ Grouped Project is being conducted based on the methodology established by the "Open Standards for the Practice of Conservation" network (version 4.0, 2020). In the strategic planning phase, the primary focus is to provide and formalize the causal relationships between the interventions prioritized by the project, aiming to generate positive impacts on its various conservation targets, while clearly outlining the underlying assumptions and opportunities identified during the process.

This phase is a crucial step for the project, as it provides a solid foundation for the subsequent stages, including the strategic monitoring plan, work plan, and tactical and operational plans.

The main goal of constructing the Theory of Change using this methodology is to ensure that the project's activities are no longer planned and executed intuitively, but are instead grounded in impact-driven approaches, meaning they are guided by the desired transformations expected from the conservation project.

During the initial discussions of the project development team, the following items were defined:

- Vision: As defined by Open Standards, vision is "a description of the desired state or ultimate condition that you are working to achieve". In other words, it is an aspirational statement outlining where a given initiative aims to be in the future.

- **Conservation Target:** An element of biodiversity (species, habitat, or ecological system) within the project area that the initiative has chosen to focus on. The set of targets should represent the relevant biodiversity of the project location.
- **Ecosystem Services:** These are services that sustain the functioning of ecosystems, species, and habitats, thereby benefiting people. They are influenced by the conservation targets.
- **Human Well-being Targets:** In the context of a conservation project, these refer to components of human well-being that are affected by the status of conservation targets. The set of these targets should represent the array of needs that are dependent on conservation targets.
- **Direct Threat:** Primarily, these are human actions that directly degrade one or more conservation targets. They can also be natural phenomena that have been altered by human activities. These represent the main threats to the health/integrity of the project.

The outcomes of this initial phase of the development process of the Theory of Change for the Vale do Rio Branco REDD+ Grouped Project are presented in Figure 2.1.17-A.

Vision: *The projects of ZEG Florestal tackle climate change and Amazon deforestation, preserve standing forests and their socio-biodiversity, reduce greenhouse gas emissions, and contribute to the economic self-sufficiency of local populations through the strengthening of their bioeconomy, offering the market secure and tangible carbon credits.*

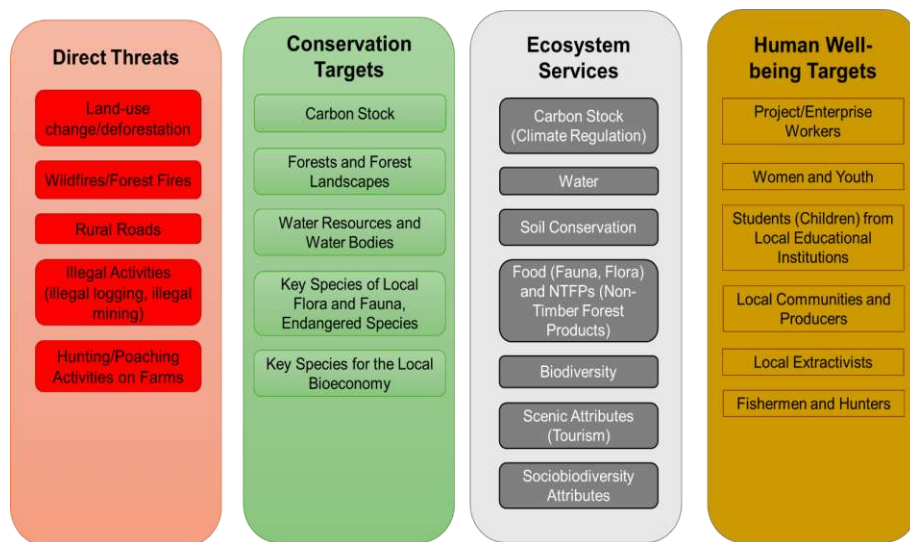


Figure 2.1.17-A: Outcomes of the initial phase of the Theory of Change development process.

The complete version of the theory of change and its details are going to be described in the project description document.

2.1.18 Sustainable Development Contributions (VCS, 3.17)

The proposed Instance #01 project activities intend to cooperate to nationally stated sustainable development priorities, such as the UN Sustainable Development Goals (SDGs). They align with Brazil's Nationally Determined Contribution (NDC) assumed by signing the Paris Agreement, which commits to reducing emissions by 59 to 67% by 2035, compared to 2005 levels, which is consistent, in absolute terms, with an emission level of 1.05 to 0.85 GtCO₂e (MMA, 2024). This alignment is due to the project's focus on reducing net greenhouse gas (GHG) emissions by preventing deforestation.

According to the Brazil's Sustainable Cities Development Index (IDSC, 2023), Caracaráí, where the project's instance #01 is located, scores 43 out of 100 in the SDG targets achievement assessment, classifying it as a municipality with a low level of sustainable development. The projects initial plans to support SDG 1 – No Poverty; 2 – Zero Hunger; 13 – Climate Action; and 15 – Life on Land. Future surveys may include other goals, depending on identified local demands.

In the municipality of Caracaráí, SDGs 1 and 2 are assessed as having a moderate level of achievement, while SDG 13 is classified as having very low achievement, and SDG 15 reaches a high level of achievement. Table 2.1.18-A shows the proposed preliminary project activities, the related SDGs and goals activities aim to contribute with.

Table 2.1.18-A: SDGs, goals and contributing project activities.

SDG	Goals	Project activities
<i>1 – End poverty in all its forms everywhere</i>	1.4	- Community needs and developmental opportunities assessments. - Bioeconomic development prospects.
<i>2 – End hunger, achieve food security, improve nutrition, and promote sustainable agriculture</i>	2.3; 2.4	- Community needs and developmental opportunities assessments. - Bioeconomic development prospects. - Potential partnerships with relevant institutions to support technical assistance and rural development.
<i>13 – Take urgent action to combat climate change and its impacts</i>	13.1; 13.3	- Forest and biodiversity monitoring. - Degradation and deforestation prevention measures. - Firefighting teams structuring. - Weather condition analysis and alerts. - Environmental education activities.

SDG	Goals	Project activities
<i>15 – Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, halt and reverse land degradation, and halt biodiversity loss</i>	15.5; 15.7	• Forest and biodiversity monitoring. • Degradation and deforestation prevention measures. • Firefighting teams structuring. • Standard procedures for unauthorized activities.

2.1.19 Implementation Schedule (CCB, G1.9)

This section is under development.

2.1.20 Risks to the Project (CCB, G1.10)

This section is under development.

2.1.21 Benefit Permanence (CCB, G1.11)

This section is under development.

2.1.22 Financial Sustainability (CCB, G1.12)

This section is under development.

2.2 Without-project Land Use Scenario and Additionality

2.2.1 Conditions Prior to Project Initiation and Land Use Scenarios without the Project (VCS, 3.13; CCB, G2.1)

2.2.1.1 Current and historical land-use

The property of Instance #01 was incorporated into Madeireira Vale Verde in 2004, with the start of Sustainable Forest Management activities in 2006. This management followed a 25-year harvest cycle, allowing each harvested area to be explored again at the end of this period. In 2020, all areas of the property were explored according to the management plan, formally concluding the harvest cycle in 2022 and initiating the fallow period.

Following this period, the property was acquired by the current owner, who also owns other rural areas dedicated to soybean production. He plans to allocate part of the area to soybean cultivation, as permitted by environmental legislation, while maintaining 50% of the property as a Legal Reserve, as described in Section 2.5.1 – National and Local Laws. This scenario reflects a common practice in the region, where the conversion of forests into

agricultural land for commodity cultivation, such as soybeans, is economically incentivized and has been widely adopted over the past ten years (for more details, see Section 3.1.5.2 – Additionality Methods).

2.2.1.2 Ecosystem Type

The Instance #01 area is located within the Amazon biome, characterized by Dense Ombrophilous Forest vegetation. This region covers a relatively flat plateau with firm ground, interspersed with areas of *cipoal* (forest with abundant lianas), small patches of rocky outcrops, and lowland forests that experience partial flooding for approximately three months each year (IFT, 2024). Figure 2.2.1-A presents a photograph of the forest within the property.



Figure 2.2.1-A: Photograph of the forest within the property of instance #01.

2.2.1.3 Vegetation

According to IBGE (2023c), the Instance #01 area is entirely located within the Amazon biome, characterized by Submontane Dense Ombrophilous Forest with Emergent Canopy. As detailed in the Technical Manual of Brazilian Vegetation (IBGE, 2012), this forest type typically occurs on the slopes of plateaus and mountain ranges, predominantly featuring tall phanerophytes (woody plants with growth points located well above the ground). The understory, the layer of vegetation beneath the forest canopy, is rich with naturally regenerating seedlings, small palms, lianas, and emergent species, creating a diverse ecosystem.

According to the ZEE (SEADI, 2022), the vegetation in the Instance #01 area is predominantly Submontane Dense Ombrophilous Forest, covering 87% of the area, with a smaller portion in the west classified as Campinarana, as shown in Figure 2.2.1-D below.

However, during the Forest Inventory survey visit by IFT (2024), no Campinarana vegetation was found within Instance #01 property area. Consultations with the long-term property employees further confirmed the absence of this vegetation type in the area. This discrepancy highlights that while indicated in the ZEE data at a broader scale, Campinarana was not observed during the on-site visit.

Additionally, according to the Forest Inventory (IFT, 2024), the vegetation in the area includes a rich diversity of species ranging from low to high density, with notable individuals such as *Bertholletia excelsa* Bonpl. (Brazil nut tree), *Pentaclethra macroloba* (Willd) Kuntze, *Vouacapoua pallidior* Ducke., *Pourouma* sp., *Eschweilera ovata* (Cambess.) Miers, *Ocotea* sp., *Inga* sp., *Geissospermum sericeum* (Sagot) Benth., and *Cecropia palmata* Willd, among others. The Figures 2.2.1-B and 2.2.1-C shows a Brazil nut tree in the instance #01.



Figures 2.2.1-B and 2.2.1-C: Photographs of Brazil nut tree in instance #01.

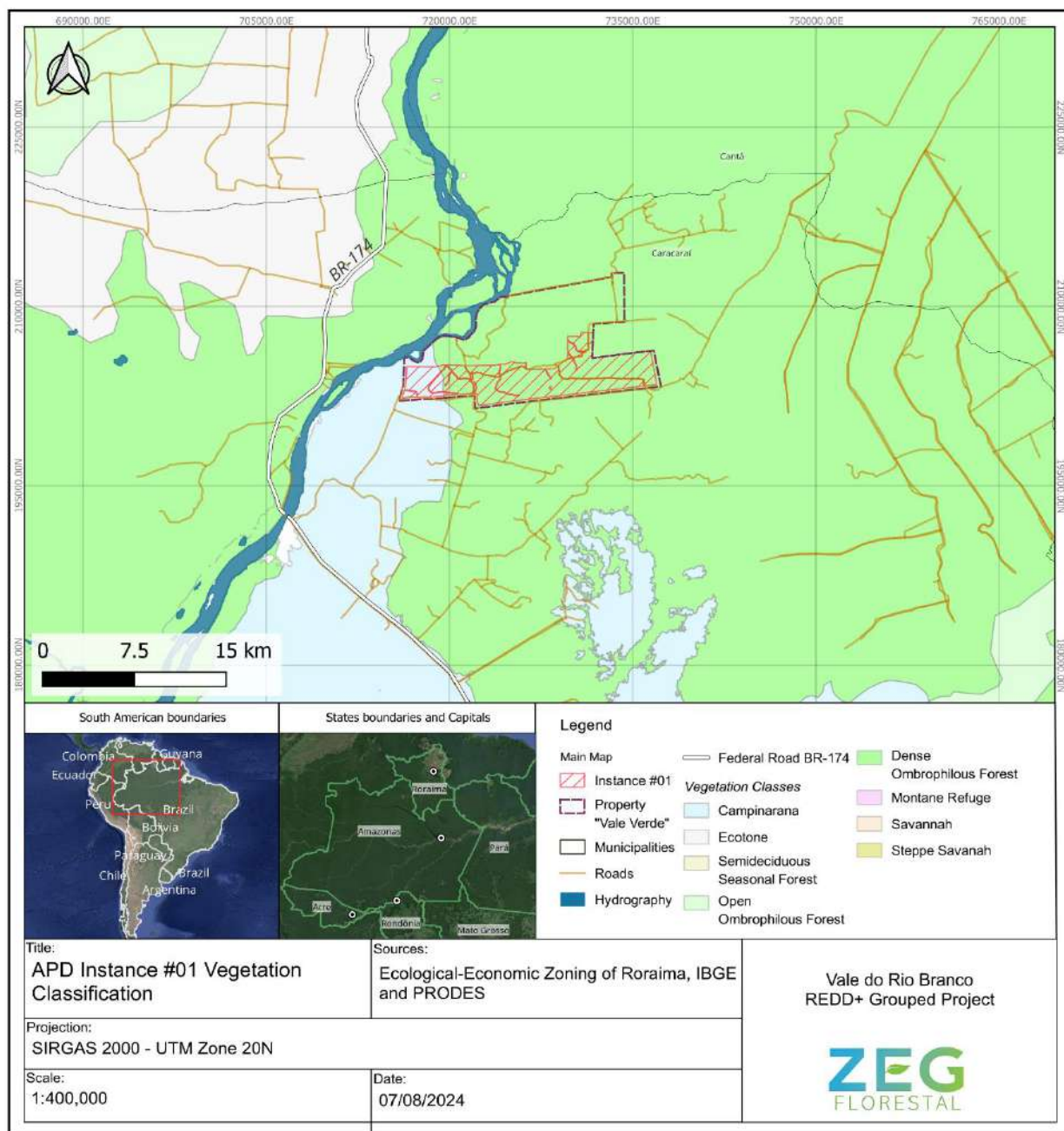


Figure 2.2.1-D: Instance #01 Vegetation Classification (SEADI, 2022).

2.2.1.4 Climate

According to the Köppen-Geiger classification, the Instance #01 project area is categorized as "Am" - Humid Tropical climate, illustrated in Figure 2.2.1-E. The "Am" climate class represents the transition zone between Super-Humid (Af) and Semi-Humid/Semi-Arid (Aw) climate zones (SEADI, 2022). It is characterized by July's (coldest month) average temperature always above 18°C and a brief dry season, which is compensated by high total annual precipitation (Golfari *et al.*, 1978 *apud* EMBRAPA, 2024).

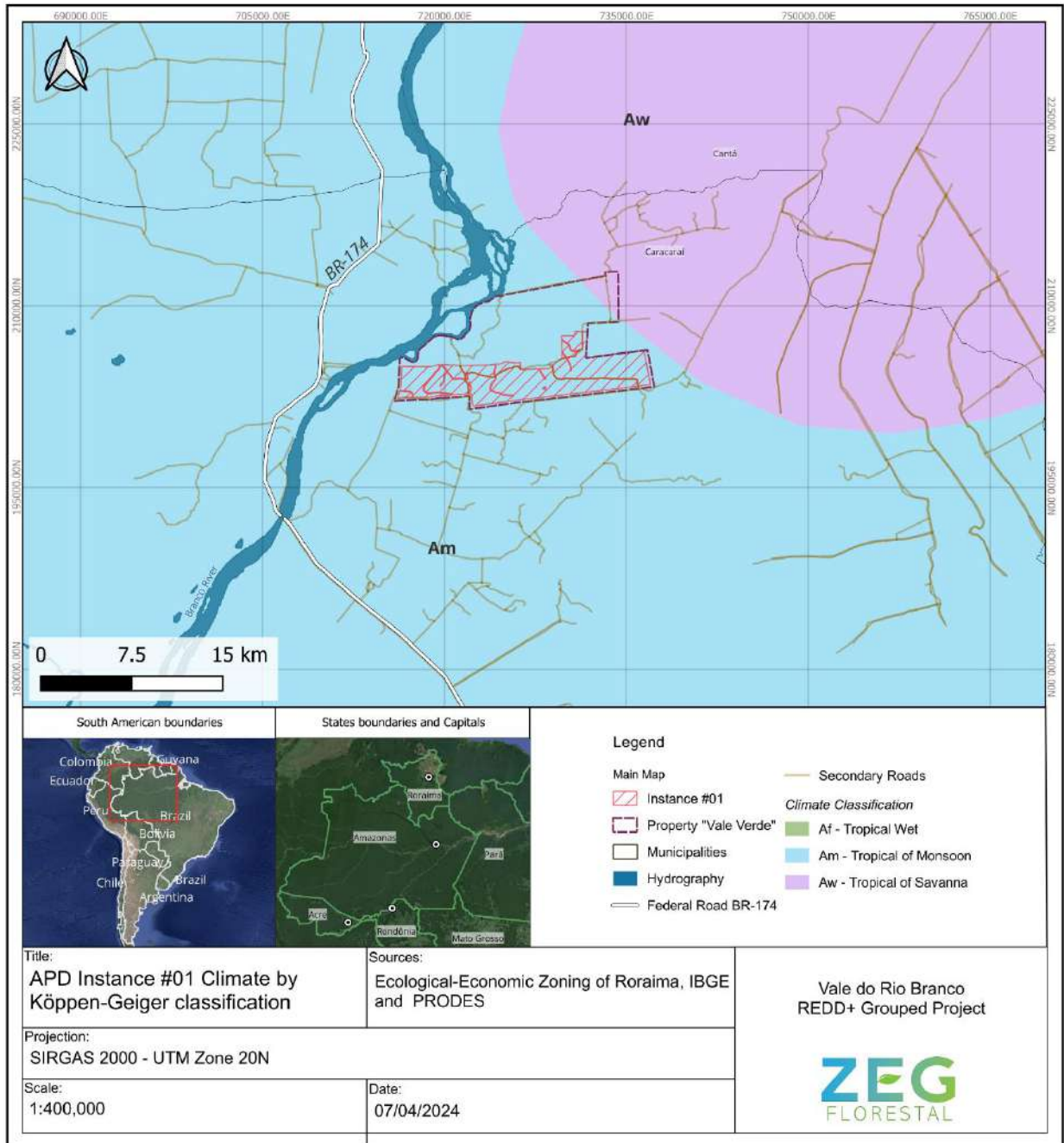


Figure 2.2.1-E: Instance #01 Köppen-Geiger Climate Classification Map (SEADI, 2022).

Figure 2.2.1-F presents the long-term mean precipitation in Instance #01 for the reference period from 1994 to 2023 (Funk, 2015), showing behavior consistent with the expectations for the Am climate of the Köppen-Geiger classification. The average annual precipitation in the Instance #01 project area over the last 30 years has been 2,070 mm. In Figure 2.2.1-G below, the spatial distribution of this precipitation is shown, considering its long-term mean, calculated between the years 1994 and 2023, in the region of Instance #01.

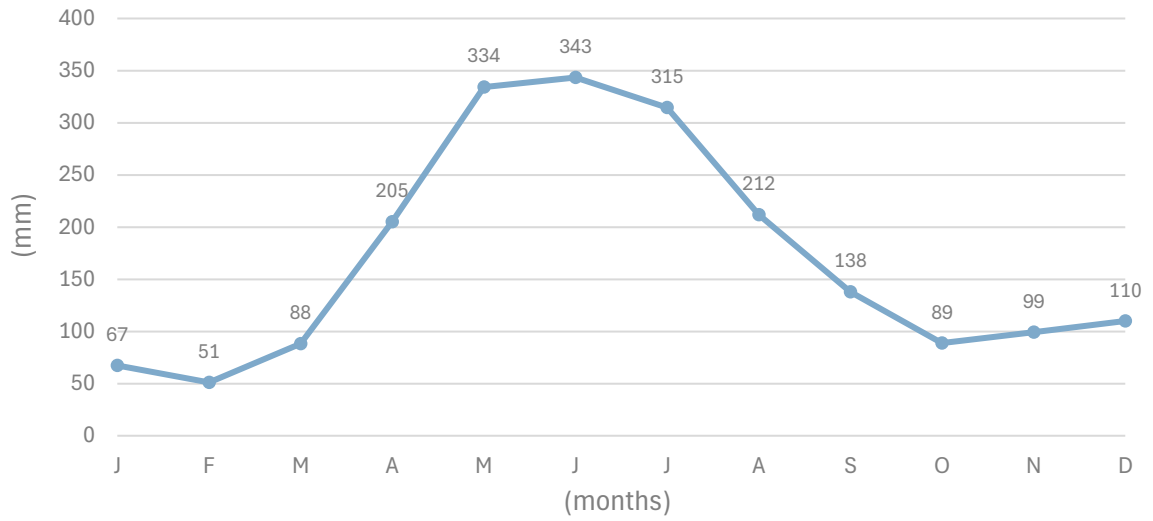


Figure 2.2.1-F: Long-term mean precipitation in Instance #01, reference period from 1994 to 2023. Source: Funk (2015).

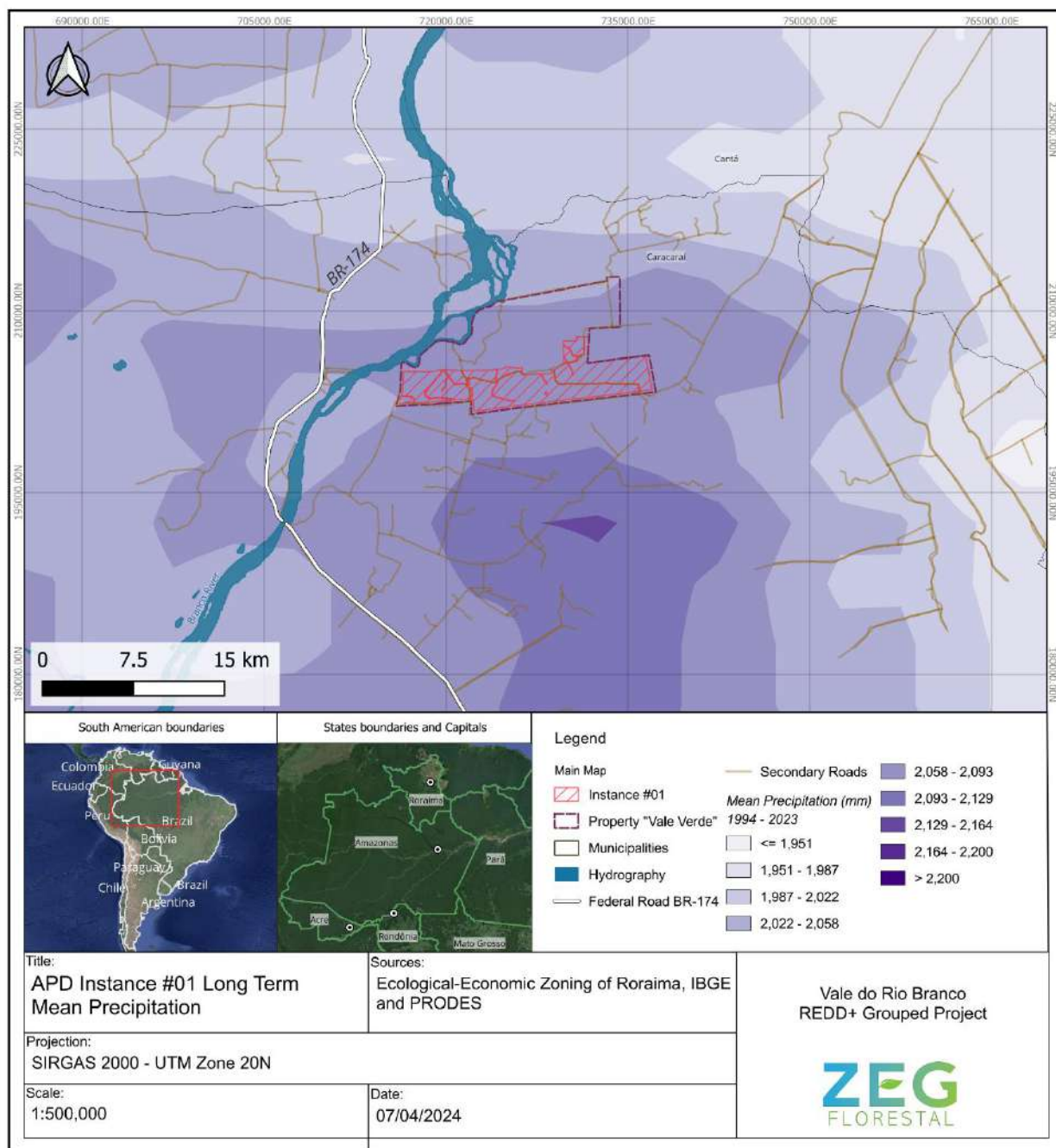


Figure 2.2.1-G: Instance #01 Long Term Mean Precipitation Map. Source: Funk (2015).

2.2.1.5 Hydrology

A small section of the Instance #01 project area, approximately 1.6 km in the western part, is bordered by a 500-meter-wide Permanent Preservation Area (APP) along the Branco River, as shown in Figure 2.2.1-I. Within the Vale Verde Property, which encompasses the Instance #01 area, there are several non-permanent watercourses known as “igarapés”, shown in Figure 2.2.1-H. These streams flow through the forest during the rainy season and

drain into the Branco River. The largest igarapés, Sumaúma and Grande, are located in the western part of the Instance #01 property.



Figure 2.2.1-H: View of one *igarapé* bed from the Branco River during the dry season (February 2024).

The hydrographic regime of the Branco River includes a rainy period from March to September, with peak floods typically occurring in July, followed by a dry period from October to February. Navigability of the river depends on the water level, therefore, navigation may be impossible at times during the dry season (Dias, 2017).

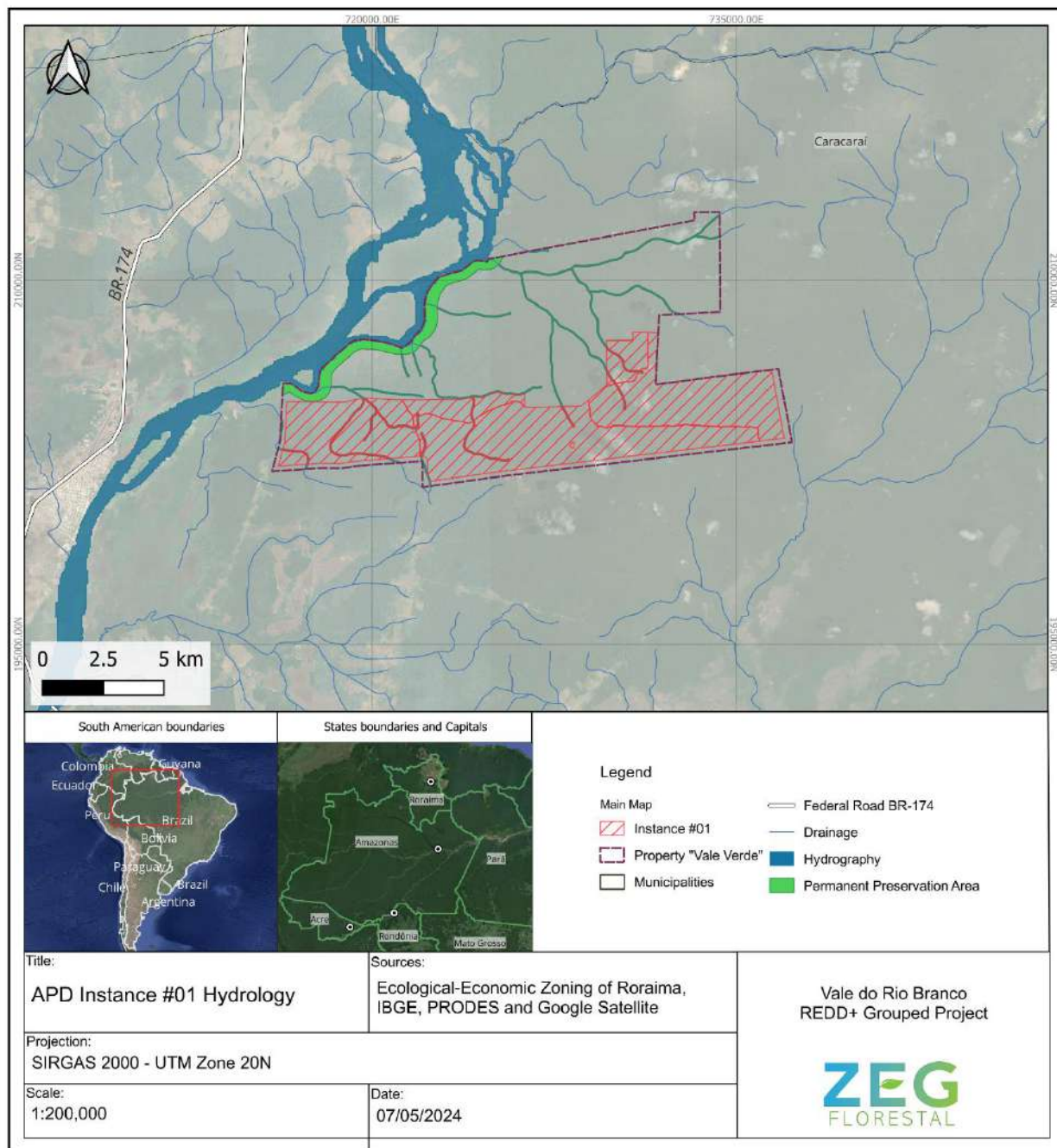


Figure 2.2.1-I: Instance #01 Hydrology and Permanent Preservation Area Map.

2.2.1.6 Topography

Analysis of the Digital Elevation Model (Laipelt *et al.*, 2024) in Instance #01 reveals an average slope of 5.8°, with gradients ranging from sea level to steep inclines of up to 60° at the most precipitous points. Most of the area exhibits predominantly flat or gentle slopes, as illustrated in Figures 2.2.1-J and 2.2.1-K below. The Forest Inventory conducted within the Instance #01 property identifies rocky outcrops as the steepest areas, with sparse tree species

presence. In contrast, the forest is predominantly distributed on relatively flat upland plateaus (IFT, 2024).

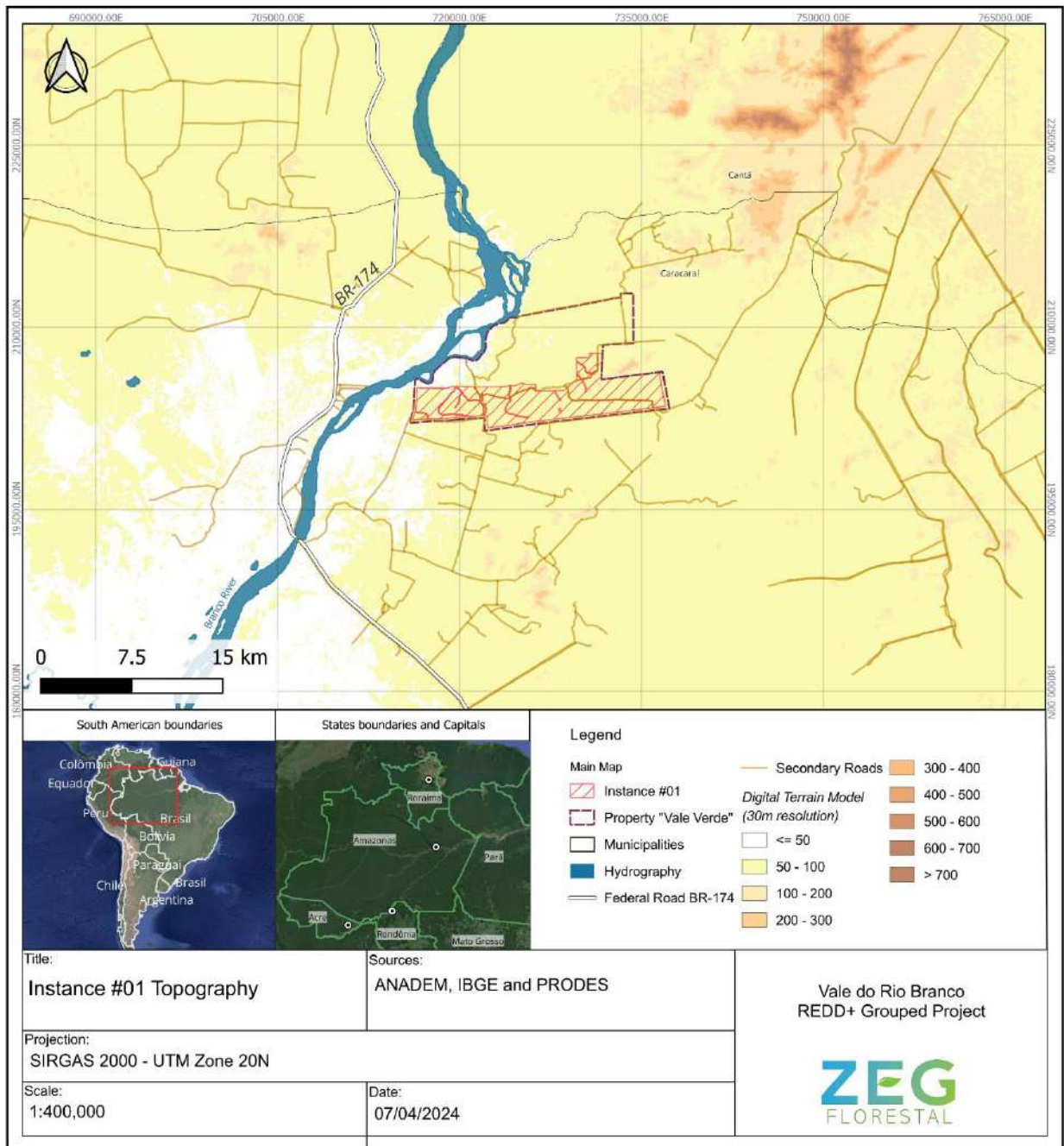


Figure 2.2.1-J: Instance #01 Topography Map.

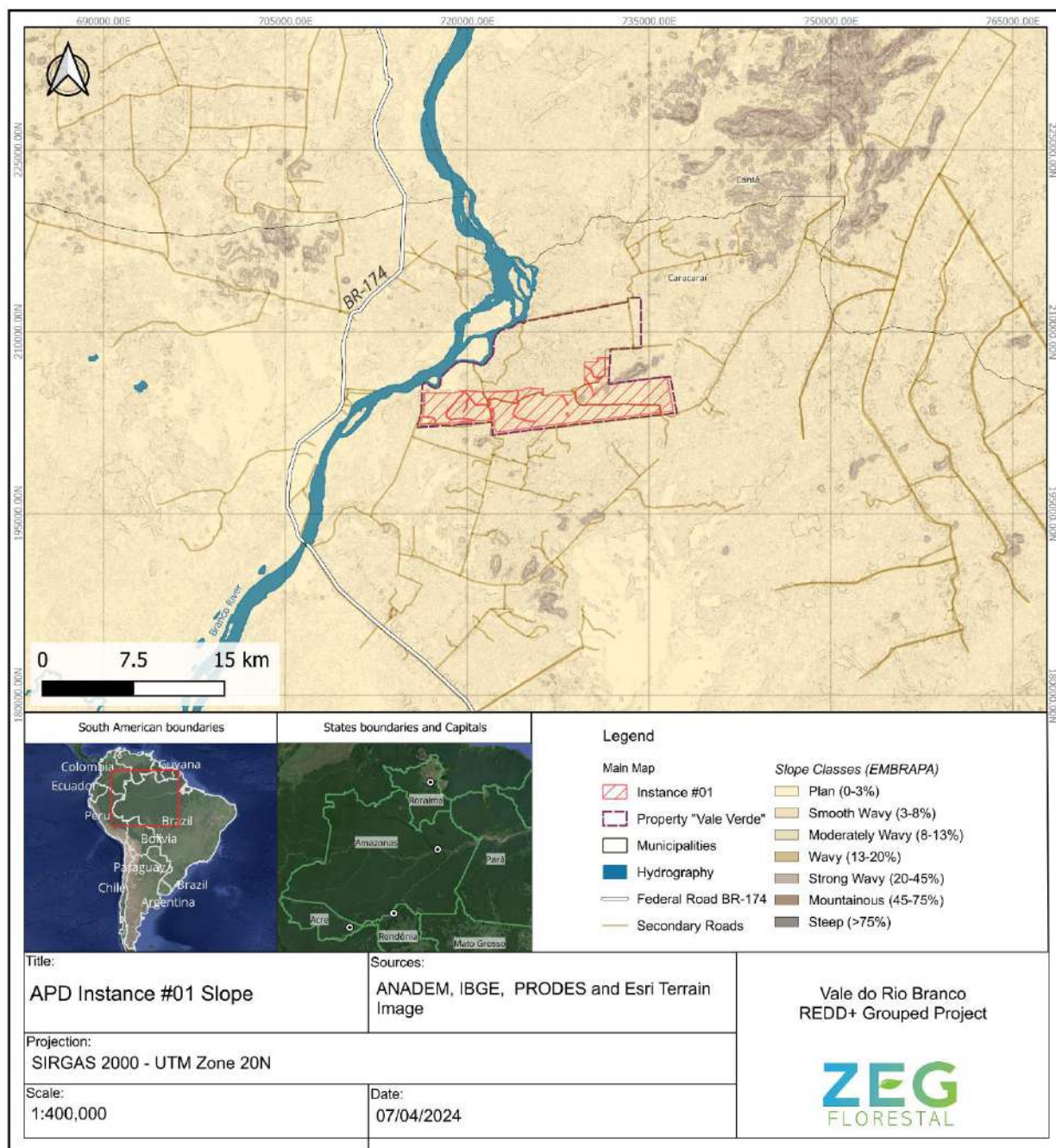


Figure 2.2.1-K: Instance #01 Topographic Slope Map.

2.2.1.7 Soils

Instance #01 area is characterized by a single soil type, classified as Latossolo Vermelho-Amarelo Distrófico Típico, according to the ZEE (SEADI, 2022), as illustrated in Figure 2.2.1-L. This soil has a texture ranging from clayey to very clayey, with low natural fertility. Latosols are deep soils, with uniformity in color, texture, and structure throughout the profile.

The Latossolo Vermelho-Amarelo is found in areas with relief ranging from plain to wavy and occurs in well-drained environments, contributing to the formation of deep and uniform soils. However, these soils have a limited amount of water available for plants. To address this limitation, it is necessary to adopt management strategies that optimize water use and increase crop resilience during drought periods. Widely common in agriculture, these soils offer favorable conditions for deep root development due to their depth and porosity. However, they may present chemical limitations, especially if they are allic, dystrophic, or acric. Additionally, the low phosphorus content found under natural conditions makes phosphorus fertilization a recommended practice. The plain or wavy terrain also facilitates agricultural mechanization (Almeida; Dos Santos; Zaroni, 2021).

According to Ramalho Filho & Beek (1995), developers of the Agricultural Land Suitability System, the implementation of advanced production technologies, including irrigation, drainage, and natural fertility correction, can significantly boost productivity in soils traditionally deemed unsuitable. In the region, this is evidenced by the land use for agricultural activities on the rural properties surrounding Instance #01.

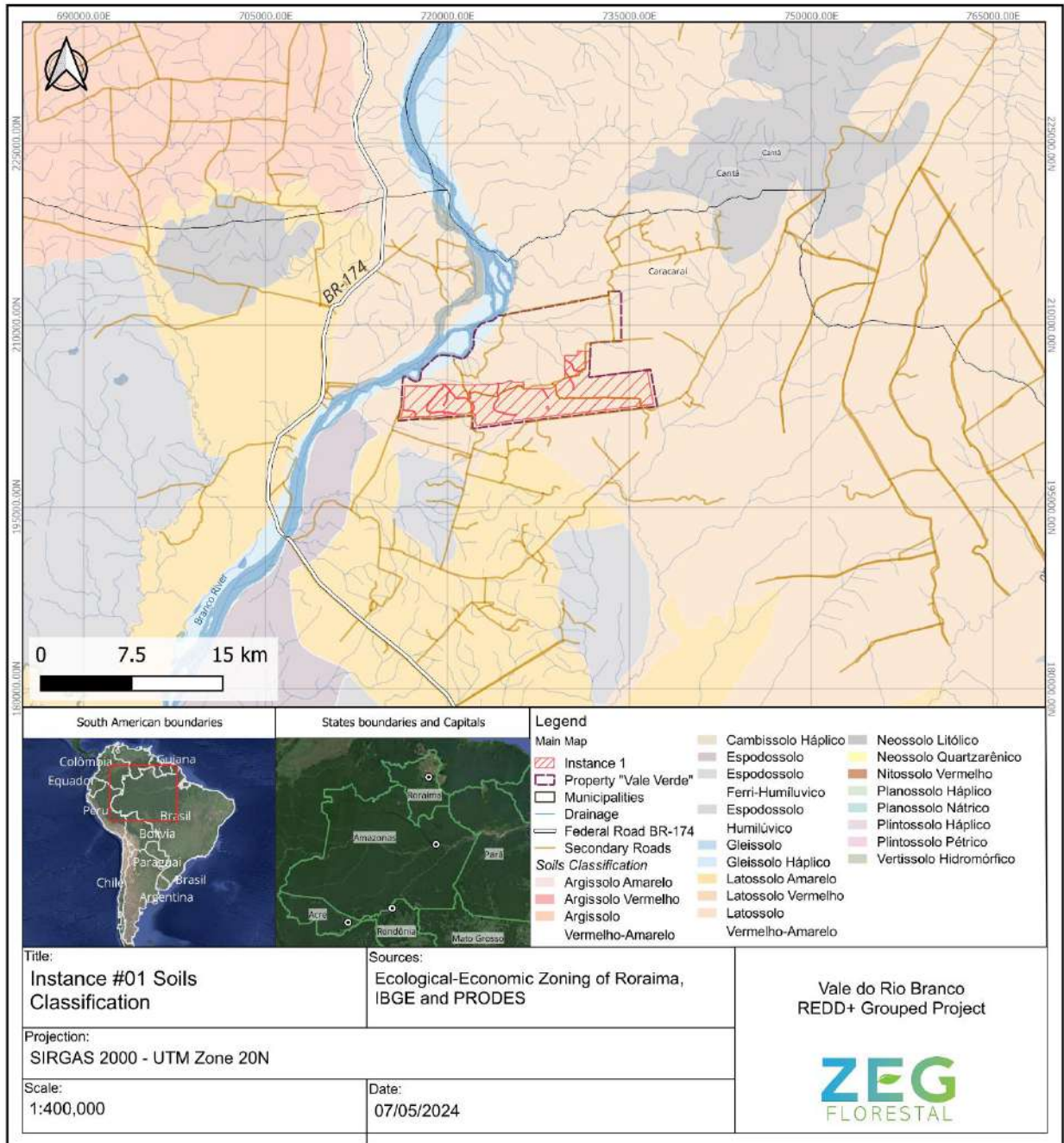


Figure 2.2.1-L: Instance #01 Soils Classification Map.

2.2.2 Most-Likely Scenario Justification (CCB, G2.1)

This section is under development.

2.2.3 Community and Biodiversity Additionality (CCB, G2.2)

This section is under development.

2.2.4 Benefits to be used as Offsets (CCB, G2.2)

This section is under development.

2.3 Safeguards and Stakeholder Engagement

This section is under development.

2.4 Legal Status and Property Rights

2.4.1 National and Local Laws (VCS, 3.1, 3.6, 3.7, 3.14, 3.18, 3.19; CCB, G5.6)

The hierarchy of laws in Brazil follows an organized and structured system, represented by the normative pyramid or legal pyramid. In this arrangement, the norms are classified according to their origin and importance, establishing a set of rules that guide life in society.

At the apex of the pyramid is the Federal Constitution, which is the country's supreme law. It defines the fundamental principles, the structure and competences of the powers, individual rights and guarantees, among other essential aspects to the national legal order. Its supremacy is unquestionable, being the basis on which all other norms must support and be in compliance.

Just below the Constitution are the complementary laws and ordinary laws. Complementary laws aim to complement specific constitutional provisions, while ordinary laws deal with various topics not directly addressed by the Constitution. Both are essential for the organization and functioning of the state, establishing norms that govern life in society.

Following the pyramid, there are legislative decrees and resolutions, which are internal norms of the Legislative Power. Legislative decrees refer to matters within the competence of the Legislature, such as authorization for international treaties, while resolutions deal with administrative and regimental issues of legislative houses.

At the base of the pyramid are the infralegal normative acts, such as decrees and ordinances. Issued by the Executive Power, these acts have the function of regulating and detailing the provisions of the superior laws, without being able to contradict them.

2.5.1.1 Environmental Legislation

Regarding environmental legislation, it occupies a prominent place in this hierarchical system. Environmental norms can be both ordinary laws and complementary laws, depending on the scope and importance of the topic addressed. They are fundamental for the protection and preservation of the environment, ensuring ecological balance and the quality of life of present and future generations. Therefore, Brazilian environmental legislation is inserted in this normative web, and should be interpreted and applied in accordance with constitutional principles and other norms of the national legal order, with concurrent competence to the Union, the member states, and the Federal District to legislate on environmental issues, nature conservation, soil protection, preservation of the landscape heritage, and responsibility for

environmental damage, while municipalities, in environmental matters, legislate on matters of local interest.

At the federal level, there are laws that establish general norms and basic guidelines for the protection of the environment throughout the national territory. In the specific case of nature-based solutions, such as REDD+ projects, it is worth highlighting the National Environmental Policy and the Forest Code. The National Environmental Policy Law (Law No. 6,938/1981) defines general principles and guidelines for environmental protection throughout Brazil and dictates its purposes and mechanisms of formulation and application. The Forest Code (Law No. 12,651/2012) calls for the protection of native vegetation and establishes specific rules for the protection and sustainable use of forests, including permanent preservation areas (APP) and legal reserves (RL). The Environmental Crimes Law (Law No. 9,605/1998), which calls for criminal and administrative sanctions resulting from conduct and activities harmful to the environment, such as illegal deforestation, complements the environmental legal framework. In addition, Law No. 12,187/2009 institutes the National Policy on Climate Change (PNMC) in Brazil, establishing principles, objectives, guidelines, and instruments for mitigation and adaptation to the country's climate change policies.

At the state level, there are specific laws and regulations that complement federal laws and address environmental issues within the oversight of the states. In the state of Roraima, the state Foundation of Environment and Water Resources (FEMARH/RR) is responsible for environmental inspections and licensing, including authorization for forest interventions.

At the municipal level, laws and regulations are implemented to solve local issues and any other matters of specific interest to the municipality. In the case of Caracaraí/RR, municipality where the Instance #01 of the Vale do Rio Branco REDD+ Grouped Project is being developed, there is only one law, Law No. 464/2008, that provides for the Policy of Protection, Conservation and Recovery of the Environment, among other provisions.

The Table 2.5.1-A below presents the compliance of the laws applicable to the project.

Table 2.5.1-A: Applicable laws, description and project compliance.

Law	Description	Project Compliance
<i>Federal</i>		
Law No. 6,938/1981	National Environmental Policy	The project's conservation activities are in line with federal environmental legislation.
Law No. 9,605/1998	Environmental Crimes Law	The project's conservation activities do not infringe upon the environmental crimes law.
Law No. 12,187/2009	National Policy on Climate Change (PNMC)	The project's conservation activities are aligned with the PNMC.
Law No. 12,651/2012	Forest Code	The project is in compliance with the Forest Code,

Law	Description	Project Compliance
		evidenced by the approval of the Legal Reserve at 50% of the property by FEMARH, the responsible environmental agency, as per the document issued on October 16, 2024.
State		
Complementary Law No. 007/1994	Institutes the Code of Environmental Protection for the Administration of Environmental Quality, Protection, Control and Development of the Environment, and appropriate use of Natural Resources of the state of Roraima	The project's conservation activities are in compliance with state environmental legislation.
Normative Instruction No. 04/2020/FEMARH/RR	Institutes the obligation of forest replacement, proper consumption of forest raw material, and other provisions	Not applicable, the project's activities do not foresee vegetation suppression and, therefore, forest replacement is not required.
Complementary Law No. 323/2022	Institutes the creation of the Ecological-Economic Zoning of the state of Roraima - ZEE-RR and other provisions	The project's activities are in accordance with the Zone in which it is located, according to the law. The project's boundaries are in compliance with the ZEE, evidenced by the approval of the Legal Reserve at 50% of the property by FEMARH, the responsible environmental agency, as per the document issued on October 16, 2024.
Decree No. 33,467-E/2022	Institutes the percentage of Legal Reserve in a property located in an area of forest formations in the state of Roraima	The project is in compliance with the law, evidenced by the approval of the Legal Reserve at 50% of the property by FEMARH, the responsible environmental agency, as per the document issued on October 16, 2024.
CEMA Resolution No. 003/2022	Institutes the procedures for registering the Legal Reserve of rural properties and	The project is in compliance with the law, evidenced by the approval of the Legal Reserve at 50% of the property by FEMARH, the responsible environmental agency, as per

Law	Description	Project Compliance
	possessions in the state of Roraima	the document issued on October 16, 2024.
Normative Instruction No. 1/2023/FEMARH	Provides for measures to promote remote inspection, monitoring, fines and embargoes	The project's conservation activities do not violate environmental crime laws. The project is subject to oversight by the relevant authorities.
<i>Municipal</i>		
Law No. 464/2008	Policy of Protection, Conservation and Recovery of the Environment	The project's conservation activities are following municipal environmental legislation.

2.5.1.2 Evaluation of Environmental Legislation Implementation in the Vale do Rio Branco REDD+ Grouped Project

According to the Brazilian Forest Code (Law No. 12,651, of 05/25/2012), all rural properties located in forest zones must have Permanent Preservation Areas (APP), which are protected spaces, covered or not by native vegetation, whose main function is to preserve water resources, geological stability, biodiversity, and the well-being of the population. They encompass river and lake margins, hilltops, steep slopes, restinga areas (the vegetation communities of sandy coastal plains of Eastern Brazil), mangroves, among others. The delimitation of the APPs varies according to the width of the watercourses and the slope of the terrain, being essential to guarantee the quality and availability of water resources, the prevention of natural disasters, and the maintenance of biodiversity.

Rural properties must also maintain the Legal Reserve (RL), areas within the rural property that must be conserved due to their native vegetation, with the purpose of ensuring biodiversity, soil protection, and the maintenance of water resources. The extent of the Legal Reserve varies according to the region of the country and the type of ecosystem present, being at least 80% for properties located in the Legal Amazon, 35% for those located in the Cerrado, and 20% for the other regions of the country. The Legal Reserve can be used sustainably, with the authorization of the responsible environmental agencies, through activities such as forest management, ecotourism and scientific research.

Paragraph 5 of the Article 12 of Forest Code states that, in cases of properties located in the Legal Amazon, the public power may reduce the Legal Reserve up to 50% when the Municipality has more than 65% of the area occupied by public domain nature conservation units and homologated indigenous lands. The analysis and execution will be the responsibility of the local Executive Power, with the participation of the state Environmental Council to define any reduction, it being understood that there is no unconstitutionality regarding these paragraphs, according to the joint judgment of the Direct Actions of Unconstitutionality (ADIs) 4901, 4902, 4903 and 4937 and the Declaratory Action of Constitutionality (ADC) 42.

The state of Roraima, based on this article 12 of the forest code, published Decree No. 33,467-E, of October 31, 2022, which reduces the legal reserve from 80% to 50% of the area of the rural property, for properties located in Zone 1 (Productive Use Zone), as provided for in Complementary Law 323/2022, which provides for the creation of the Ecological-Economic Zoning of the state of Roraima and describes the state zoning categories, without prejudice to the application of the rules on Permanent Preservation Areas.

The Vale do Rio Branco REDD+ Grouped Project is in compliance with this state legislation and therefore 50% of its area qualifies as a Legal Reserve, since it is located in the Roraima's productive use zone referred to in Complementary Law 323/2022, as observed in the Figure 2.5.1-A, which is based on data provided by the state government and Complementary Law 323/2022, and presents the Productive Use Zone (ZUP), Urban Zone (ZU), and Special Use Zones (ZUE) boundaries.

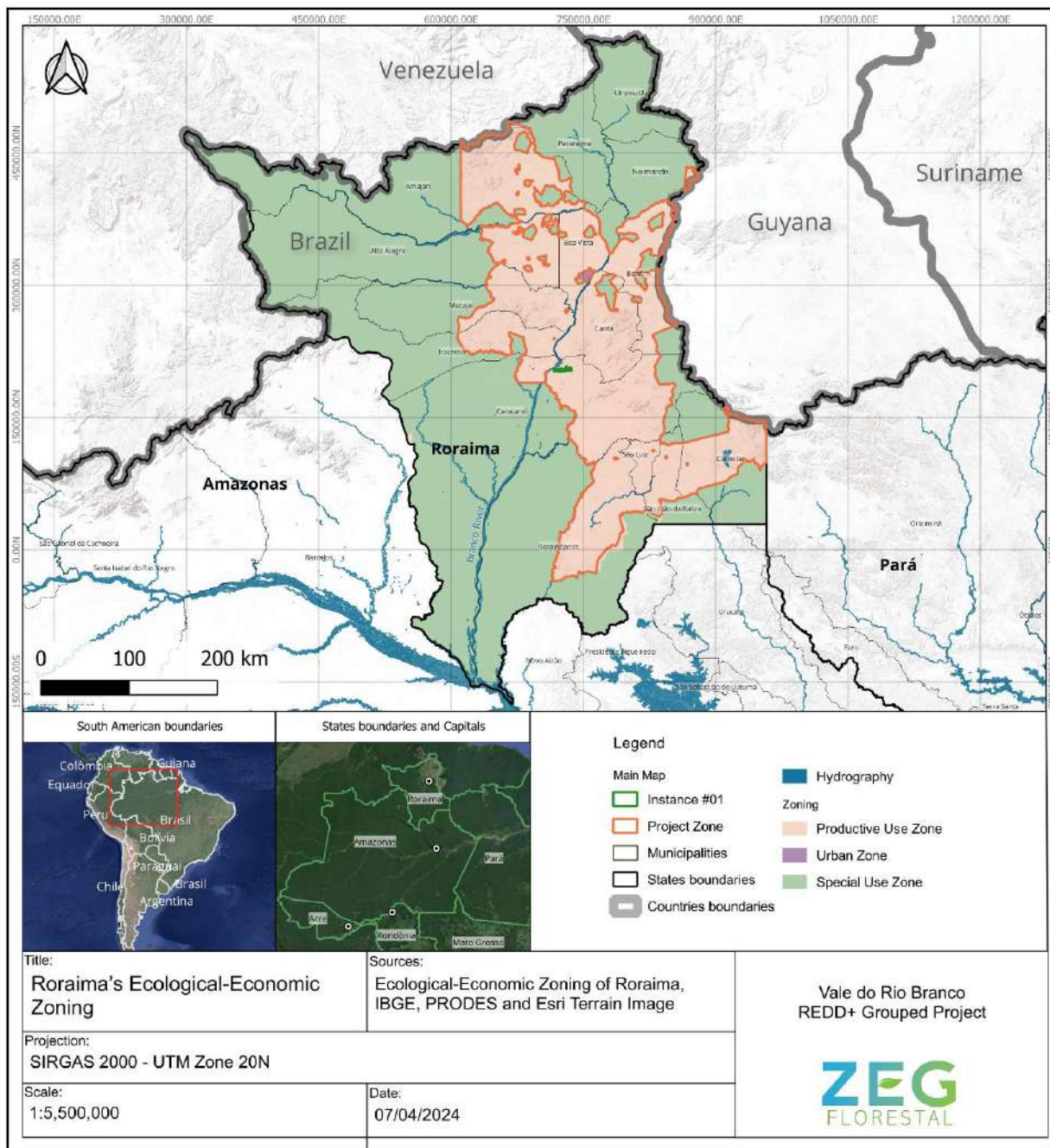


Figure 2.5.1-A: Roraima's Ecological-Economic Zoning.

2.5.1.3 Legislation on Climate Change and Carbon Market

Regarding other regulatory frameworks existing in Brazil, Federal Decree 11,548/2023 was published on June 5, 2023, which establishes the National Commission for the Reduction of Greenhouse Gas Emissions from Deforestation and Forest Degradation, Conservation of Forest Carbon Stocks, Sustainable Forest Management and Increase of Forest Carbon Stocks – REDD+. However, the focus is on the National Strategy for REDD+ (ENREDD+) and not on

REDD+ projects with a market-oriented bias, as is the case with the Vale do Rio Branco REDD+ Grouped Project.

In short, Decree 11,548/2023 seeks to delimit the understanding of the Federal Government about what will be accounted for and presented to the United Nations Framework Convention on Climate Change to fulfill mitigation commitments of the Warsaw Framework for REDD+ and Paris Agreement, notably the Brazilian NDC. Thus, this Decree does not impose a barrier or obstacles to the implementation of market-oriented REDD+ projects, nor to the commercialization of carbon assets generated from these projects. It means: this Decree does not affect or interfere with the voluntary, domestic, or international carbon market.

The National Policy on Climate Change (PNMC), established by Law No. 12,187/2009, in turn, is a comprehensive legislation that establishes the principles, objectives, guidelines, and instruments for addressing climate change in Brazil. It recognizes the country's responsibility in reducing greenhouse gas emissions and in promoting adaptation to the impacts of climate change. The PNMC provides for the elaboration of sectoral plans, emission reduction programs, financing mechanisms, and incentives for the adoption of sustainable practices. In addition, it establishes the creation of a national system of information on climate change, for monitoring and evaluating the results of the policies implemented. Its validity does not impact the Vale do Rio Branco REDD+ Grouped Project.

The National Plan on Climate Change, regulated by Decree No. 9,578/2018, is an instrument of implementation of the PNMC, which will be integrated into the action plans for the prevention and control of deforestation in affected biomes and sectoral plans of mitigation and adaptation to climate change. It defines strategies to reduce greenhouse gas emissions in different sectors of the economy, such as energy, agriculture, transportation and industry. In addition, it provides for the creation of financing mechanisms and incentives to promote the adoption of clean technologies and sustainable practices.

Brazilian legislation, in general, does not restrict or prohibit the execution of REDD+ projects or any types of commercial transaction of assets resulting from REDD+ projects. Thus, there is no contradiction, or irregularity, between the Vale do Rio Branco REDD+ Grouped Project and all relevant legislation.

2.4.2 Relevant Laws and Regulations Related to Worker's Rights (VCS, 3.18.2; CCB, G3.11)

This section is under development.

2.4.3 Human Rights (VCS, 3.19)

This section is under development.

2.4.4 Indigenous Peoples and Cultural Heritage (VCS, 3.18, 3.19)

This section is under development.

2.4.5 Statutory and Customary Property Rights (VCS, 3.18, 3.19; CCB, G5.1)

This section is under development.

2.4.6 Recognition of Property Rights (VCS, 3.7, 3.18, 3.19; CCB, G5.1)

This section is under development.

2.4.7 Free, Prior and Informed Consent (VCS, 3.18; CCB, G5.2)

This section is under development.

2.4.8 Benefit Sharing Mechanisms (VCS, 3.18, 3.19;)

This section is under development.

2.4.9 Property Rights Protection (VCS, 3.18, 3.19; CCB, G5.3)

This section is under development.

2.4.10 Illegal Activity Identification (VCS, 3.19; CCB, G5.4)

This section is under development.

2.4.11 Ongoing Disputes (VCS, 3.18, 3.19; CCB, G5.5)

This section is under development.

2.4.12 Approvals (CCB, G5.7)

This section is under development.

2.4.13 Double Counting and Participation under Other GHG Programs (VCS, 3.23; CCB G5.9)

2.4.13.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program, or any other form of community, social, or biodiversity unit or credit?

☐ Yes ☒ No

2.4.13.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

2.4.13.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

2.4.14 Double Claiming, Other Forms of Credit, and Scope 3 Emissions (VCS, 3.24)

2.4.14.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

2.4.14.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

2.4.14.3 Supply Chain (Scope 3) Emissions

Do the project activities affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

2.5 Additional Information Relevant to the Project

This section is under development.

3 CLIMATE

3.1 Application of Methodology

3.1.1 Title and Reference of Methodology (VCS, 3.1)

Table 3.1.1-A contains the methodology, modules and tools reference applied in this project.

Table 3.1.1-A: Reference of methodology, modules and tools applied in this project.

Type (methodology, tool, module)	Reference ID (if applicable)	Title	Version
Methodology	VM0007	VM0007 REDD+ Methodology Framework (REDD+MF),	1.8
Module	VMD0001	VMD0001 Estimation of carbon stocks in the above- and belowground biomass in live tree and non-tree pools (CP-AB)	1.2
Module	VMD0002	VMD0002 Estimation of carbon stocks in the dead-wood pool (CP-D)	1.1
Module	VMD0003	VMD0003 Estimation of carbon stocks in the litter pool (CP-L)	1.1
Module	VMD0006	VMD0006 Estimation of baseline carbon stock changes and greenhouse gas emissions from planned deforestation and planned degradation (BL-PL)	1.4
Module	VMD0009	Estimation of emissions from activity shifting for avoiding planned deforestation/forest degradation and avoiding	1.3

Type (methodology, tool, module)	Reference ID (if applicable)	Title	Version
		planned wetland degradation (LK-ASP)	
Module	VMD0011	VMD0011 Estimation of emissions from market-effects (LK-ME)	1.2
Module	VMD0013	VMD0013 Estimation of greenhouse gas emissions from biomass and peat burning (E-BPB)	1.3
Module	VMD0016	VMD0016 Methods for stratification of the project area (X-STR)	1.3
Module	VMD0017	VMD0017 Estimation of uncertainty for REDD project activities (X-UNC)	2.2
Tool	VT0001	VT0001 Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities	3.0

3.1.2 Applicability of Methodology (VCS, 3.1)

Table 3.1.2-A contains the applicability conditions of the methodologies applied in the project and the justification of conformance.

Table 3.1.2-A: Reference of methodology, applicability conditions and justification of conformance.

Reference ID/Title	Applicability condition	Justification of conformance
VM0007 (REDD+MF)	1) Land in the project area qualified as forest (following the definition used by the	According to the VCS Program Definitions v4.5, forest is the " <i>land with woody vegetation that meets an internationally accepted definition (e.g.,</i>

Reference ID/Title	Applicability condition	Justification of conformance
	VCS; in addition, see Section 5.1.2) for at least 10 years prior to the project start date. Mangrove forests are excluded from any tree height requirement in a forest definition, as they consist of 95 percent mangrove species, which often do not reach the same height as other tree species. Mangroves occupy contiguous areas and their functioning as a forest is independent of tree height.	<i>UNFCCC, FAO, or IPCC) of what constitutes a forest, which includes threshold parameters, such as minimum forest area, tree height, and level of crown cover, and may include mature, secondary, degraded, and wetland forests". Land in the project area qualifies as forest 10 years prior to the project start date (2013) following FAO (2023) definition: "Land spanning more than 0.5 hectares with trees higher than 5 meters and a canopy cover of more than 10 percent, or trees able to reach these thresholds in situ. It does not include land that is predominantly under agricultural or urban land use". The Brazilian Forest Service (SFB, 2019) considers forests to be: the types of woody vegetation that come closest to the FAO definition, including 13 vegetation categories described by the Brazilian Institute of Geography and Statistics (IBGE) Classification System. One of them is Dense Ombrophilous Forest, which covers the entire Instance #01 project area. There are no mangrove forests in the project area.</i>
VM0007 (REDD+MF)	2) Where land within the project area is peatland or tidal wetlands and emissions from the SOC pool are deemed significant, the relevant WRC modules (see Table 3) are applied alongside other relevant modules.	Not applicable since the land within the project area is not peatland or tidal wetland.
VM0007 (REDD+MF)	3) Baseline deforestation in the project area falls within either of the following categories: a) Unplanned	Baseline deforestation falls within the Planned deforestation/degradation (VCS category APDef).

Reference ID/Title	Applicability condition	Justification of conformance
	deforestation (VCS category AUDef). b) Planned deforestation (VCS category APDef).	
VM0007 (REDD+MF)	REDD activity types are not applicable under the following condition: 4) Leakage prevention, activities include: a) Flooding agricultural lands to increase production (e.g., rice paddies); and/or b) Intensifying livestock production through use of feed-lots and/or manure lagoons.	Leakage avoidance activities do not include flooding agricultural land to increase production or creating feed-lots and/or manure lagoons for intensifying livestock production. Such activities are not common practice in the project region.
VM0007 (REDD+MF)	Avoiding planned deforestation activities are applicable under the following condition: 7) Where conversion of forest lands to a deforested condition is legally permitted.	The conversion of forest land to a deforested condition is legally permitted by the Brazil's Forest Code, Law no. 12,651 of May 25, 2012 (Native Vegetation Protection Law). In general, it states that 80% of a private property located in a forest area of Legal Amazon must be preserved as Legal Reserve, along with the Permanent Preservation Areas. The remaining unprotected area of the property is available for other land uses, which can involve legally permitted deforestation for production purposes. The Forest Code also states that the Legal Reserve can be reduced up to 50% when the state has approved Ecological-Economic Zoning (ZEE in Portuguese) and more than 65% of its territory is occupied by duly regularized public domain Conservation Units and homologated indigenous lands, which is the case of the Roraima State where the project is located. The Decree No. 33,467-E, of

Reference ID/Title	Applicability condition	Justification of conformance
		October 31, 2022, approves the reduction of the Legal Reserve by 50% of the rural property area, without prejudice to the application of the rules on Permanent Preservation Areas, in properties located in forest areas in the Roraima State. As a result, it is legally permitted to deforest up to 50% of the property where the project is located.
VMD0001 (CP-AB)	This module is applicable to all forest types and age classes.	Inclusion of this module is mandatory by the REDD+MF and it is applicable to all forest types and age classes.
VMD0002 (CP-D)	This module is applicable to all forest types and age classes.	The pool was not included as part of the project boundary, "Appendix 1: Testing Significance of GHG Emissions" was applied to determine the significance of the pool. It is applicable to all forest types and age classes.
VMD0003 (CP-L)	This module is applicable to all forest types and age classes.	The pool was not included as part of the project boundary. "Appendix 1: Testing Significance of GHG Emissions" was applied to determine the significance of the pool. It is applicable to all forest types and age classes.
VMD0006 (BL-PL)	The module is applicable for estimating the baseline emissions on forest lands (usually privately or government owned) that are legally authorized and documented to be converted to non-forest land. Where, pre-project, unsustainable fuelwood collection is occurring within the project boundaries Modules BL-DFW and LK-	The conversion of forest land to a deforested condition is legally permitted by the Brazil's Forest Code, Law no. 12,651 of May 25, 2012 (Native Vegetation Protection Law). In general, it states that 80% of a private property located in a forest area of Legal Amazon must be preserved as Legal Reserve, along with the Permanent Preservation Areas. The remaining unprotected area of the property is available for other land uses, which can involve legally

Reference ID/Title	Applicability condition	Justification of conformance
	DFW must be used to determine potential leakage.	permitted deforestation for production purposes. The Forest Code also states that the Legal Reserve can be reduced up to 50% when the state has approved Ecological-Economic Zoning (ZEE in Portuguese) and more than 65% of its territory is occupied by duly regularized public domain Conservation Units and homologated indigenous lands, which is the case of the Roraima State where the project is located. The Decree No. 33,467-E, of October 31, 2022, approves the reduction of the Legal Reserve by 50% of the rural property area, without prejudice to the application of the rules on Permanent Preservation Areas, in properties located in forest areas in the Roraima State. As a result, it is legally permitted to deforest up to 50% of the property where the project is located. Unsustainable fuelwood collection is not occurring within the project boundaries.
VMD0011 (LK-ME)	This module is applicable for calculating market-effects leakage from projects that are anticipated to reduce levels of wood harvest substantially and permanently. When project activities result in reductions in wood harvest, it is likely that production could shift to other areas of the country to compensate for the reduction, including activity shifting to forested peatland that is drained as a consequence of project implementation. This tool shall be used in countries	This module is mandatory for project activities where the process of deforestation involves timber harvesting for commercial markets, as will be described in Section 3.2.3 Leakage Emissions.

Reference ID/Title	Applicability condition	Justification of conformance
	where wood harvest happens on forested peatland regardless of the absence of peatland within the project boundary.	
VMD0013 (E-BPB)	This module is applicable to REDD project activities with emissions from biomass burning and REDD-WRC project activities with emissions from biomass and/or peat burning. This module is also applicable to RWE and ARR-RWE project activities with emissions from peat burning.	This module is applied in the project as there are activities with emissions from biomass burning in the baseline scenario, as will be described in the Section 3.2.1 Baseline Emissions.
VMD0017 (X-UNC)	This module is mandatory when using methodology REDD+ MF. It is applicable for estimating the uncertainty of estimates of emissions and removals of CO _{2e} generated from REDD and WRC project activities. The module focuses on the following sources of uncertainty: Determination of rates of deforestation and degradation; Uncertainty associated with estimation of stocks in carbon pools and changes in carbon stocks; Uncertainty associated with estimation of peat emissions; Uncertainty in assessment of project emissions.	This project applies REDD+MF, thus it is mandatory to calculate uncertainty for emissions and removals. As will be described in the Section 3.2.1 Baseline Emissions, the estimation of stocks in carbon pools and changes in carbon stocks presents the uncertainty associated with it.
VT0001	This tool is applicable under the following conditions: a)	The AFOLU project activities include monitoring and conservation of the

Reference ID/Title	Applicability condition	Justification of conformance
	AFOLU activities, the same or similar to the proposed project activity on the land within the proposed project boundary performed with or without being registered as the VCS AFOLU project, shall not lead to violation of any applicable law even if the law is not enforced.	forest area, avoiding forest planned deforestation. These activities do not lead to violation of any applicable law, even if the law is not enforced, as per the Section 2.5.1 National and Local Laws.
VT0001	b) The use of this tool to determine additionality requires the baseline methodology to provide for a stepwise approach justifying the determination of the most plausible baseline scenario. Project proponent(s) proposing new baseline methodologies shall ensure consistency between the determination of a baseline scenario and the determination of additionality of a project activity.	The REDD+MF applied in the project provides a stepwise approach to justify the determination of the most plausible baseline scenario (VM0007, Section 6.1). Project proponent is not proposing a new baseline methodology.

3.1.3 Project Boundary (VCS, 3.12)

This section is under development.

3.1.4 Baseline Scenario (VCS, 3.13)

This section is under development.

3.1.5 Additionality (VCS, 3.14)

This section is under development.

3.1.5.1 Regulatory Surplus (VCS, 3.14)

This section is under development.

3.1.5.2 Additionality Methods (VCS, 3.14)

This section is under development.

3.1.6 Methodology Deviations (VCS, 3.20)

This section is under development.

3.2 Quantification of Estimated GHG Emission Reductions and Removals

This section is under development.

3.3 Monitoring

This section is under development.

3.4 Optional Criterion: Climate Change Adaptation Benefits

3.4.1 Regional Climate Change Scenarios (CCB, GL1.1)

The Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report (AR6) provides an in-depth evaluation of regional climate change scenarios, highlighting regional disparities regarding climate impact. Specifically, we analyze the Northern South America (NSA) region, where the project is located. Initially, essential terms such as Coupled Model Intercomparison Project (CMIP), Radiative Forcing, Representative Concentration Pathways (RCP), and Shared Socioeconomic Pathways (SSP) will be clarified for better understanding. Then, the AR6's information on climatic impact in the NSA region (Working Group I – Physical Science Basis) will be presented and the climate projections for Roraima State, as provided by the Climate Change Knowledge Portal (CCKP) of the World Bank, will be discussed to illustrate localized climate changes.

The Coupled Model Intercomparison Project (CMIP) is an international climate modeling project from the World Climate Research Programme (WCRP), designed to better understand past, present, and future changes in the climate (WCRP, 2024). It establishes a coordinating framework for research by defining a series of model experiments for coupled atmosphere-ocean general circulation and Earth system models (CCKP, 2024). In addition to CMIP, the Coordinated Regional Climate Downscaling Experiment (CORDEX) is another key initiative of the WCRP, focusing on providing information on smaller scales, supporting more detailed impact, adaptation assessments, and planning (CORDEX, 2024).

CMIP and its associated data infrastructure have become essential to the IPCC and other international climate assessments. Despite being independent of the regularly produced IPCC Assessment Reports, CMIP results directly contribute to these assessments. Organized into distinct phases, each introducing new and enhanced climate model experiment protocols, standards, and data distribution mechanisms, CMIP has evolved over time. CMIP6 is the most recent phase to release its modeling output data for general use, whilst its latest phase, CMIP7, is in its earliest organizational stages. Notably, CMIP5 provided the foundation for the 5th Assessment Report released in 2013 and 2014, and CMIP6 similarly contributed to the 6th Assessment Report released in 2021 and 2022 (CCKP, 2024; WCRP, 2024).

One suite of experiments within CMIP consistently explores the climate response to various plausible future societal development storylines and contrasting emission pathways,

known as scenarios. The goal of these scenarios is to outline how future emissions and land use changes could translate into responses from the climate system (CCKP, 2024).

One key concept about these climate change scenarios is “radiative forcing,” which is employed in climate science to quantify the change in the energy balance of the Earth–atmosphere system. It is influenced by anthropogenic factors, such as changes in concentrations of greenhouse gases and aerosols, as well as alterations in land use that can impact surface albedo. It also encompasses natural factors, such as variations in solar irradiance. According to IPCC (2021a), radiative forcing is defined as “The change in the net, downward minus upward, radiative flux (expressed in W/m^2) due to a change in an external driver of climate change, such as a change in the concentration of carbon dioxide (CO_2), the concentration of volcanic aerosols or the output of the Sun.”

The CMIP5 projections employ Representative Concentration Pathways (RCPs) to outline future scenarios of anthropogenic forcing. The RCP scenarios are named according to the possible range of radiative forcing values in the year 2100. As a result, the RCPs range from a low emission scenario with significant mitigation efforts (RCP 2.6), to two intermediate scenarios (RCP 4.5 and RCP 6.0), to a high emission scenario (RCP 8.5).

In the IPCC’s AR6 the original pathways are complemented with the Shared Socioeconomic Pathways (SSPs), which specify pathways to qualify and quantify factors of future population, economic growth, education, urbanization, and the rate of technological development. According to O’Neill, *et al.* (2017), the five SSPs are SSP1 (Sustainability—Taking the green road), SSP2 (Middle of the Road), SSP3 (Regional rivalry—A rocky road), SSP4 (Inequality—A road divided) and SSP5 (Fossil-fueled development—Taking the highway). The SSPs can be paired with RCPs to create various SSP-RCP combinations, including the addition of RCP 1.9, RCP 3.4, and RCP 7.0. However, some combinations are not feasible; for example, RCP2.6 and achieving lower levels of future global mean surface temperature rise (such as 1.5°C) are not possible under SSP3 in modeled pathways (IPCC, 2019). Throughout IPCC’s AR6, the five illustrative scenarios are shown in Table 3.4.1-A.

Table 3.4.1-A: Illustrative scenarios on IPCC’s Sixth Assessment Report.

Scenario	SSP	RCP (W/m^2)	Description
SSP1-1.9	Sustainability— Taking the green road	1.9	The most optimistic scenario aiming to limit global temperature rise to 1.5°C by 2100 as per the Paris Accord.
SSP1-2.6	Sustainability— Taking the green road	2.6	Targeting a 2°C rise with severe emission cuts.
SSP2-4.5	Middle of the Road	4.5	A moderate path in which emissions remain around

Scenario	SSP	RCP (W/m ²)	Description
			current levels before declining around mid-century.
SSP3-7.0	Regional rivalry—A rocky road	7.0	A scenario depicting a divided world with increasing emissions.
SSP5-8.5	Fossil-fueled development—Taking the highway	8.5	A high-emission future driven by extensive fossil fuel exploitation and technological progress.

The Figure 3.4.1-A displays the global surface temperature change in °C between 1950-2100 relative to the 1850–1900 reference under the five scenarios.

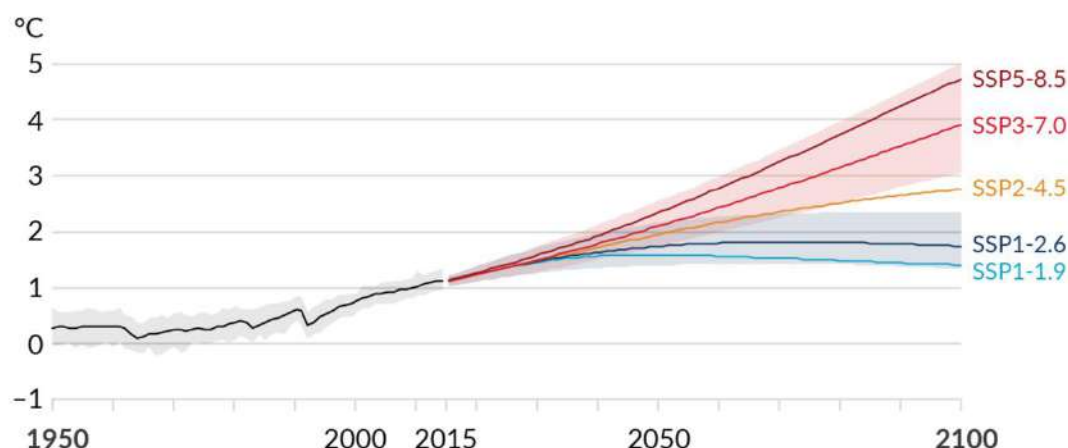


Figure 3.4.1-A: Global surface temperature change. Adapted from: *Figure SPM.8 in IPCC, 2021: Summary for Policymakers. In: Climate Change 2021: The Physical Science Basis.*

3.4.1.1 Regional Information on Changing Climate

The following description of Climatic Impact-Drivers (CID), the physical climate system conditions that affect an element of society or ecosystems, are focused on Northern South America (NSA) geographical scope and are based on IPCC's AR6 Working Group I – "Chapter 12: Climate Change Information for Regional Impact and for Risk Assessment" (IPCC, 2021b). Figure 3.4.1-B shows the boundaries of the reference regions used in AR6 of the IPCC Working Group I and the location of the Project Zone.



Figure 3.4.1-B: IPCC WGI Reference Regions and Project Zone.

Climate change projections point to a virtually certain (> 99% probability) warming across all Central and South America, with the most significant increases expected in the Amazon basin (NSA and South America Monsoon – SAM). Under scenario SSP2-4.5, there is high confidence of an increase in mean air temperatures and extreme heat in the NSA, alongside a decreasing trend in cold spells (IPCC, 2021b).

The Amazon basin is projected to experience the largest increases in the frequency of hot days (maximum temperatures, $T_x > 35^\circ\text{C}$), exceeding 200 days annually under SSP5-8.5 by the end of the century, shown in Figure 3.4.1-C (b). Increases under SSP1-2.6, however,

are more moderate, ranging from 50 to 100 days, shown in Figure 3.4.1-C (a). Similarly, frequencies exceeding the dangerous threshold of Heat Index > 41°C are expected to rise comparably, shown in Figure 3.4.1-C (c) for SSP1-2.6 and Figure 3.4.1-C (d) for SSP5-8.5 (Coppola *et al.*, 2021; Schwingshackl *et al.*, 2021).

Figure 3.4.1-C depicts median projected changes based on CMIP6 models. It shows the number of days per year with maximum temperatures exceeding 35°C (a and b) and the mean number of days per year with the NOAA Heat Index exceeding 41°C (c and d). The Project Zone is indicated by the yellow spot.

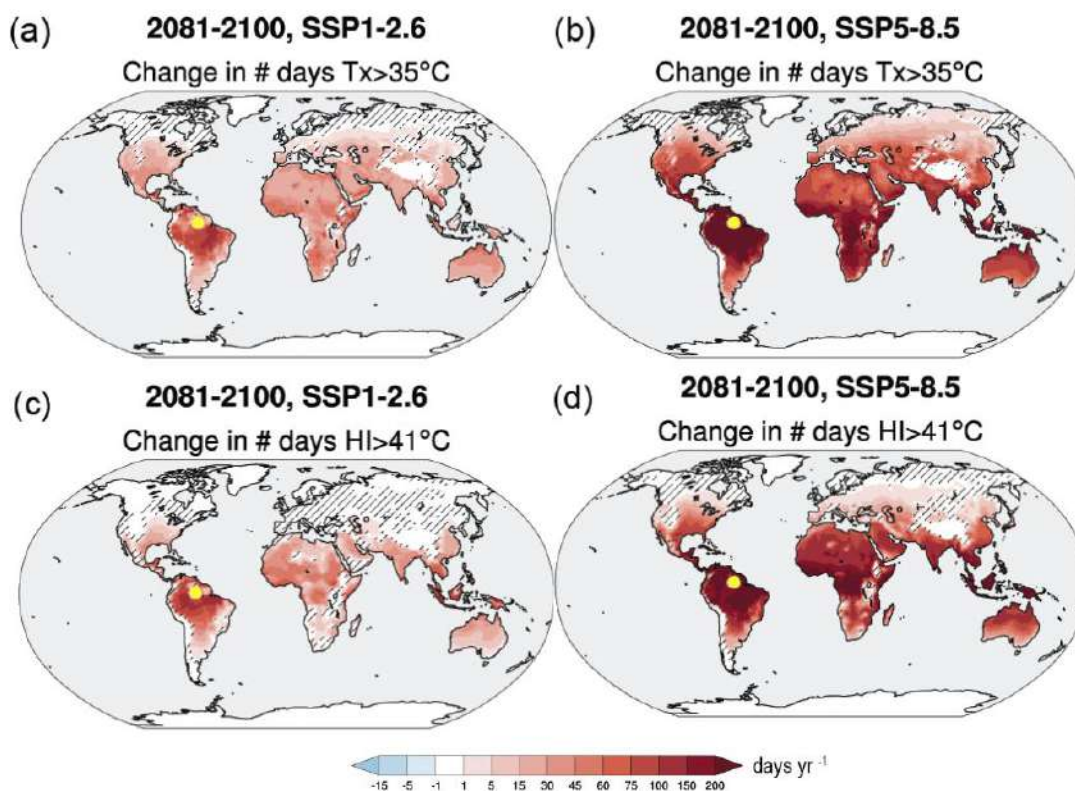


Figure 3.4.1-C: Projected Temperature Extremes and Heat Index: CMIP6 Models and Project Zone. Adapted from: *Figure 12.4 in IPCC, 2021: Chapter 12. In: Climate Change 2021: The Physical Science Basis.*

Projections indicate decreasing precipitation in the Amazon (NSA) with medium confidence, consistent by mid and end of century for both RCP4.5 and RCP8.5 (Jones and Carvalho, 2013). While there is limited evidence of extreme precipitation trends in almost all Central and South America, there is medium confidence in increased heavy precipitation in the NSA region under a Global Warming Level (GWL) of 4°C (IPCC, 2021b).

Flooding events have been reported in recent decades in the tributaries of the Amazon River or along its mainstream (Sena *et al.*, 2012; Espinoza *et al.*, 2013; Marengo *et al.*, 2013; Filizola *et al.*, 2014). In the Amazon basin, the increase in extreme flow is associated with the

strengthening of the Walker circulation, a large-scale atmospheric circulation in the equatorial Pacific Ocean (Barichivich *et al.*, 2018). However, under SSP2-4.5, there is low confidence in predicting whether river floods in the NSA region will increase or decrease (IPCC, 2021b).

In contrast, the AR6 points to the Amazon basin as one of the most concerning drying hotspots in South America with long-term soil moisture decline and precipitation decline, as indicated in Figure 3.4.1-D (IPCC, 2021b). Regional studies indicate increasing trends in the frequency and length of droughts in the NSA region (Marengo *et al.*, 2016; Cunha *et al.*, 2019). Among regions worldwide, NSA and SAM are projected to experience the strongest signals of increasing number of dry days (NDD) and drought frequency (DF) (Coppola *et al.*, 2021). By the end of this century under RCP8.5, the NSA averages 43 (CORDEX), 32 (CMIP5), and 27 (CMIP6) dry days annually. Regarding drought frequency, the averages are approximately 4.6 (CORDEX), 3.4 (CMIP5), and 3.8 (CMIP6) drought events per decade.

Figure 3.4.1-D below presents median projected changes derived from CMIP6 models, focusing on several key indicators. These include the frequency of negative precipitation anomaly events per decade, assessed using the six-month Standardized Precipitation Index (e and f). Additionally, it examines the average soil moisture levels as a percentage (g and h). The Project Zone is visually indicated by a yellow spot on the map.

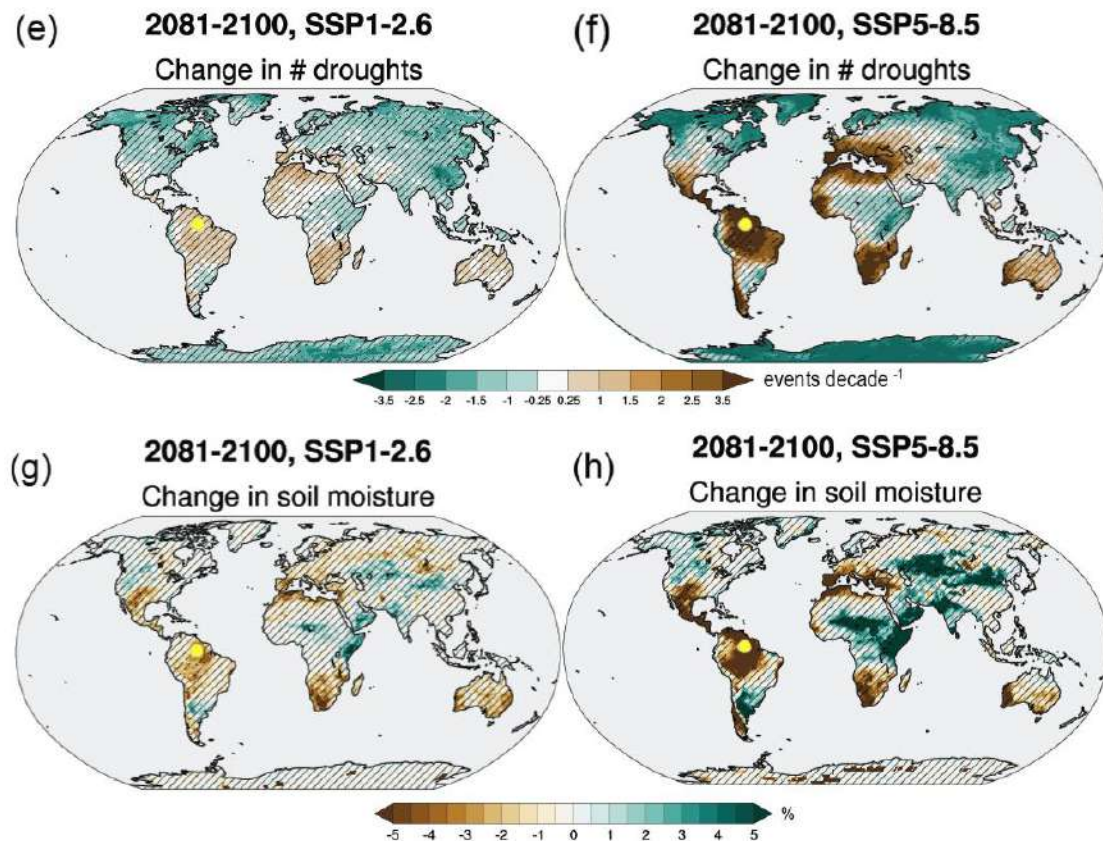


Figure 3.4.1-D: CMIP6 Model Projections and Project Zone Analysis. Adapted from: *Figure 12.4 in IPCC, 2021: Chapter 12. In: Climate Change 2021: The Physical Science Basis.*

The drought severity is projected to increase with high confidence in NSA under the three GWLs of 1.5°C, 2°C and 4°C. Seasonal changes are projected to decrease in June–July–August (JJA) rainfall for NSA by the end of the century under RCP 8.5 (Marengo *et al.*, 2016). The global climate projections indicate that the Amazon will be one of the regions in the world with the highest increase in fire weather indices (weather conditions conducive to triggering and sustaining wildfires) throughout the 21st century, across all scenarios, with high confidence (Betts *et al.*, 2015; Abatzoglou *et al.*, 2019; Sun *et al.*, 2019). This is consistent with the joint occurrence of extreme hot and dry day increase, projected to increase by a 2°C warming level or more (Vogel *et al.*, 2020).

There is limited evidence of current trends in observed wind speed and windstorms in Central and South America. However, climate models project a medium confidence increase in wind speeds under all future scenarios (IPCC, 2021b).

In summary, the climate changes projected in Northern South America are the increase in mean temperatures, extreme heat, fire weather indices, frequency of hot and dry days, drought frequency, and wind speed, along with a decrease in mean precipitation and cold spells.

3.4.1.2 Climate Projection Scenarios in Roraima

The climate projection scenarios for Roraima State are provided by the Climate Change Knowledge Portal (CCKP) of the World Bank to illustrate localized climate changes. Data presented is from CMIP6, at a 25km x 25km resolution, with the selection of scenarios SSP1-1.9 (optimistic scenario), SSP2-4.5 (intermediate scenario) and SSP5-8.5 (high emission scenario). The Figure 3.4.1-E displays the projected annual average surface air temperature for Roraima State in °C (with the reference period of 1950 – 2014 in black) under the scenarios SSP1-1.9 (in blue), SSP2-4.5 (in orange) and SSP5-8.5 (in red).

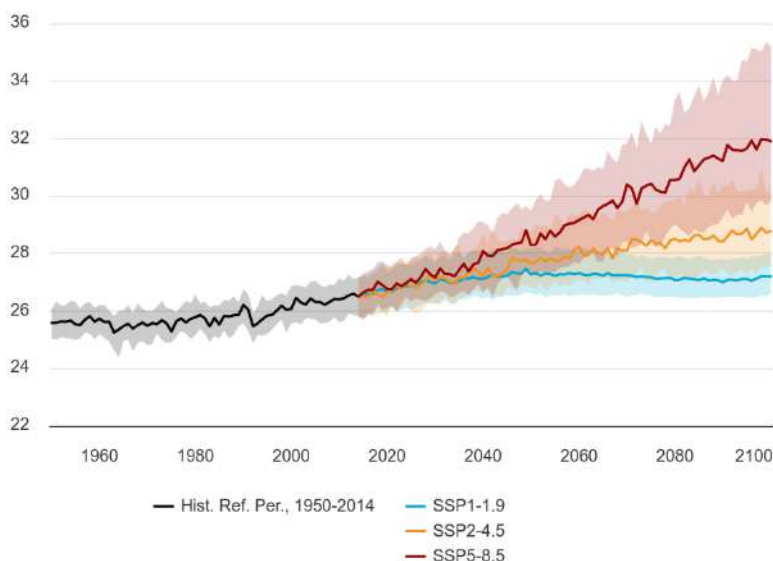


Figure 3.4.1-E: Projected Annual Average Surface Air Temperature for Roraima State.

Source: CCKP (2024)

SSP1-1.9 projects a warming with mean annual temperatures not exceeding 27.50°C until 2100. SSP2-4.5 and SSP5-8.5 project a higher increase, reaching a mean annual temperature of 28.78°C and 31.91°C respectively until 2100.

Considering the monthly projected mean temperature in °C for the period of 2020–2039, as displayed in Figure 3.4.1-F, all scenarios demonstrate higher temperature increases from August through October. This is consistent with the end of the wet season, as described in Section 2.1.14 Physical Parameters. Under SSP1-1.9, the projected increase in September is 0.94°C, while SSP5-8.5 projects an increase of 1.35°C in the same month. From April through July, the increase is less pronounced, but an overall temperature rise is observed across all months.

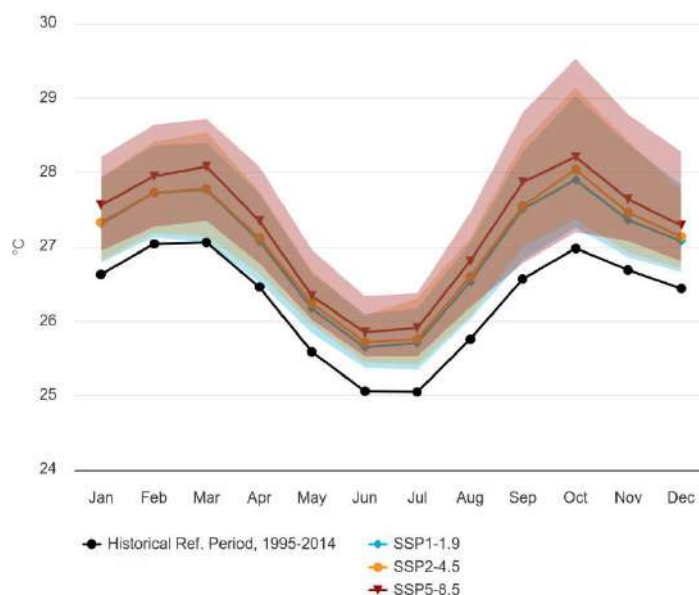


Figure 3.4.1-F: Projected Monthly Average Surface Air Temperature for Roraima State.
Source: CCKP (2024).

The Figure 3.4.1-G displays the projected annual precipitation for Roraima State (with the reference period of 1950 – 2014 in black) under the scenarios SSP1-1.9 (in blue), SSP2-4.5 (in orange) and SSP5-8.5 (in red).

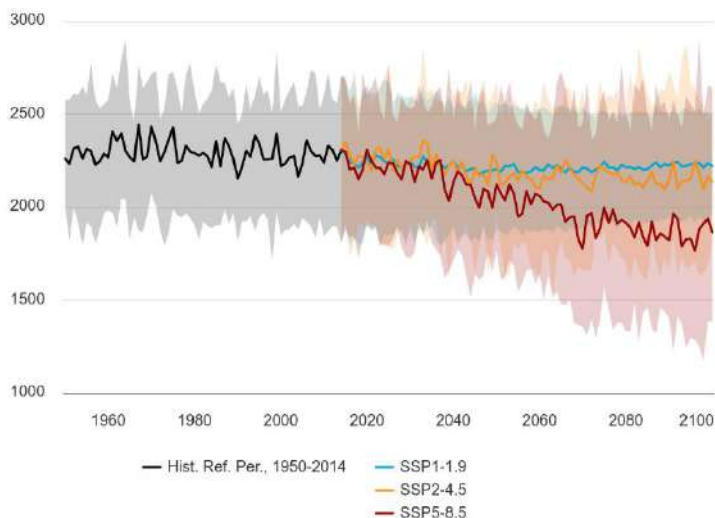


Figure 3.4.1-G: Projected Annual Precipitation in mm for Roraima State. Source: CCKP (2024).

As described in AR6, the precipitation is projected to decrease in the region. Under scenario SSP5-8.5, the decrease is more intense, reaching an annual mean of 2,121 mm by 2050 and 1,866 mm by 2100. In the intermediate scenario (SSP2-4.5), project's mean precipitation will reach 2,243 mm and 2,135 mm by 2050 and 2100 respectively. Meanwhile, the optimistic scenario (SSP1-1.9) project's mean precipitation will reach 2,203 mm by 2050 and 2,219 mm by 2100. Considering the monthly projected mean precipitation for the period of 2020 – 2039, as displayed in Figure 3.4.1-H, the decrease is more pronounced in August through October under all scenarios. As an example, September reaches a decrease in mean precipitation of 8% under SSP1-1.9, 8% under SSP2-4.5, and 15% under SSP5-8.5, compared to the reference period of 1995 – 2014 (mean precipitation of 128 mm in September).

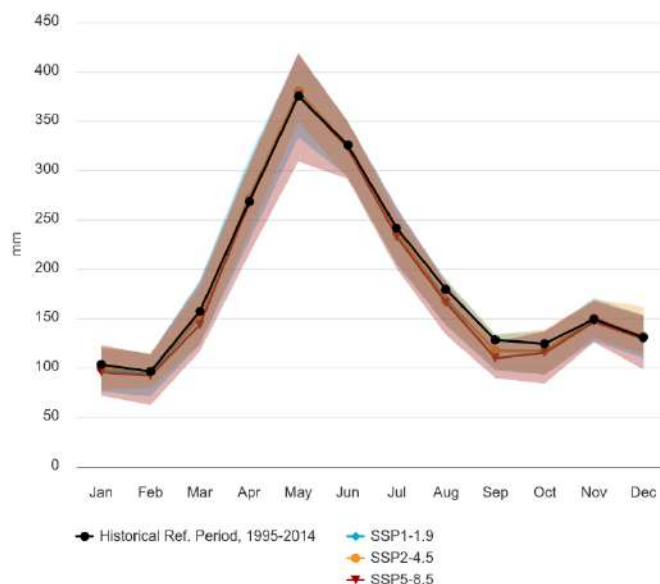


Figure 3.4.1-H: Projected Monthly Precipitation in mm for Roraima State for the period of 2020 – 2039. Source: CCKP (2024).

3.4.1.3 Forest Fires Risk in Roraima

As discussed, there has been an increase in the intensity and frequency of extreme climatic events in the Amazon. These events have not only led to significant forest fires (Maeda *et al.*, 2009; Aragão *et al.*, 2007; Anderson *et al.*, 2015) but have also impacted pristine forests by increasing tree mortality (Phillips *et al.*, 2009; Brien *et al.*, 2015; Rowland *et al.*, 2015). Natural forest fires in tropical forests are rare, paleoecological studies have shown thousands of years without any indications of fire occurrence in the Amazon (Bush *et al.*, 2007). However, due to the colonization of this region, fire has become recurrent, typically ignited by anthropogenic activities (Anderson *et al.*, 2017).

Fire in the Amazon region is strongly associated with land-use changes, land cover, and climatic conditions. Additionally, an ignition source is required to trigger wildfires (Fearnside; Barbosa; Pereira, 2013). The primary ignition source is slash-and-burn practices to clear land and use of fire for maintenance of agricultural and pasture areas, leading the fire to escape into the forested areas (Dos Reis *et al.*, 2021).

The significant increase in the occurrence of fires and wildfires in the Amazon is associated with years of extreme drought (Barbosa & Fearnside, 1999; Alencar *et al.*, 2006; Aragão *et al.*, 2007). In Roraima, extreme droughts are also linked to El Niño events, which have increased in frequency and are predicted to become even more prevalent, thereby heightening the forest's susceptibility to fires (Cai *et al.*, 2014).

Prolonged drought events driven by the increasing frequency of El Niño have a direct impact on the spread of forest fires in the Amazon (Barni *et al.*, 2021). Uncontrolled fires can escalate into large-scale wildfires, exemplified by the major fires in Roraima in 1998 and 2003. The "Great Fire" of 1997-98 affected approximately 12,000 km² of primary forests, releasing 4 – 4,7 million of tons C-CO₂ into the atmosphere (Fearnside; Barbosa; Pereira, 2013). Similarly, the years 2016 and 2019 were marked by significant El Niño events. In 2016, large-scale and severe forest fires were recorded, while 2019 experienced a high occurrence of hotspots, though with fewer burned areas (Barni *et al.*, 2023).

The Oceanic Niño Index (ONI) is NOAA's primary indicator for monitoring El Niño and La Niña, which are opposite phases of the climate pattern called the El Niño-Southern Oscillation (ENSO). The ONI is the difference between a three-month running average of the sea surface temperature, averaged over an area of the ocean from 120 West to 170 West longitude along the Equator and the long-term average for the same three months. NOAA considers El Niño conditions to be present when the Oceanic Niño Index is +0.5 or higher, indicating the east-central tropical Pacific is significantly warmer than usual. La Niña conditions exist when the Oceanic Niño Index is -0.5 or lower, indicating the region is cooler than usual (NASA, 2023). Figure 3.4.1-I displays the occurrence of El Niño (in orange) and La Niña (in blue) events (NOAA, 2024) along with the number of hotspots (in red) in Roraima detected by AVHRR sensor on NOAA-12 (1998–2002) and MODIS sensor on Aqua satellites since 2002 (INPE, 2024).

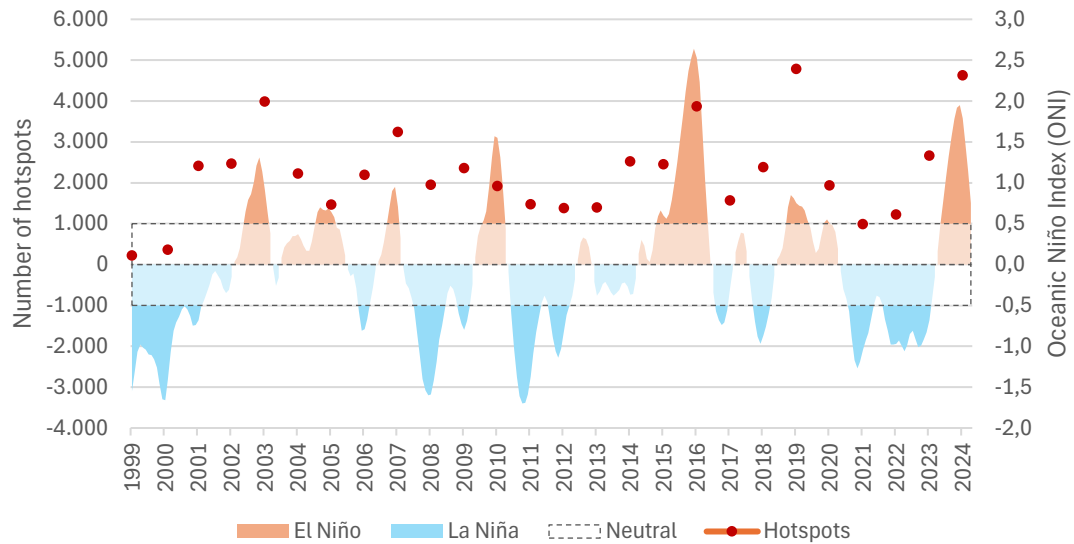


Figure 3.4.1-I: Occurrence of El Niño and La Niña events along with annual hotspots in Roraima State in 1999 – 2024 (NOAA, 2024; INPE, 2024).

The correlation between El Niño-induced droughts and forest fire susceptibility in the Amazon highlights the forest's vulnerability to climate change. Figure 3.4.1-J illustrates the spatial distribution of the hotspots with high fire risk index in Roraima State in the first semester of 2024, highlighting their incidence within the Project Zone. During the period from January through June of 2024, the municipalities of Caracaraí, Mucajaí, and Amajari registered the highest occurrence of hotspots (INPE, 2024).

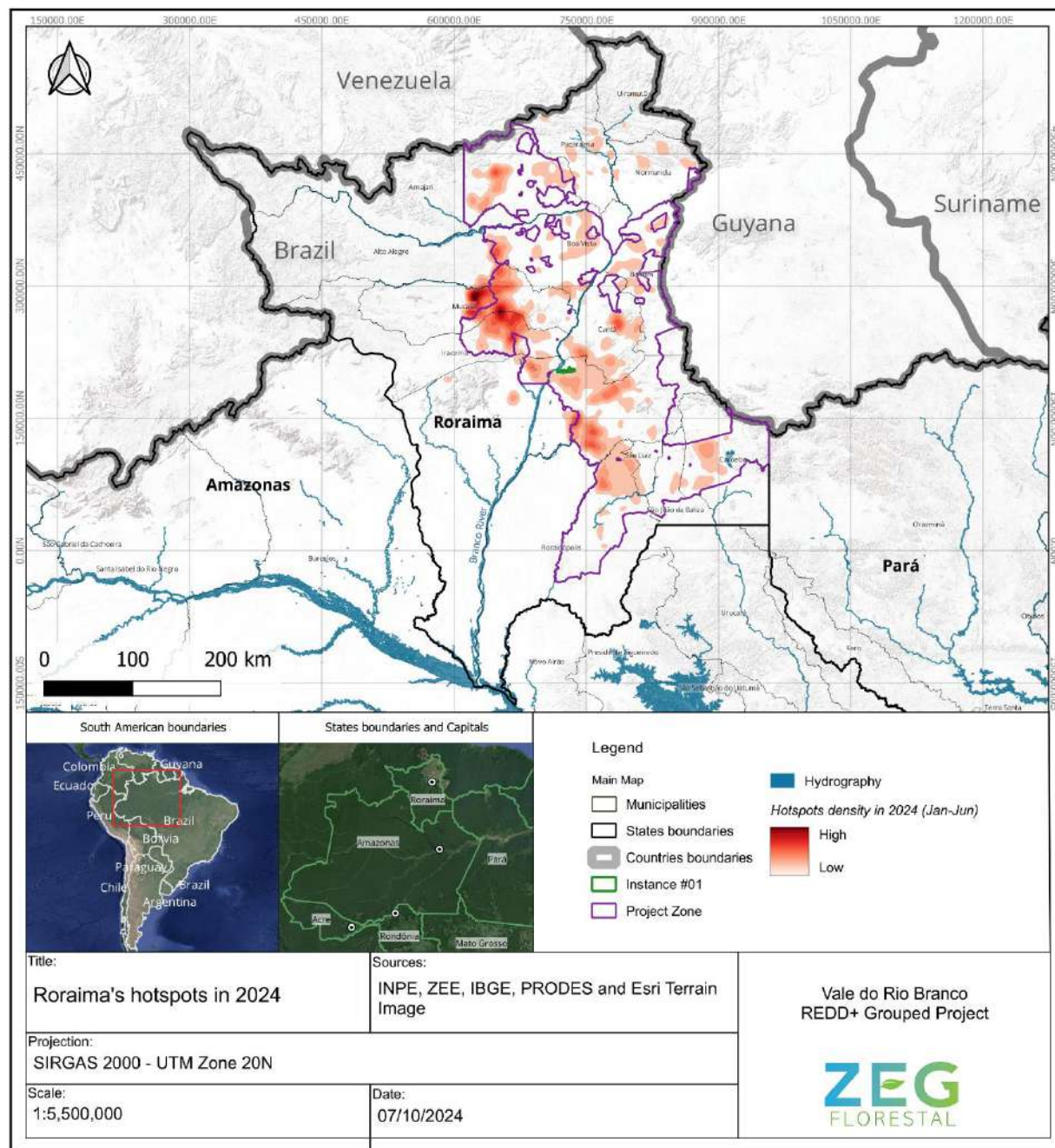


Figure 3.4.1-J: Spatial distribution of hotspots in Roraima State and its relative density for 2024.

3.4.2 Climate Change Impacts (CCB, GL1.2)

Central and South America are highly exposed, vulnerable, and strongly impacted by climate change. This situation is amplified by inequality; poverty; population growth; high population density; economy's high dependence on natural resources for commodities production; and land use change, particularly deforestation, that contributes to biodiversity loss and soil degradation. The region is already experiencing numerous extreme events that are projected to intensify. These events include warming temperatures and dryness, as well as

more frequent and severe droughts in some regions, leading to reduced water supplies, which impact agricultural production, traditional fishing, food security, and human health (IPCC, 2022).

Climate change impacts do not affect men and women equally. Women, particularly the poorest, are more vulnerable and disproportionately affected by these impacts. For several economic and social reasons, women often have less capacity to adapt, further widening structural gender gaps (Arana Zegarra, 2017; Olivera, 2021). In a rural community vulnerable to drought, short-term coping strategies were more common among women, especially female heads of households, while adaptive actions were more prevalent among men. This is due to gendered inequalities in access to and control over various forms of capital, resulting in a gender-differentiated capacity to adapt (Segnestam, 2017). However, while in a broad sense climate change impacts women more severely, there are situations where women have reacted, adapted better, or shown greater resilience (IPCC, 2022).

Northern South America (NSA) is one of the most vulnerable in the region, following Central America. This vulnerability is highlighted across several sectors assessed by the IPCC's AR6. It attributes very high vulnerability in "Food systems", "Cities and infrastructure", "Population in poverty and their livelihoods", and "Water"; along with high vulnerability in "Health" and "Terrestrial and freshwater ecosystems". It also identifies key risks in the region, such as food insecurity due to droughts; risk to life and infrastructure due to floods and landslides; risk of severe health effects due to increasing epidemics (in particular vector-borne diseases); systemic risks of surpassing infrastructure and public service systems; and risk of large-scale change and biome shifts in the Amazon (IPCC, 2022).

Scientific evidence, since the IPCC's AR5 increased the confidence in synergy among fire, land use change (particularly deforestation), and climate change, suggests these changes will have a direct impact on human health, ecosystem functioning, forest structure, food security, and the livelihoods of resource dependent communities (IPCC, 2022).

3.4.2.1 Impacts on Community Well-Being

In the Amazon basin, approximately 80% of the population is concentrated in cities. People have migrated there in search of improvements in education, job opportunities, healthcare, and access to goods and services (Eloy *et al.*, 2015; Pinho *et al.*, 2015). The migration of Indigenous Peoples and rural resource-dependent communities to cities forces them to abandon their livelihoods to seek temporary jobs, often leading to a life of poverty and social exclusion on the periphery of the city (Cardoso *et al.*, 2018). Amazon populations residing in remote urban centers with limited or no access to roads are more vulnerable to extreme events compared to more connected urban centers (Parry *et al.*, 2018).

Extreme events alter the region's natural hydrology dynamics, increasing the sensitivity of community food and transport systems (Pinho *et al.*, 2015). Over the past decade, Indigenous Peoples and resource-dependent rural communities in the Amazon have faced significant impacts by extreme droughts and floods, affecting various aspects of their livelihoods (Pinho *et al.*, 2015). Food security has been particularly affected since it is based on fishing and small-scale agriculture, both of which are highly vulnerable to climate change.

During extreme events, fishing activities decrease due to limited access to fishing grounds (Pinho *et al.*, 2015; Camacho Guerreiro *et al.*, 2016). Overfishing, deforestation, and dam construction are a threat to fishing in Northern South America (Lopes *et al.*, 2019) and therefore contribute to exacerbating climate change impacts.

Small-scale agriculture practices, such as floodplain agriculture and slash-and-burn techniques, are closely coupled with natural hydrological cycles and therefore severely affected by extreme events (IPCC, 2022). According to Evangelista-Vale *et al.* (2021), climate change may affect the geographic distribution of 18 species of palms and trees that are extracted and used as resources by traditional populations living in extractive reserves in the Amazon. Livelihoods are also impacted by disruptions in land and river transport, restrictions in drinking water access, increased incidence of forest fires and disease outbreaks (Marengo *et al.*, 2013; Pinho *et al.*, 2015; Marengo and Espinoza, 2016; Marengo *et al.*, 2018).

As reported by Mongabay (2023), communities in Roraima were affected by the repercussions of the 2023 drought. According to the article, the national coordinator for the Indigenous Committee on Climate Change remarked, *“Our communities are already suffering from the drought. Schools have no water for the children, there’s no water for animal husbandry, the creeks that have dried up are affecting fishing and food supply. Water is life, and when it’s lacking, it affects us on many levels, including the social and cultural life of the communities”*. Indigenous communities across Roraima convened in assemblies to strategize mitigation efforts, and the measures included preventive fires conducted by volunteer brigades in collaboration with Brazil’s environmental protection agency, IBAMA. Instituto Socioambiental (ISA) highlighted dire conditions for artisanal fishermen, stating, *“Roraima has nearly 5,000 families who rely on fishing, and prolonged droughts have harsh economic impacts on their lives”*.

The Climate Change Impact Information and Analysis System (AdaptaBrasil MCTI) provides comprehensive climate change impact risk data for Brazilian municipalities, incorporating the three integrated dimensions of climate threat, vulnerability, and exposure, in accordance with the IPCC’s framework. Future scenarios for 2050 use CORDEX-CMIP5 values for the period 2041-2060, with optimistic scenarios based on RCP 4.5 and pessimistic scenarios on RCP 8.5 (AdaptaBrasil, 2024).

In the optimistic scenario of the Drought Impact Risk Index for Food Security, four municipalities in the northwest Roraima exhibit low risk. In the southern and eastern regions, eight municipalities show moderate risk. Three high-risk municipalities—Pacaraima, Uiramutã, and Normandia—are located in the northern state, as shown in Figure 3.4.2-A (left).

In the pessimistic scenario, additional municipalities such as Boa Vista, Cantá, Caracaraí, São Luiz, and São João da Baliza are categorized as high risk. Iracema and Amajari maintain low risk, while the remaining five municipalities present moderate risk, as illustrated in Figure 3.4.2-A (right).

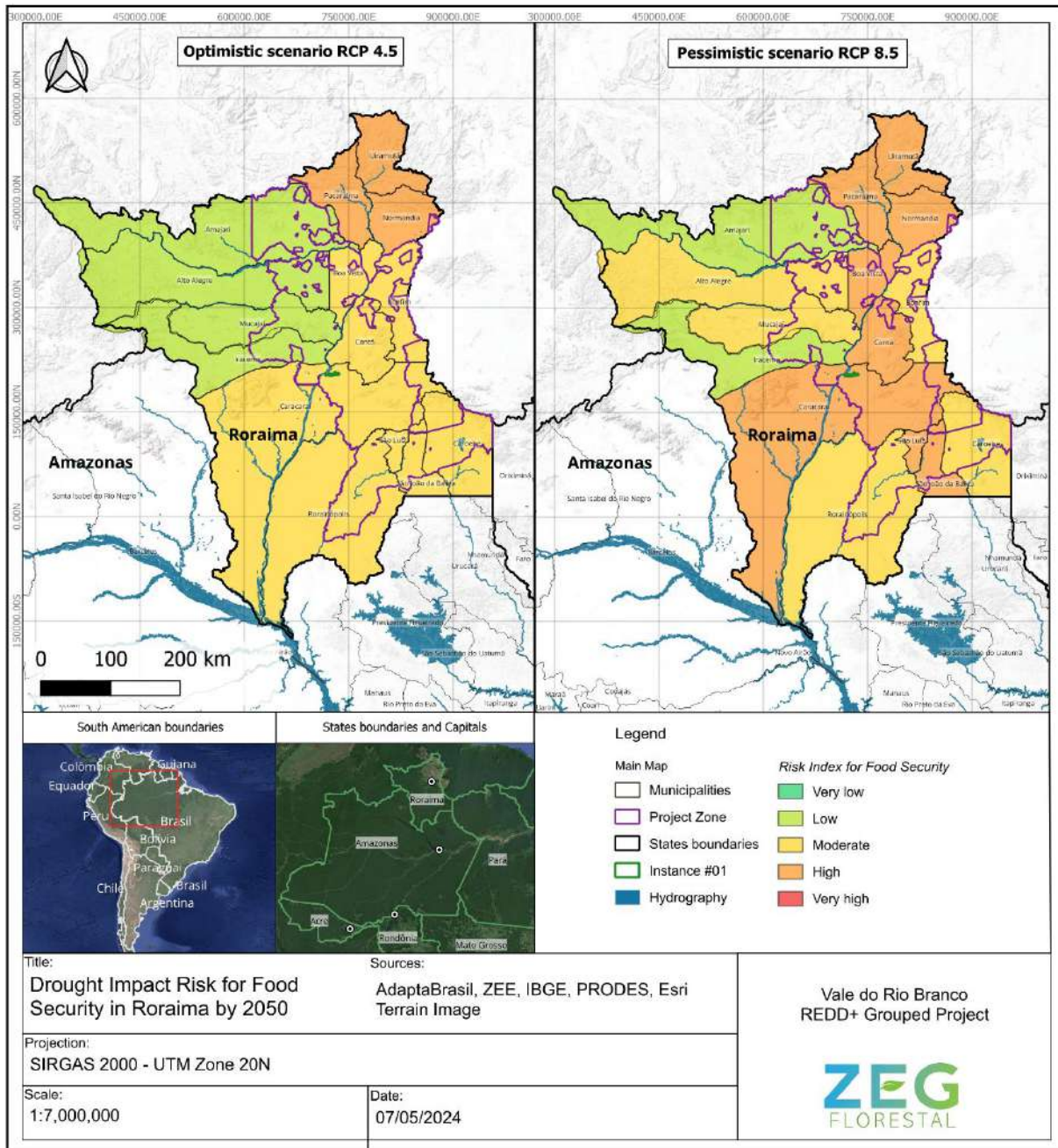


Figure 3.4.2-A: Drought Impact Risk for Food Security in Roraima by 2050.

Source: AdaptaBrasil (2024).

For Water Security, in the optimistic scenario of the Drought Impact Risk Index, five municipalities are classified as moderate risk and ten as high risk, as shown in Figure 3.4.2-B (left). In the pessimistic scenario, Pacaraima and Normandia are classified as very high risk, Rorainópolis as moderate risk, with the others categorized as high risk, as shown in Figure 3.4.2-B (right).

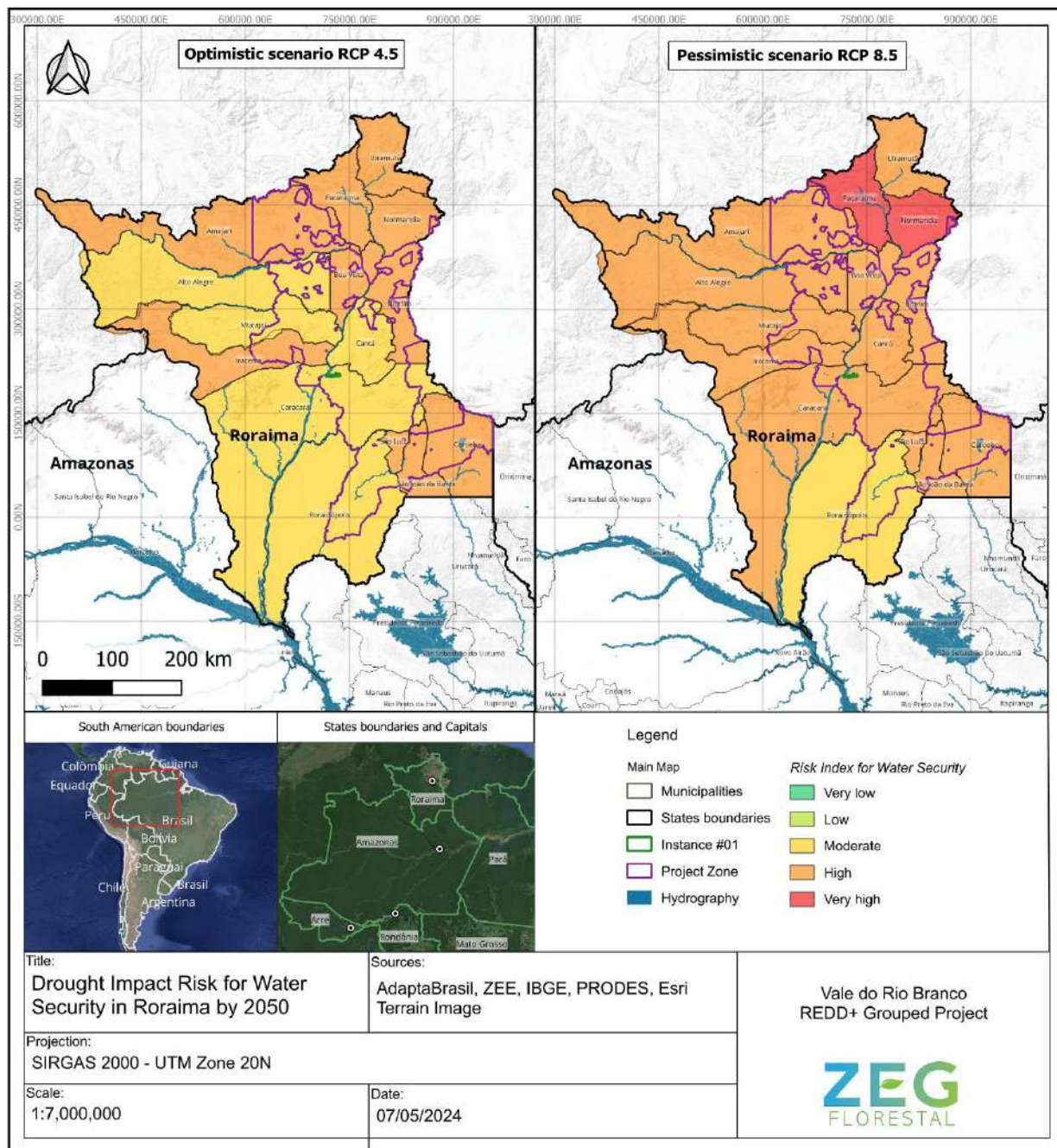


Figure 3.4.2-B: Drought Impact Risk for Water Security in Roraima by 2050.

Source: AdaptaBrasil (2024).

Several vector-driven diseases such as malaria and leishmaniasis are endemic in the Amazon region, but their natural dynamics are being altered by socio-environmental changes (Confalonieri *et al.*, 2014). An important relationship between the outbreak of infectious diseases and changes in climatic events (e.g., droughts, floods, heat waves, ENSO) or environmental events (e.g., deforestation, dam construction and habitat fragmentation) has been found to exist for the Brazilian Amazon (Pan *et al.*, 2014; Filho *et al.*, 2016; Nava *et al.*,

2017; Ellwanger *et al.*, 2020). These impacts are particularly severe among impoverished populations with limited access to healthcare services (Pan *et al.*, 2014).

Regarding health impacts, AdaptaBrasil provides risk data for malaria, visceral leishmaniasis, and American cutaneous leishmaniasis. In the 2050 optimistic scenario, malaria risk is classified as very high in seven municipalities, high in three, and moderate in five, as shown in Figure 3.4.2-C (left). The pessimistic scenario classifies Bonfim as high risk (with a slight index difference of 0.01), and the others the same as the optimistic scenario, as illustrated in Figure 3.4.2-C (right).

For leishmaniosis, the future scenarios show consistent findings in both optimistic and pessimistic scenarios. American cutaneous leishmaniasis is classified as high risk in Cantá, São Luiz, Bonfim, and Boa Vista, with the remaining areas showing very low risk, except for Normandia, which has moderate risk. Regarding visceral leishmaniasis, nine municipalities exhibit very low risk, Cantá and Pacaraima have low risk, and Boa Vista, Bonfim, Normandia and São Luiz are categorized with moderate risk (AdaptaBrasil, 2024).

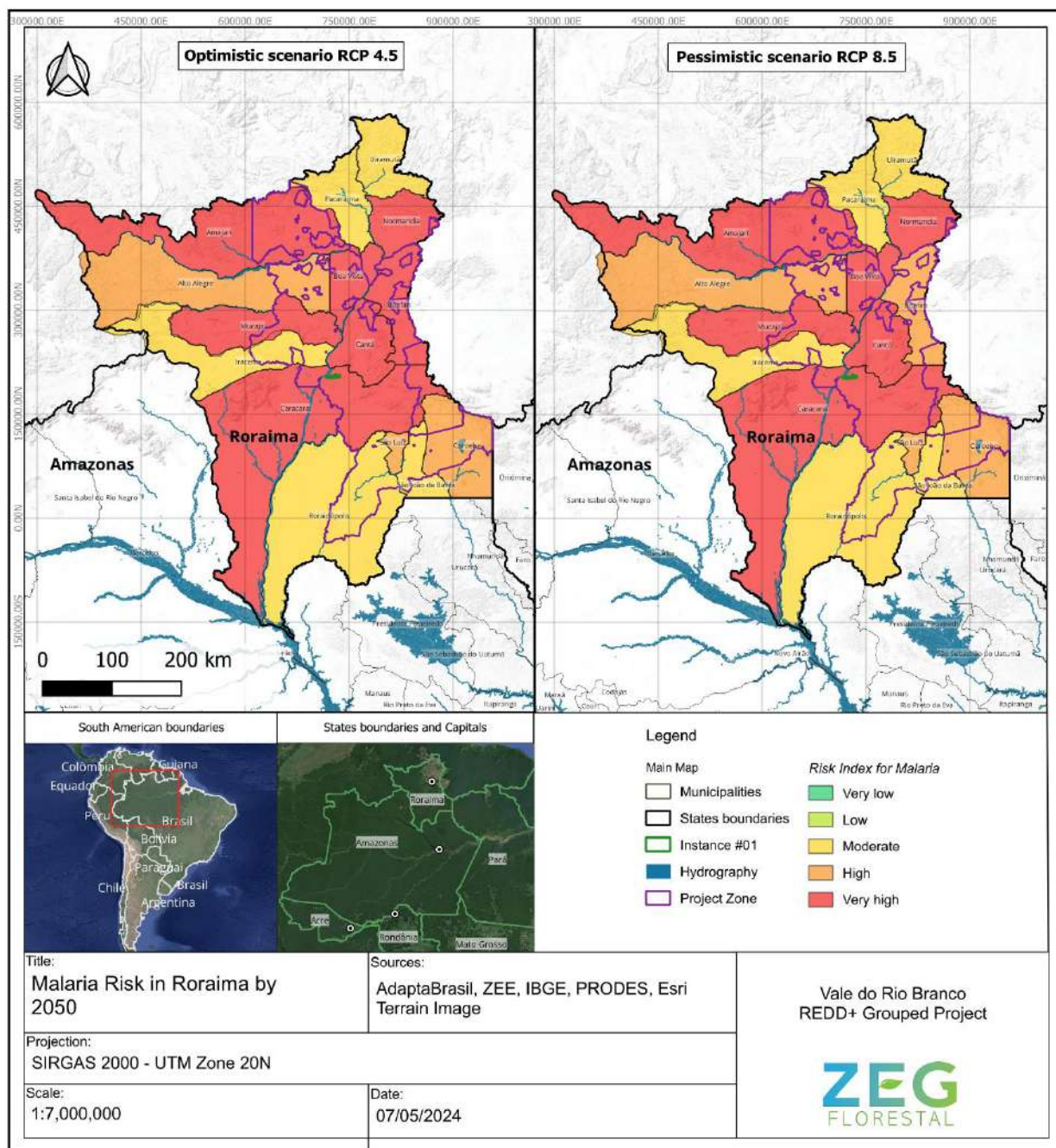


Figure 3.4.2-C: Malaria Risk in Roraima by 2050.

Source: AdaptaBrasil (2024).

Furthermore, forest fires constitute a major threat to public health in the region because they relate to an increase in hospital admissions for respiratory problems, particularly among children and the elderly. Air pollutant levels detected can surpass those found in major urban areas, especially during dry seasons when biomass burning increases (Aragão *et al.*, 2016; de Oliveira Alves *et al.*, 2017; Paralovo *et al.*, 2019).

3.4.2.2 Impacts on Biodiversity Conservation Status

There is robust scientific evidence of the Amazon rainforest's high vulnerability to increasing temperature and repeated extreme drought events (Brienen *et al.*, 2015; Olivares *et al.*, 2015; Feldpausch *et al.*, 2016; Zhao *et al.*, 2017; Anderson *et al.*, 2018; Anjos and De Toledo, 2018; Yang *et al.*, 2018; Barkhordarian *et al.*, 2019; Sampaio *et al.*, 2019; Rammig, 2020; Sullivan *et al.*, 2020). These events, observed in 1998, 2005, 2010, and 2015/2016, are partly attributed to climate change and resulted in high tree mortality rates and basin wide reductions in forest productivity (IPCC, 2022). Exposure to severe and extreme drought has doubled from 8% in 2004/2005 to 16% in both 2009/2010 and 2015/2016. During the extreme drought of 2015/2016, more than 10% of the area showed negative anomalies of the minimum cumulative water deficit (Anderson *et al.*, 2018). This also caused an increase in the occurrence and spread of fires in the basin (Aragão *et al.*, 2018; Lima *et al.*, 2018; Silva Junior *et al.*, 2019; Bilbao *et al.*, 2020).

Additionally, recent studies have aimed to differentiate the impacts of climate change across different regions of the Amazon, considering regional characteristics. For instance, Chen *et al.* (2024) highlighted the varying resilience and vulnerability to drought across the Amazon's biogeography. In the higher-fertility southern Amazon, the response to drought was influenced by water-table depth. In contrast, in the lower-fertility northern Amazon, forests with slower-growing but hardier trees, as well as tall forests with deep-rooted water access, were more drought-resilient regardless of water-table depth. This highlights the importance of a framework for conservation decisions and improved predictions of heterogeneous forest responses to future climate changes.

The combined effect between climate change and land use change, particularly deforestation, is projected to exacerbate threats to the Amazon forest (IPCC, 2022). This includes increased occurrences of fires, forest degradation, and long-term loss of forest structure, all of which contribute to a long-term decline in carbon stocks within the forest biomass. This compromises Amazon's role as a carbon sink. Changes in deforestation rates, temperature increase, and other factors affecting forest dynamics will have an impact on atmospheric CO₂ concentrations and, hence, on the global climate (Ruiz-Vásquez *et al.*, 2020; Sullivan *et al.*, 2020). Over the past decade, the southern portion of the Amazon has become a net carbon source to the atmosphere (IPCC, 2022).

In February 2024, NASA satellites detected unusually widespread and intense fire activity in Roraima. This observation was supported by multiple fire monitoring platforms, including the SERVIR Amazon Fire Dashboard, the Brazilian Space Agency's Queimadas program, and NASA's Fire Information for Resource Management System. Figure 3.4.2-D shows the False-color (bands 7-5-2) image of a burn scar created by a large fire burning near Boa Vista on February 23, 2024, captured by the OLI-2 on Landsat 9.

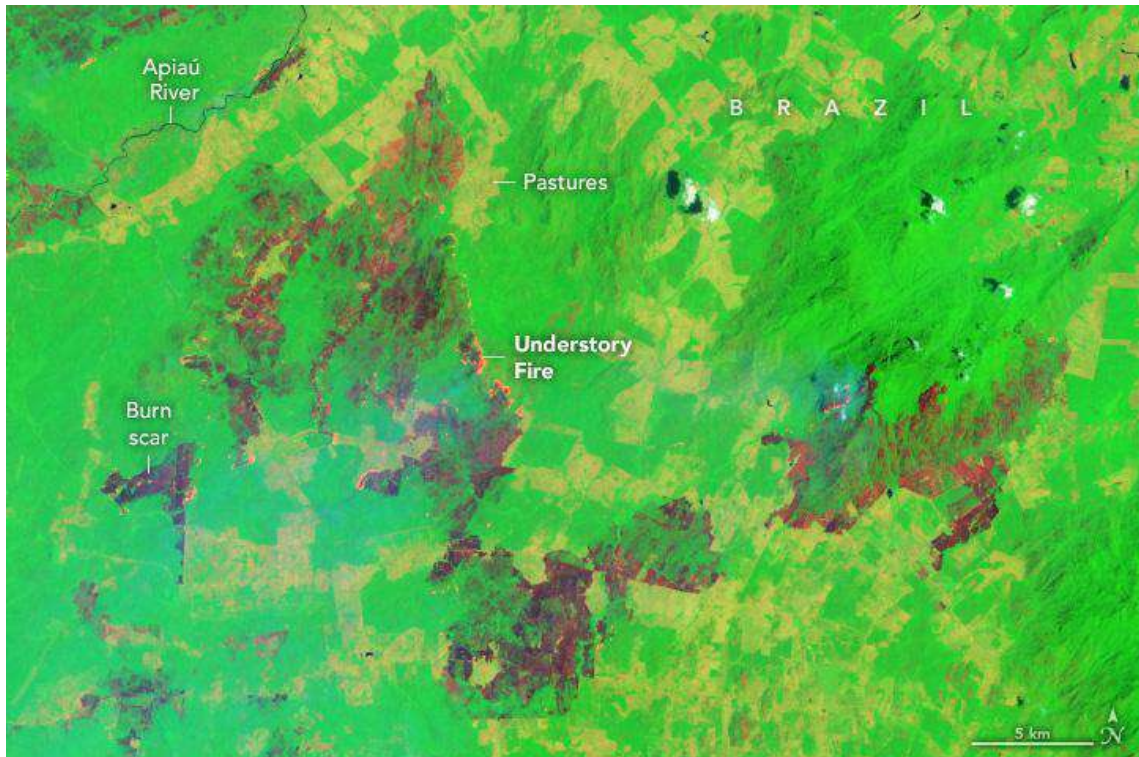


Figure 3.4.2-D: Burn scar created by a large fire burning near Boa Vista.

Source: NASA (2024).

According to the European Commission's Copernicus Atmosphere Monitoring Service, fires in Brazil released more than 4.1 megatons of carbon into the atmosphere in February 2024, the highest February monthly total on record since 2003. Roughly half of those carbon emissions originated from the fires in Roraima. Figure 3.4.2-E below shows smoke streaming from fires southwest of Boa Vista on February 22, 2024, acquired by the MODIS on NASA's Terra satellite. Typically, extreme fire activity in the northern Amazon early in the year precedes an increase in fire activity in the southern Amazon from July through September, often occurring months after the end of El Niño conditions in the tropical Pacific Ocean (NASA, 2024).

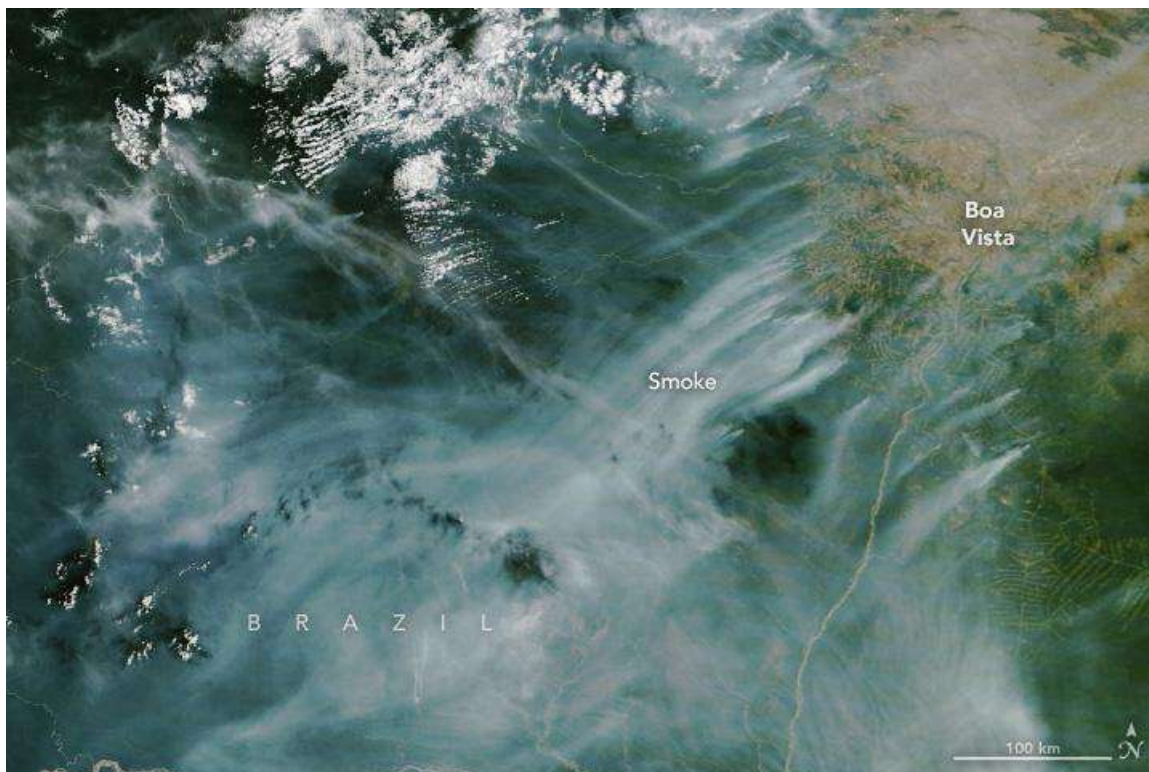


Figure 3.4.2-E: Smoke streaming from fires southwest of Boa Vista. Source: NASA (2024).

Exposure to anomalous fires in ecosystems such as savannahs (found in Central and Northeastern Roraima), which are more fire-prone, increases the exposure and vulnerability of adjacent forest ecosystems not adapted to fire, such as seasonally flooded forests (Bilbao *et al.*, 2020; Flores and Holmgren, 2021). Species in freshwater ecoregions (e.g., the Orinoco and Amazon Rivers and their flooded forests) are predicted to suffer a decrease in range and climatic suitability (Manes *et al.*, 2021). A 4°C rise in temperature is projected to cause a significant 90% reduction in climate refugia for various vertebrate and plant species in the region, with considerable benefits of mitigation and reducing risks to 40% for a 2°C scenario (Warren *et al.*, 2018).

The combined effects of deforestation, fire, expansion of the agricultural frontier, infrastructure development, extractive activities, climate change, and extreme events may exacerbate the risk of converting rainforest to savannah (Nobre *et al.*, 2016; Bebbington *et al.*, 2019; Sampaio *et al.*, 2019; Rammig, 2020). Under the RCP8.5 scenario for 2070, increased drought is expected to accelerate the “*savannisation*”, leading to biodiversity loss and changes in ecosystem functions and services (Anadón *et al.*, 2014; Olivares *et al.*, 2015; Sampaio *et al.*, 2019).

Figure 3.4.2-F summarizes the climatic conditions projected to change and its impacts on communities’ well-being and biodiversity conservation in the Northern South America region. On the left, the climatic conditions indicate projected increases in temperature, extreme heat, fire weather, frequency of hot and dry days, drought frequency, and wind speed, along

with the decrease in mean precipitation and cold spells. On the right, the impacts of these climatic changes are described, including effects on human health, food and water security, and livelihoods of the communities. Additionally, it describes the impacts on forest ecosystems, species, and habitats related to biodiversity conservation.

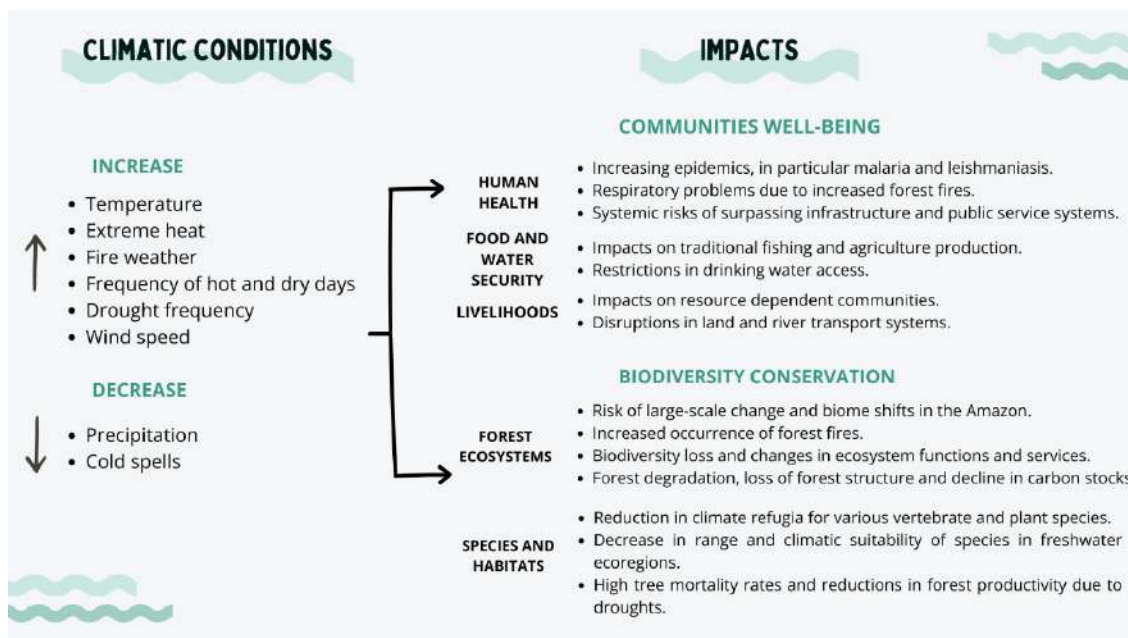


Figure 3.4.2-F: Northern South America expected changes on climatic conditions and impacts on communities and biodiversity.

3.4.3 Measures Needed and Designed for Adaptation (CCB, GL1.3)

This section is under development.

4 COMMUNITY

4.1 Without-Project Community Scenario

This section is under development.

4.2 Net Positive Community Impacts

This section is under development.

4.3 Other Stakeholder Impacts

This section is under development.

4.4 Community Impact Monitoring

This section is under development.

4.5 Optional Criterion: Exceptional Community Benefits

This section is under development.

5 BIODIVERSITY

5.1 Without-Project Biodiversity Scenario

This section is under development.

5.2 Net Positive Biodiversity Impacts

This section is under development.

5.3 Offsite Biodiversity Impacts

This section is under development.

5.4 Biodiversity Impact Monitoring

This section is under development.

5.5 Optional Criterion: Exceptional Biodiversity Benefits

This section is under development.

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